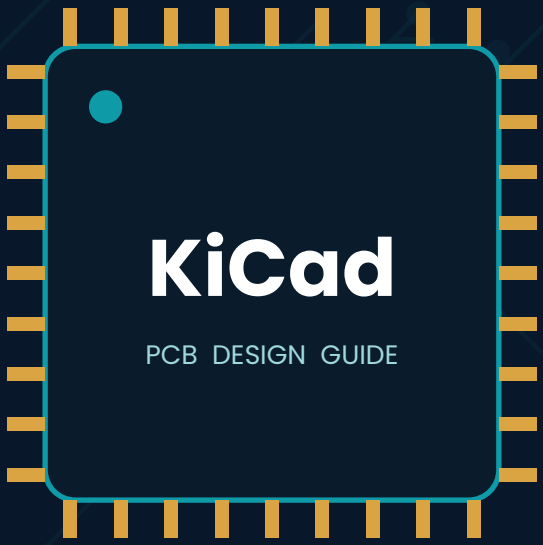


KiCad PCB Design

Complete Beginner's Guide

From the first idea to the factory files,
step by step, in simple English.

Schematic | Footprints | Layout | Routing | Gerber

"Knowledge grows when it is shared."

DESIGNED BY

Engr. Abdul Basit

First Edition · 2026 · Made for first-semester electronics students

Welcome, Future PCB Designer!

This book will teach you how to design a real printed circuit board (PCB) with **KiCad**. KiCad is free. You do not need any money or any special licence. You only need a computer, patience, and the wish to learn.

The book is written in **very simple English**. Sentences are short. Hard words are explained in blue "Easy Words" boxes. Every big idea has a picture. You do not need to be an expert in electronics. If you know what a resistor and a capacitor are, you can start.

How this book is organised

Part	Chapters	What you learn
1. Basics	1 – 2	What a PCB is, the KiCad workflow, and the "why" behind good design: signal integrity, power integrity, EMI, impedance, reflections.
2. Circuit	3 – 4	The parts we use (capacitors, ferrite beads, …) and how to read a datasheet and take a circuit from it.
3. Schematic	5 – 7	New project, schematic, ERC, footprints (download, import, design), package sizes (THT, SMD, SOT, SOP, QFN, DFN).
4. PCB layout	8 – 12	Board setup, stackup, net classes, placement, trace width, impedance, differential pairs, routing, vias, copper pours, and the final Gerber files.
5. Extras	Appendix	Shortcut keys, quick tables, common mistakes, glossary, practice projects.

Boxes you will see

Easy Words

Blue boxes explain difficult words in simple English.

Pro Tip

Green boxes give you a good habit that saves time or avoids mistakes.

Warning

Red boxes show common mistakes. Read them carefully!

Remember

Orange boxes are the most important points. Learn them by heart.

Why?

Purple boxes explain the reason. When you know the reason, you will never forget the rule.

What you need

- A computer (Windows, Linux or macOS) with a mouse. A mouse with a wheel makes zooming easy.
- KiCad version 8 or newer. Download it free from `kicad.org`. Menus in newer versions are almost the same.
- A PDF reader, because you will open many datasheets.
- Optional: a pen and paper. Draw your circuit first. It helps a lot!

> **How to study this book**
>
> - Do not only read. **Open KiCad and do every step** with your own hands.
> - Finish one chapter, then answer the “Check yourself” questions at its end.
> - Make a small practice board (see the Appendix). Real practice is the best teacher.

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01 Introduction to PCB Design and the KiCad Workflow

What is a PCB, what is KiCad, and what are the steps?

In this chapter you will learn

- What a PCB is and why we use it
- The parts of a PCB: copper, pads, traces, vias, solder mask, silkscreen
- What KiCad is and its main programs
- The 10-step workflow from idea to factory files

1.1 What is a PCB?

PCB means **Printed Circuit Board**. It is a flat board that holds electronic parts and connects them with thin copper lines. The copper lines are called **traces**. They work like wires, but they are printed (etched) on the board.

Open any mobile phone, TV remote or computer. You will see a green (or black, blue, white) board full of small parts. That is a PCB. It is the “skeleton and the nerves” of every electronic product.

Easy Words

- **Trace** = a copper line on the board. It carries current or a signal.
- **Pad** = a small copper spot where we solder a part.
- **Via** = a small plated hole that connects one copper layer to another.
- **Component** = any part: resistor, capacitor, chip, connector.
- **Net** = all pins that are connected together, for example all GND pins.

1.2 Why not use wires or a breadboard?

A breadboard is good for learning. But for a real product, a PCB is much better.

	Breadboard / wires	PCB
Connections	Loose, can fall out	Fixed, strong, no loose wire
Size	Big	Small and neat
Speed of signals	Poor above a few MHz	Good, even at GHz (if designed well)
Noise	Very noisy	Low noise with good design

	Breadboard / wires	PCB
Making 100 copies	Impossible	Easy and cheap
Looks	Messy	Professional

1.3 Parts of a PCB

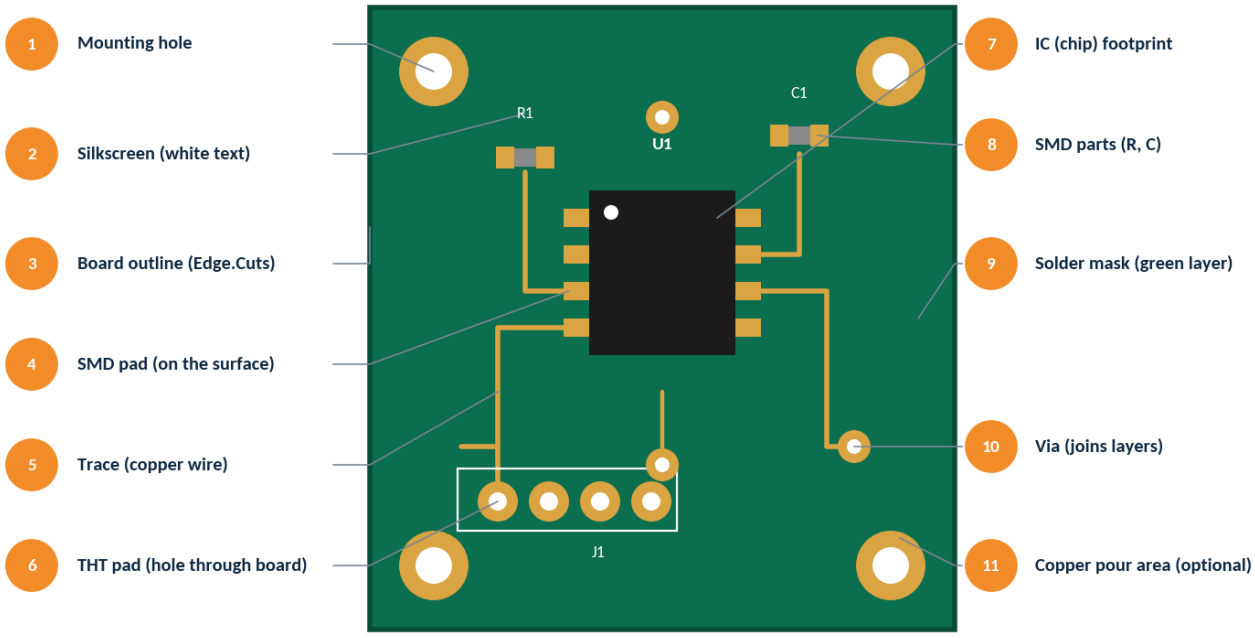


Figure 1. *A simple PCB seen from the top. The numbers show the most common parts.*

Now let us look inside the board. A PCB is like a sandwich.

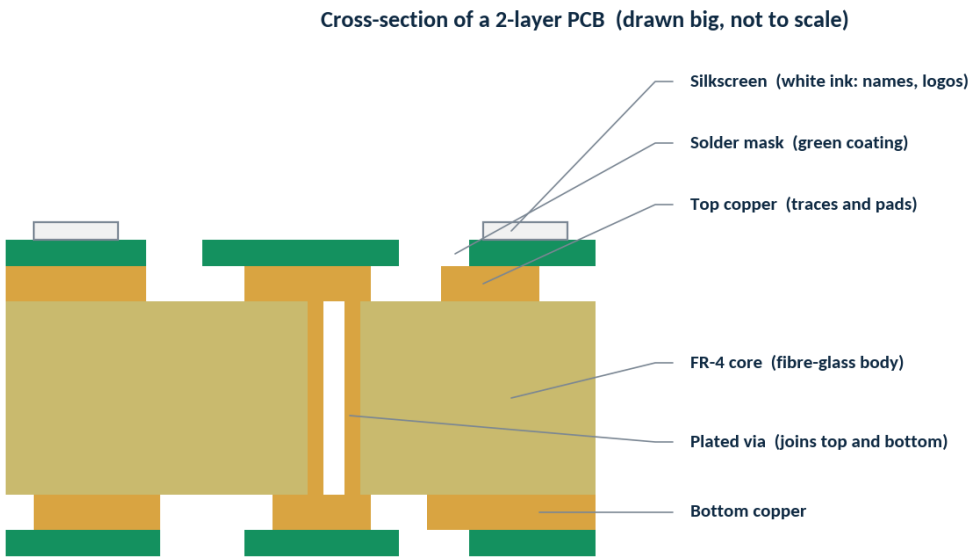


Figure 2. *Cross-section of a 2-layer board. The via is a copper tube that joins top and bottom copper.*

Layer / part	What it is	Why we need it
FR-4 core	Fibre-glass with epoxy. The board body.	Holds everything and isolates the copper layers.
Copper	Thin copper foil, usually 35 µm (1 oz) thick.	Makes traces, pads and planes.
Solder mask	Green (or other colour) coating.	Protects copper and stops solder from joining wrong pads.
Silkscreen	White ink print.	Shows part names (R1, C2), logos, pin 1 marks.
Via	Plated hole.	Joins traces on different layers.
Plane / pour	Big area of copper (often GND).	Gives a low-resistance ground and a clean return path.

1.4 How many layers?

Layers	Used for	Comment
1 layer	Very simple, cheap products (toys, LED lamps)	All traces on one side. Many jumpers.
2 layers	Arduino-type boards, small microcontroller boards	Most common for beginners. Top and bottom traces.
4 layers	USB, HDMI, Ethernet, DDR, fast microcontrollers	Two inner layers: GND and POWER. Much better signal quality.
6+ layers	Phones, laptops, FPGA, high-speed boards	More routing space and more planes. Higher cost.

1.5 What is KiCad?

KiCad is a **free and open-source** program to design electronics. Professionals and companies use it. It is not a toy. It can design boards with many layers and high-speed signals. KiCad has many small programs inside one package. The main window is called the **Project Manager**.

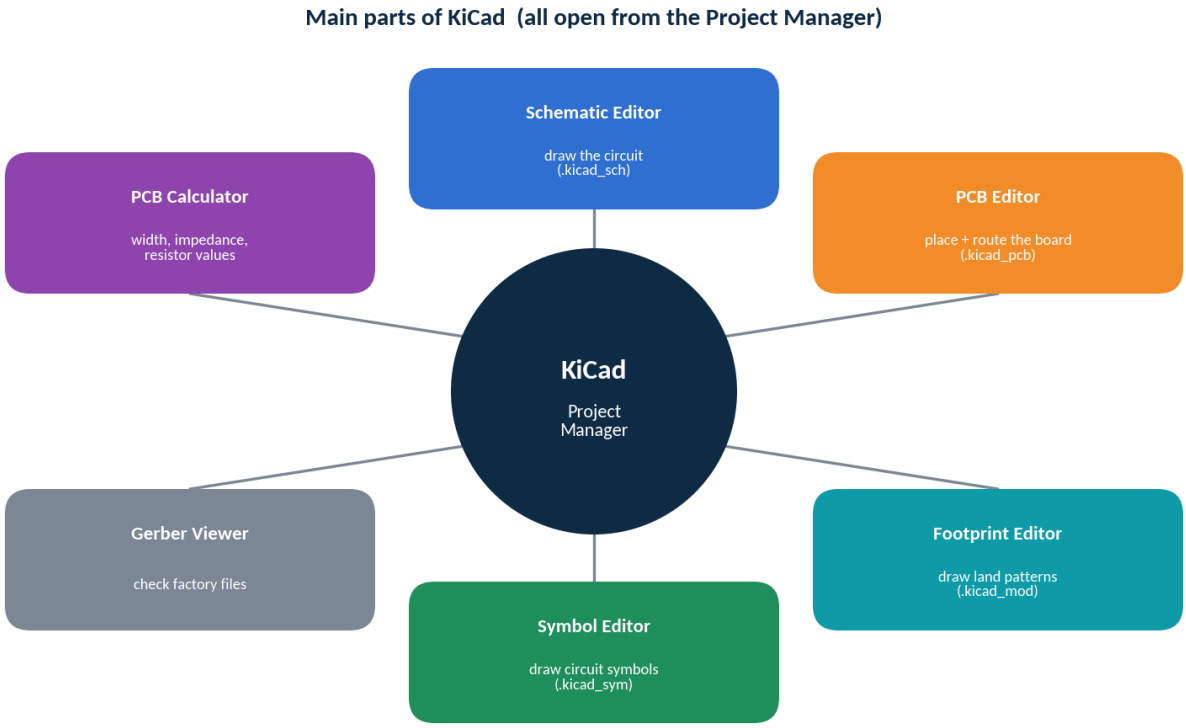


Figure 3. KiCad has separate programs for each job. All are opened from the Project Manager.

Program	Job	File made
Schematic Editor	Draw the circuit with symbols and wires	.kicad_sch
Symbol Editor	Make or edit circuit symbols	.kicad_sym
Footprint Editor	Make or edit the real shapes (pads) of parts	.kicad_mod
PCB Editor	Place parts and route traces	.kicad_pcb
Gerber Viewer	Check the files before sending to factory	.gbr
Calculator Tools	Trace width, impedance, resistor values and more	—

1.6 The KiCad workflow

Every board follows the same path. Learn this path well. In this book, each chapter follows one or more of these steps.

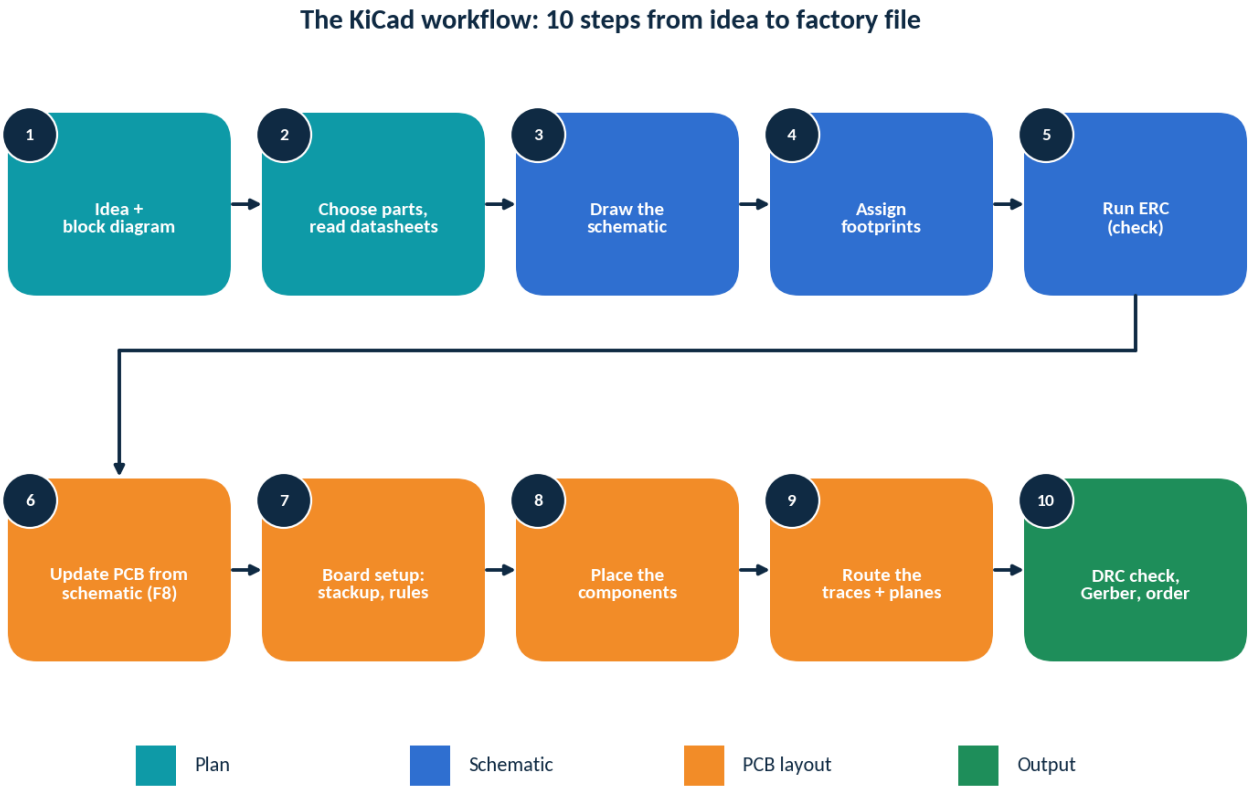


Figure 4. The complete KiCad workflow in 10 steps.

1. **Idea and block diagram.** What must the board do? Draw boxes: power, microcontroller, sensors, connectors.
2. **Choose parts and read datasheets.** Find the chips. Read how to connect them (Chapters 3 and 4).
3. **Draw the schematic.** Place symbols, connect with wires and labels (Chapter 5).
4. **Assign footprints.** Tell KiCad the real shape of every part: 0603, SOIC-8, QFN... (Chapters 6 and 7).
5. **Run ERC** (Electrical Rules Check). KiCad finds mistakes in your circuit.
6. **Update PCB from schematic.** All parts and connections go to the PCB editor (Chapter 8).
7. **Board setup.** Choose layers (stackup), clearances, trace widths and net classes (Chapter 8).
8. **Place the components** in smart places (Chapter 9).
9. **Route the traces** and make planes (Chapters 10 and 11).
10. **DRC, finishing and Gerber files** for the factory (Chapter 12).

1.7 Your project folder

KiCad saves all files of one project in one folder. Keep this folder clean.

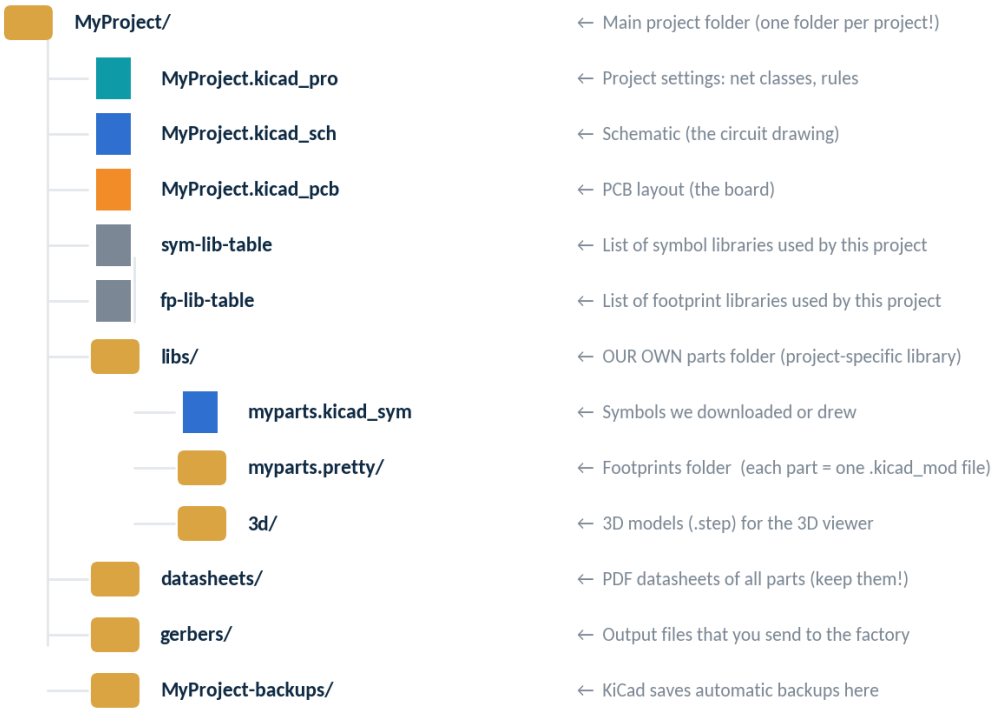


Figure 5. A good project folder. The "libs" folder holds the project-specific library.

Warning

- Make **one folder for each project**. Never mix two projects in one folder.
- Do not rename or move project files with Windows Explorer. Use KiCad's own menu: **File › Save As**
- Keep a backup copy (USB or cloud) after every working day.

Pro Tip: use clear names

Use simple names without spaces, for example "led_blinker_v1". Spaces and special letters can create problems on some computers.

Check yourself

1. What is the difference between a trace and a via?
2. Why is a 4-layer board better than a 2-layer board for fast signals?
3. Name three programs inside KiCad and what each one does.
4. Write the 10 workflow steps in your own words.

02 Theory: Why Good PCB Design Matters

Signal integrity, power integrity, EMI, impedance and reflections in simple words

In this chapter you will learn

- Why a trace is not "just a wire" at high speed
- Signal integrity (SI), power integrity (PI) and EMI
- Controlled impedance and reflections
- Return current, crosstalk and the 3W rule

Many beginners think: "If the schematic is correct, the board will work." This is **not always true**. A wire on a PCB has resistance, capacitance and inductance. When signals are slow, we can ignore this. When signals are fast, the trace becomes part of the circuit. This chapter explains why.

2.1 When does a trace matter?

The important number is the **rise time**: how fast a signal goes from 0 to 1. Fast edges contain very high frequencies, even if the clock is slow.

A signal travels on an outer layer of FR-4 at about **170 mm every nanosecond** (ns). This is a little more than half the speed of light. Use this simple rule:

Rule of thumb

A signal edge of **1 ns** travels about **170 mm** during its rise.

If the trace is longer than **1/6 of that length** (about **28 mm** for 1 ns), treat the trace as a **transmission line**.

Faster edge → shorter limit. A 0.2 ns edge → only 5 mm!

Signal	Typical rise time	Trace becomes a transmission line when longer than
LED, button, slow sensor	> 100 ns	Never (ignore)
I2C 400 kHz	≈ 100 ns	Never (ignore)
SPI 10 MHz, 3.3 V logic	≈ 2 ns	≈ 55 mm
USB 2.0 High-Speed (480 Mb/s)	≈ 0.5 ns	≈ 14 mm
PCIe, USB 3, HDMI (Gb/s)	< 0.1 ns	< 3 mm → always

2.2 Signal Integrity (SI)

Signal integrity means: the signal arrives at the receiver clean and on time, with the correct shape. Bad signal integrity gives you "strange" problems: random resets, wrong data, Wi-Fi that drops, USB that does not enumerate.

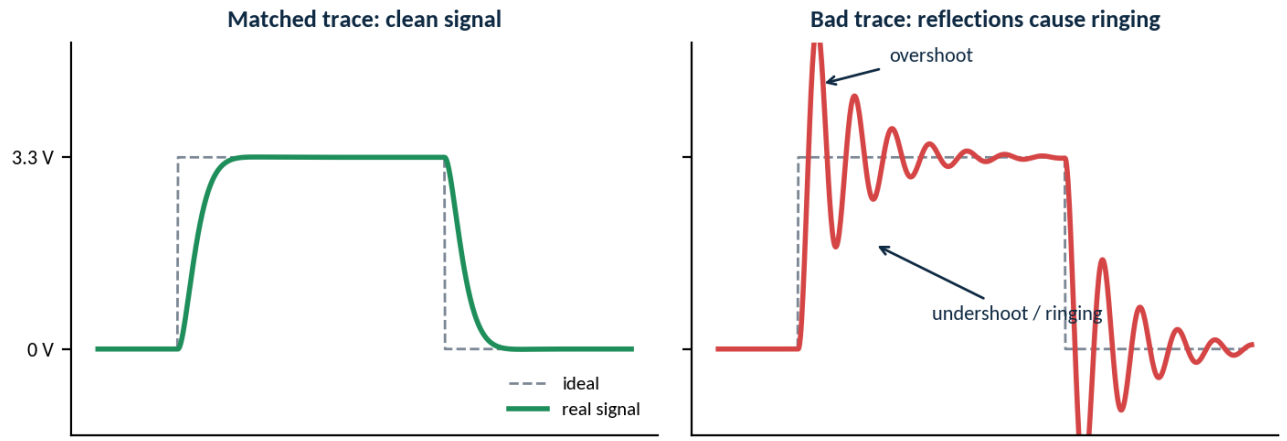


Figure 6. Left: a clean signal. Right: the same signal with reflections. The receiver may read wrong values.

Problem	Simple meaning	Main cure
Reflection / ringing	Signal bounces back and forth on the trace	Controlled impedance, correct termination, short stubs
Crosstalk	A neighbour trace "talks" into your trace	More spacing (3W), ground plane close by
Attenuation (loss)	Signal gets weaker on long traces	Short traces, good material
Skew	Two signals of a pair arrive at different times	Length matching
Ground bounce	The ground voltage moves when chips switch	Solid ground plane, decoupling capacitors

2.3 Controlled impedance

Impedance (symbol Z, unit ohm Ω) is how a trace "resists" a fast signal. Think of a water pipe. If the pipe suddenly becomes thinner or wider, the water splashes back. In the same way, if the impedance of a trace changes suddenly, the signal is reflected.

For a trace over a ground plane, the impedance is called **Z0** (say "Z-zero"). It depends only on the **geometry** (trace width, height above plane, copper thickness) and the **material** (εr of the board). It does NOT depend on trace length.

Common target impedances

- **50 Ω** single-ended: RF, clocks, fast single traces.
- **90 Ω** differential: USB.
- **100 Ω** differential: Ethernet, HDMI, LVDS, many others.
- PCIe: 85 Ω or 100 Ω, depending on the chip guide.

Controlled impedance means: we choose the trace width and the layer stack so that the impedance has the right value everywhere along the trace. This is the main job when we route USB, HDMI, Ethernet and PCIe.

2.4 Reflections

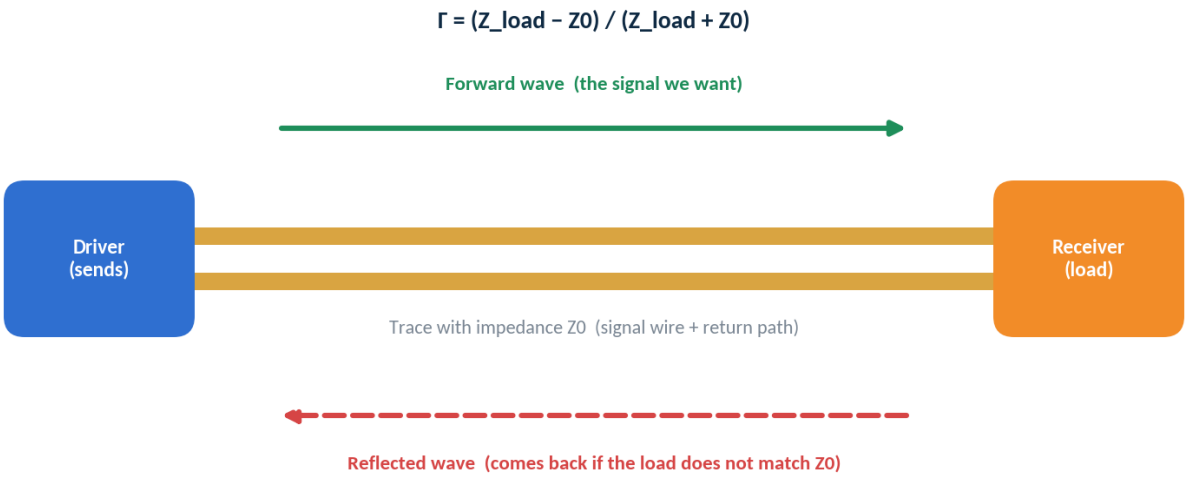


Figure 7. A reflection happens when the load does not match the impedance Z0 of the line.

The reflection coefficient Γ (gamma) tells us how big the reflected wave is:

Formula

Γ = (Z_load − Z0) / (Z_load + Z0)

Load	Γ	What happens
Z_load = Z0 (matched)	0	No reflection. Perfect!
Open end (very high Z)	+1	Full reflection, same sign. Overshoot.
Short to ground (Z = 0)	−1	Full reflection, opposite sign.
Z_load = 100 Ω on 50 Ω line	+0.33	33 % of the wave comes back.

How to avoid reflections

- Keep the impedance of the trace the same from start to end (same width, same layer, same reference plane).
- Avoid stubs (short branches) and big pads on fast lines.
- Use a **series resistor** (22 – 33 Ω) near the driver of fast single-ended signals when the chip guide says so.
- Terminate the line at the end when the interface requires it (for example 100 Ω across LVDS lines).

2.5 The return current: the hidden wire

Current always flows in a **loop**. The signal goes out on the trace. It must come back to the source. At high frequency, the return current chooses the path of **least impedance**: this is directly **under the trace** on the ground plane.

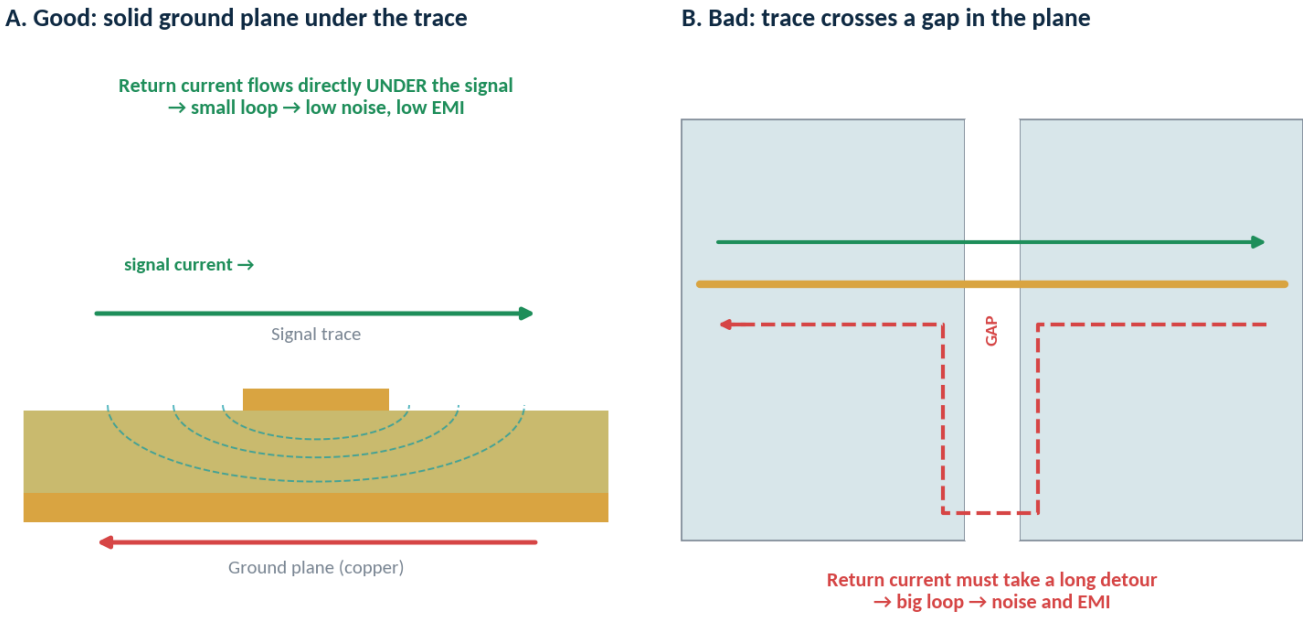


Figure 8. The return current flows under the signal trace. If the plane has a gap, the current must go around it.

> **Golden rule**
>
> Never let a fast trace cross a gap or slot in its reference plane. Give every fast trace an unbroken ground plane below it.

2.6 Crosstalk and the 3W rule

Two traces that run side by side form a small capacitor and a small transformer. A fast signal on one trace creates a noise signal on the other. This is **crosstalk**. The closer and the longer the traces run in parallel, the stronger it is.

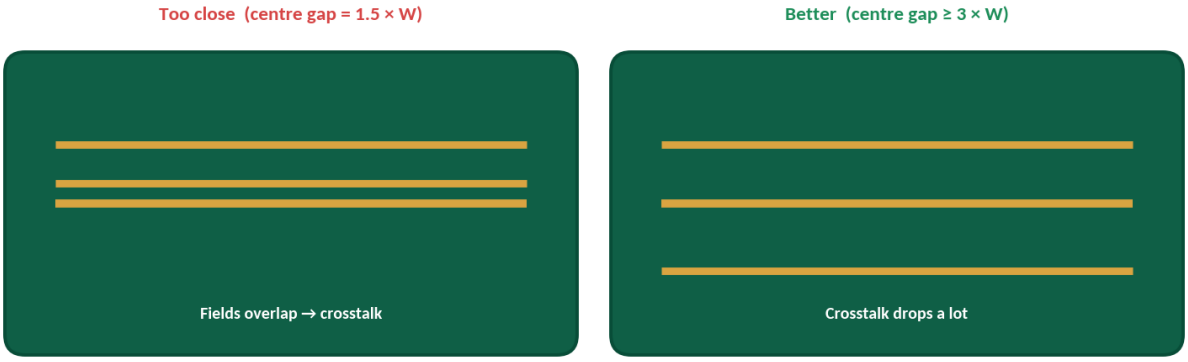


Figure 9. The 3W rule: keep the centre-to-centre distance at least 3 times the trace width (W).

The **3W rule** says: the distance from the centre of one trace to the centre of the next trace should be at least 3 × W. It cuts crosstalk by a large amount. For very noisy or very sensitive lines, use more space. A ground plane close under the traces also reduces crosstalk.

2.7 Power Integrity (PI)

Power integrity means: every chip gets a clean and steady supply voltage. Digital chips take current in very short, sharp pulses. If the supply path has resistance or inductance, the voltage at the chip pin drops and makes noise.

- **Voltage drop (IR drop):** too thin traces waste voltage. Use wide traces or copper planes.
- **Inductance:** long thin power wires act like inductors. They fight sudden current changes. Use short and wide connections and planes.
- **Decoupling:** small capacitors placed close to each chip give the fast current locally (Chapter 3).
- **Ground bounce:** when many outputs switch together, the ground at the chip lifts a little. A solid ground plane makes this small.

2.8 EMI: Electromagnetic Interference

EMI means that your circuit sends out radio noise (or is hurt by noise from outside). A PCB with large current loops behaves like a small antenna. Products must pass EMC tests before they can be sold (for example CE or FCC). Good layout is the cheapest way to pass.

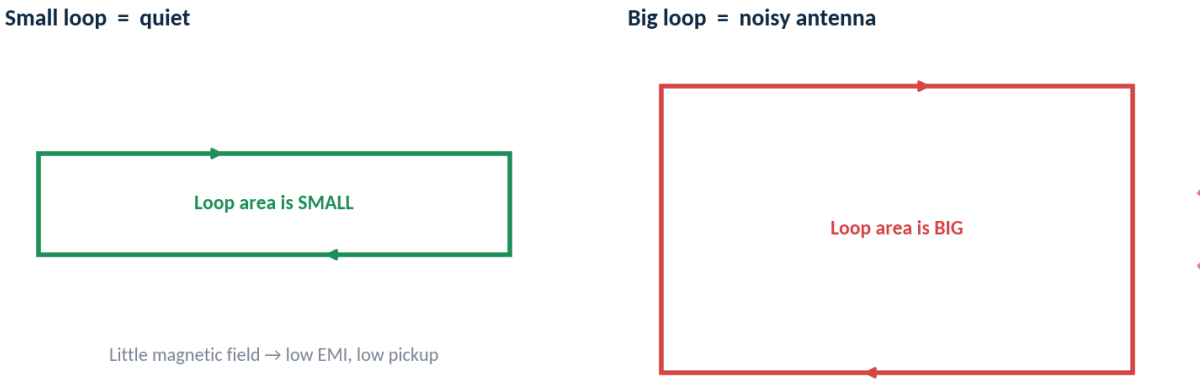


Figure 10. A small loop makes little EMI. A big loop acts as an antenna.

Five simple EMI rules

- Keep **loop areas small**: signal and its return close together.
- Use a **solid ground plane**.
- Keep fast traces **short**.
- Place **decoupling capacitors** right at the chip pins.
- Do not leave floating copper. It can act as an antenna. Stitch it with ground vias or remove it.

2.9 Summary of the golden rules

Topic	Rule
Impedance	Fix width and layer for USB, HDMI, PCIe, Ethernet and RF lines. Do not change them on the way.
Return path	Solid ground plane directly below fast traces. No gaps.
Crosstalk	Keep 3W spacing. Do not run fast lines in parallel for long.
Power	Wide traces or planes. Decoupling capacitor at every power pin.
EMI	Small loops, short traces, no antennas.

Check yourself

1. A signal has a rise time of 2 ns. After which trace length should you worry about reflections?
2. What is the reflection coefficient when the load is equal to Z0?
3. Why must a fast trace not cross a gap in the ground plane?
4. Explain the 3W rule in your own words.
5. Name three causes of bad power integrity.

03 Understanding the Components

What each part does: capacitors, ferrite beads, diodes, crystals and more

In this chapter you will learn

- The job of resistors, capacitors, inductors, diodes and crystals
- Decoupling capacitors, big electrolytic caps and small ceramic caps
- AC coupling capacitors
- Ferrite beads as high-frequency noise suppressors

Before drawing a schematic, you must know what each part is for. If you know the job of a part, you will also know where to place it on the PCB.

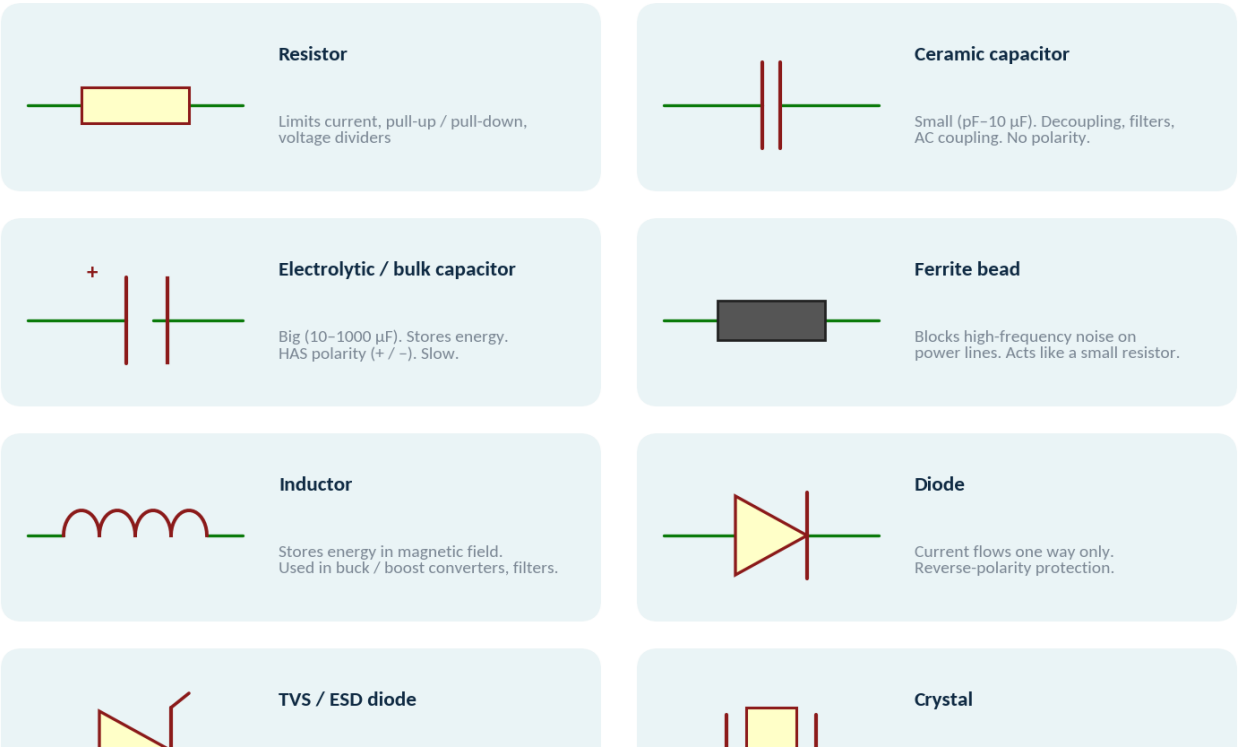


Figure 11. *The most common parts and their jobs.*

3.1 Resistors

- **Current limiting:** an LED needs a resistor. R = (Vsupply − Vled) / I. Example: (3.3 − 2.0) / 0.005 = 260 Ω → use 270 Ω.
- **Pull-up / pull-down:** keeps an input at a known level (1 or 0) when nothing drives it. Typical 10 kΩ. For I2C use 2.2 kΩ – 4.7 kΩ pull-ups on SDA and SCL.
- **Voltage divider:** two resistors make a lower voltage. Used for regulator feedback and ADC inputs.
- **Series termination:** 22 – 33 Ω in series with a fast output reduces ringing.
- **Tolerance:** 1 % resistors are cheap and good. Use them for dividers.

3.2 Capacitors: the most important parts on your board

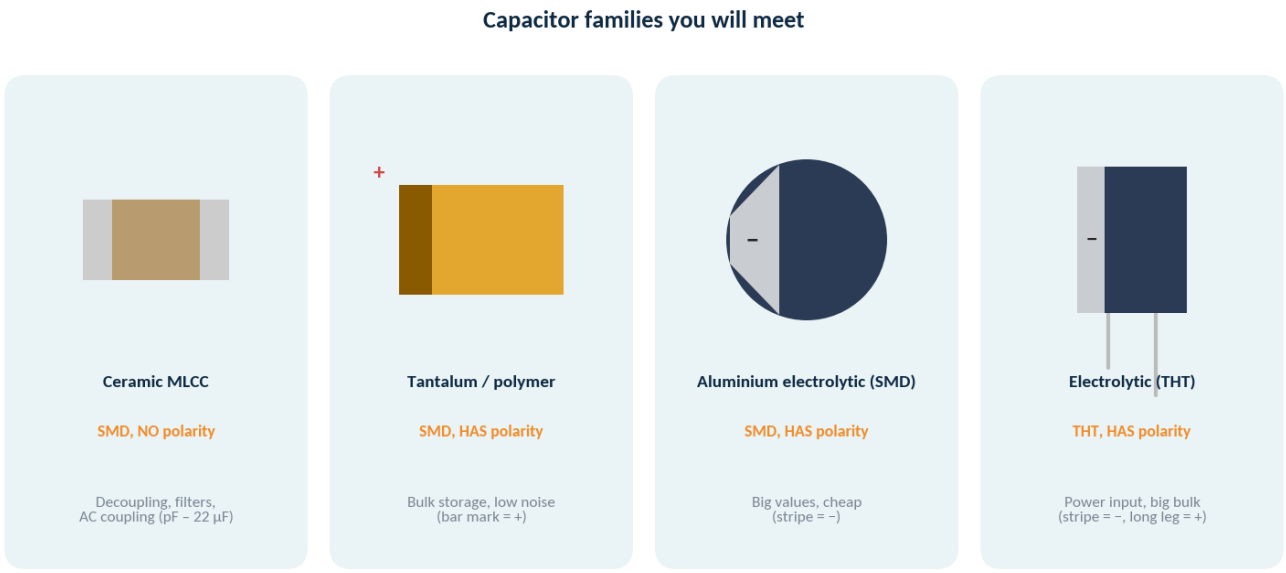


Figure 12. *Capacitor families. Always check the polarity of tantalum and electrolytic capacitors.*

A capacitor is like a tiny rechargeable battery. It can give or take current very quickly. But a real capacitor also has a little inductance (ESL) and resistance (ESR). Because of ESL, **every capacitor works well only in a certain frequency range**.

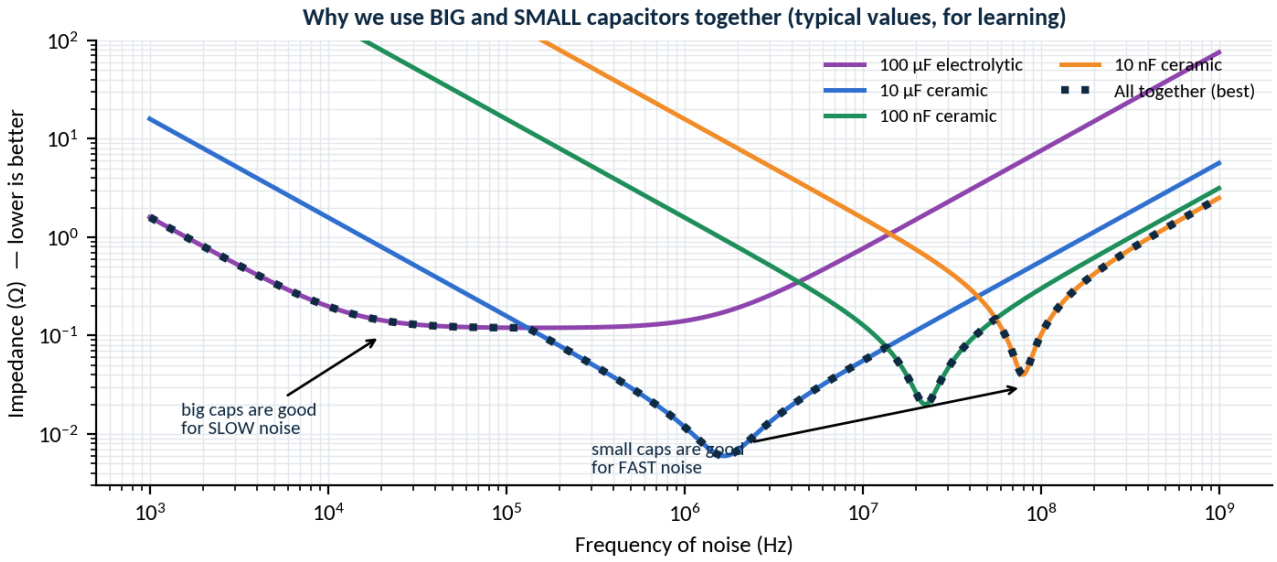


Figure 13. Impedance of different capacitors. Big capacitors are good at low frequency. Small ones are good at high frequency. We use both.

Capacitor	Typical value	Where we use it	Remember
Big electrolytic / polymer	47 – 1000 µF	Power input, near connectors, motors, regulators output	Slow. Stores lots of energy. Check polarity and voltage rating.
Ceramic (MLCC) medium	1 – 22 µF	Regulator input / output, local bulk near ICs	Voltage rating: use ≥ 2× the working voltage. Real value drops with DC voltage!
Ceramic small	100 nF (0.1 µF)	At EVERY power pin of every IC	The classic decoupling capacitor. Place as close as possible.
Ceramic very small	1 – 10 nF, 10 – 100 pF	High-speed ICs, RF, crystals	Handles very fast noise.

3.3 Decoupling (bypass) capacitors

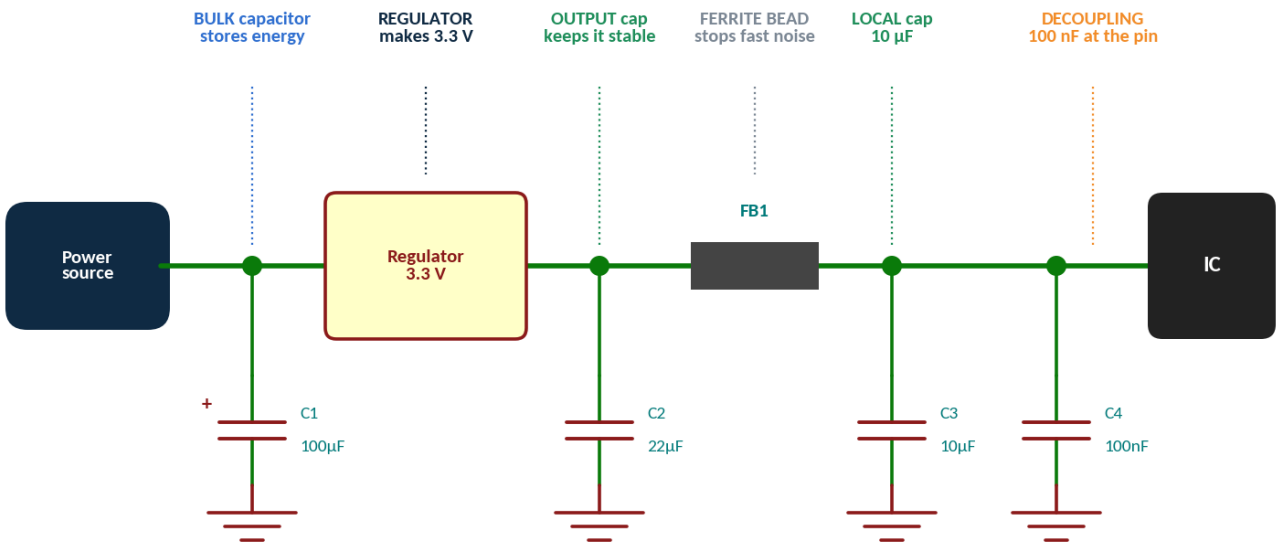
When a digital chip switches, it needs a pulse of current **right now**. The power supply is far away. The wire between them has inductance and cannot deliver the current fast enough. The solution: a small capacitor **right next to the chip pin** gives that pulse of current locally.

> **Why 100 nF at every pin?**
>
> The capacitor is a small local energy tank. If it is close, the current path is very short and the inductance is very small. If it is far, the long wire has high inductance and the capacitor becomes useless.

Decoupling rules

- **One 100 nF capacitor for each power pin** of an IC (VDD / VCC).
- Add one bigger capacitor (1 – 10 µF) near the IC or group of ICs.
- Place the capacitor **between the power source and the pin**, as close to the pin as possible.
- Connect its ground side to the ground plane with a via placed right at the pad.
- Use small packages (0402 / 0603): smaller package = smaller inductance.

3.4 Bulk capacitors and the power chain



Power path: each part has one job. Together they give clean power to the IC.

Figure 14. A good power chain: bulk capacitor → regulator → ferrite bead → local capacitors → IC.

Big electrolytic or polymer capacitors are **bulk capacitors**. They store a lot of energy and handle slow changes. They also help when the cable from the power source is long (for example a USB cable or battery wires). Place them where power enters the board.

3.5 AC coupling capacitors

An **AC coupling capacitor** is placed in series with a signal line. It blocks DC (steady voltage) and passes AC (changing signal). It is used when two chips have different DC levels, or when we want to remove a DC offset.

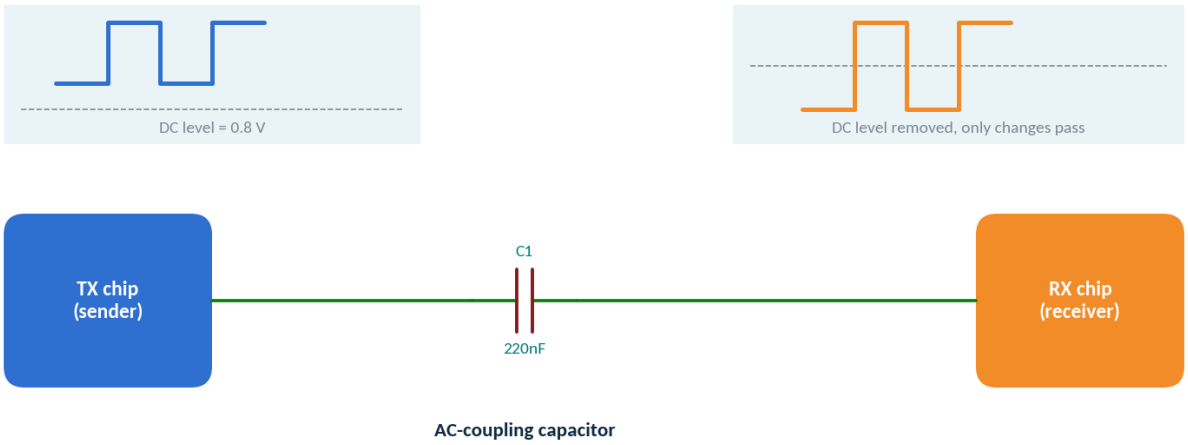


Figure 15. An AC coupling capacitor blocks the DC level and lets the fast signal pass.

- **Where:** high-speed serial links (PCIe, SATA, HDMI inputs, USB 3 TX lines), audio inputs, RF signals.
- **Typical value:** 100 nF to 220 nF for high-speed serial pairs. For example, many PCIe reference designs put 100 – 220 nF capacitors in series with the TX pairs. Audio needs bigger values (1 – 10 µF) so low tones can pass.
- **Placement:** put the two capacitors of a differential pair side by side, same size, same orientation, close to the connector or chip. Keep the pair symmetric.
- **Choose:** a small package (0402) with a good dielectric (X7R). Do not use a big package: it adds extra inductance.

> **Do not confuse**
>
> AC coupling capacitors (in the signal path) and decoupling capacitors (between power and ground) look the same in the schematic but have different jobs and different placement rules.

3.6 Ferrite beads: high-frequency noise suppressors

A **ferrite bead** is a small part that looks like a resistor. At low frequency (DC) it is almost a wire (very low resistance). At high frequency (10 MHz – 1 GHz) it becomes a resistor and turns noise into a little heat. It is used in series with the power line to **stop high-frequency noise** from going from one part of the board to another.

- **Use it for:** separating a noisy digital supply from a clean analog supply, PLL and USB supply lines, power entering sensitive ICs.
- **How to choose:** look at the **impedance at 100 MHz** (for example 600 Ω), the **rated current** (choose at least 2 × your current) and the **DC resistance** (should be small, to avoid voltage drop).
- **Ferrite bead + capacitor** make a simple low-pass filter. Put a bead in the line and a capacitor to ground after it.

- **Ferrite bead ≠ inductor:** a bead is lossy (it removes noise as heat). An inductor stores energy and is used for DC-DC converters.

3.7 Inductors, diodes and crystals

- **Power inductor:** used with a buck or boost converter. Choose the value and saturation current from the converter datasheet. Use a shielded type for less EMI.
- **Schottky diode:** low voltage drop (0.2 – 0.4 V). Used for reverse-polarity protection, OR-ing two supplies and in converters.
- **TVS / ESD diode:** protects against static spikes. Place it **at the connector**, on USB, HDMI, buttons and other lines touched by people. For fast lines choose a **low-capacitance** type (often below 0.5 pF).
- **Zener diode:** keeps a voltage at a fixed level.
- **Crystal:** gives an exact clock. It needs two load capacitors. Formula: C_load_each = 2 × (CL − C_stray). Example: CL = 12 pF and stray ≈ 3 pF → C = 2 × 9 = 18 pF. Place the crystal very close to the chip and keep the traces short.

3.8 Regulators

	LDO (linear regulator)	Buck converter (switching)
Principle	Burns extra voltage as heat	Switches fast, uses an inductor
Efficiency	Low when Vin is much higher than Vout	High (85 – 95 %)
Noise	Very quiet	Switching noise (needs careful layout)
Parts	2 capacitors	Inductor, diode (or integrated), capacitors, resistors
Use	Low current, clean supply, small drop	High current, big drop, battery products

3.9 Other useful parts

- **Fuse / PTC resettable fuse:** protects the board from over-current at the input.
- **Level shifter / MOSFET shifter:** connects a 5 V signal to a 3.3 V chip, for example for HDMI DDC lines (I2C).
- **Test points:** small pads so you can measure voltages (3.3 V, 5 V, GND) with a probe.
- **Mounting holes and fiducials:** for screws and for the assembly machine.
- **Galvanic isolation (transformer):** Ethernet uses a magnetics transformer to separate the board from the cable electrically.

Real example: a high-speed carrier board

A professional carrier board that provides USB 3, HDMI, PCIe and Ethernet uses many of the parts above:

- **AC coupling capacitors** (about 220 nF) in series with PCIe TX pairs.
- **Low-capacitance ESD diodes** at the USB 3 and HDMI connectors.
- **Schottky diodes** to stop one supply from feeding another.
- **Level shifters and pull-ups** on the HDMI DDC and hot-plug lines.
- **100 nF decoupling capacitors** on every power pin, plus bulk capacitors at the input.
- **Fiducials and mounting holes** for assembly.

Check yourself

1. Why do we use both a 100 nF and a 10 µF capacitor near an IC?
2. What is the difference between a decoupling capacitor and an AC coupling capacitor?
3. What does a ferrite bead do at high frequency?
4. Calculate the LED resistor for a 5 V supply, 2.0 V LED, 10 mA.
5. Why must an ESD diode be placed near the connector?

04 Circuit Design and Reading Datasheets

How to find a circuit, read a datasheet, and copy a working design

In this chapter you will learn

- How to go from idea to circuit
- How to find the datasheet and what each section means
- How to read tables and graphs
- A worked example: a 3.3 V regulator

4.1 From idea to circuit

Nobody designs a whole circuit from zero. Engineers **use the circuits that the chip makers give**. Your job is to choose the parts, read the datasheets, and connect them correctly.

1. **Write the requirements.** Input voltage? Output voltage? Current? Which connectors? Size of board?
2. **Draw a block diagram.** Boxes: power → microcontroller → sensor → connector.
3. **Choose one main chip for each box.** Check price, stock (for example on LCSC or DigiKey) and package.
4. **Download every datasheet** and save it in your project folder “datasheets”.
5. **Copy the typical application circuit** from the datasheet. Change values with the formulas in the datasheet.
6. **Check power:** total current, heat, voltage of each rail.

4.2 Where do I find circuits?

Source	What you get	Quality
Datasheet: “Typical Application”	A small circuit for that chip	Very good. Start here.
Evaluation board schematic (EVM)	Complete working board, drawn by the chip maker	Excellent. Study it.
Reference design / application note	Full system, layout tips	Excellent.
Open-source hardware (GitHub)	Real boards from other engineers	Good. Check it carefully.
Random web forums	Unchecked circuits	Be careful!

4.3 What is a datasheet?

A **datasheet** is the "user manual" of an electronic part. It is written by the manufacturer. It tells you what the part does, its pins, its limits and how to use it. Download datasheets from the manufacturer's website (best), or from LCSC, DigiKey, Mouser. Always check that the **part number matches exactly**, including the package letters at the end.



Figure 16. *The parts of a typical datasheet. Learn where to find each one.*

4.4 Reading a datasheet: section by section

#	Section	What to look for
1	Front page	Part name, short description, main numbers (voltage, current), package.
2	Features	Quick check: is this chip right for my project?
3	Pin configuration	Which pin does what. Pin 1 position. Pin types: power, input, output, NC (not connected).
4	Absolute maximum ratings	**Never exceed these.** The chip may be damaged. They are NOT working values.
5	Recommended operating conditions	The range in which the chip works correctly. Design inside this range.
6	Electrical characteristics	Min / Typ / Max values under stated conditions. Use the worst case.
7	Typical performance graphs	How the chip behaves with temperature, load, voltage.
8	Typical application circuit	The circuit to copy, with part values.
9	Layout guidelines	**Very important for PCB.** Where to place capacitors, how to route, where to put vias.
10	Package information	Drawing with dimensions and the **recommended land pattern** (footprint).

Absolute maximum is NOT a working limit

If the absolute maximum input voltage is 7 V and you apply 6.9 V all day, the chip may live for a short time but it will not be reliable. Stay below the **recommended** range, with a safety margin of at least 20 %.

4.5 Min, Typ and Max

In the electrical table each value has up to three numbers. **Min** = the lowest value you can expect. **Typ** = typical (average). **Max** = the highest value. Use **Typ** only for estimation. For safe design use the **worst case**: Min for “at least” values, Max for “not more than” values. Always read the **test conditions** (temperature, voltage, load) next to the number.

4.6 Reading a graph

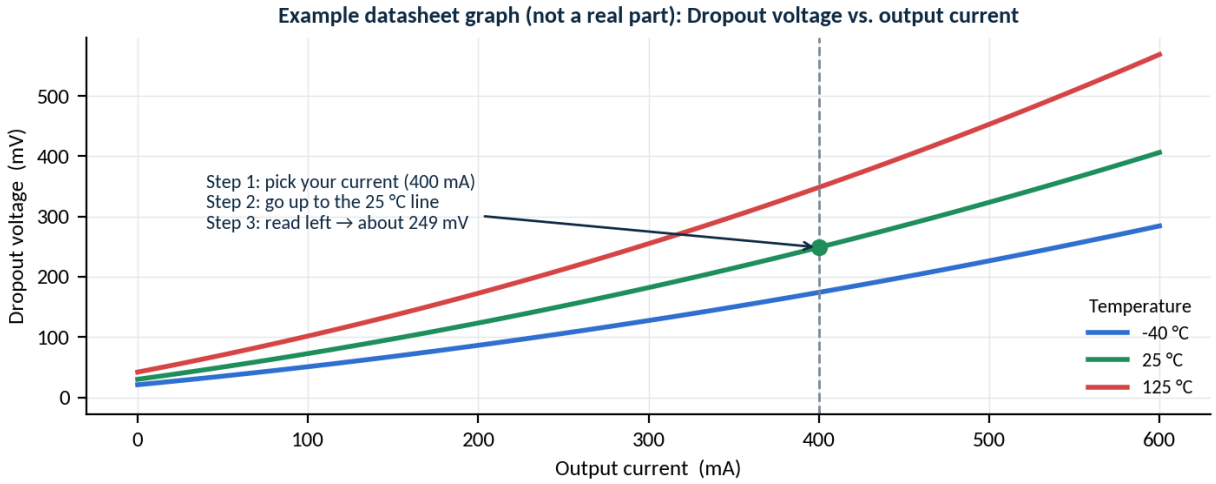


Figure 17. Reading a graph: find the condition on the x-axis, go up to the correct curve, then read the y-axis.

4.7 Words you will meet in datasheets

Short name	Meaning
Vin, Vout	Input and output voltage
Iout / Iload	Output (load) current
Iq	Quiescent current: the current the chip uses itself, doing nothing
Vdo (dropout)	Minimum difference Vin − Vout for a linear regulator to work
EN / SHDN	Enable / shutdown pin. Turns the chip on or off
FB	Feedback pin of a regulator. Connected to a resistor divider
ESR	Equivalent Series Resistance of a capacitor. Some LDOs need a certain ESR
Tj, Ta	Junction temperature (inside the chip), ambient (air) temperature
θJA	Thermal resistance in °C per watt. Tells how hot the chip gets
Fsw	Switching frequency of a DC-DC converter
NC	Not connected. Leave it open (follow the datasheet)
EP / PAD	Exposed pad under the chip. Often connects to GND and gives heat sinking

4.8 Worked example: a 3.3 V supply

Task

We have a 5 V USB supply. We need 3.3 V at 300 mA for a microcontroller. We choose an LDO regulator with fixed 3.3 V output (the part name AP2112K-3.3 is used here only as an example; always read the real datasheet).

1. **Check limits:** Vin = 5 V is inside the input range 2.5 – 6 V. Output current 300 mA is below the maximum 600 mA. OK.
2. **Check dropout:** Vin – Vout = 1.7 V, much more than the dropout (about 0.25 V at full load). OK.
3. **Check heat:** P = (Vin – Vout) × I = (5 – 3.3) × 0.3 = **0.51 W**. The datasheet gives θJA. In a small SOT-23-5 package 0.51 W may be too hot! Use a bigger copper area, or a buck converter instead.
4. **Copy the typical circuit:** input capacitor and output capacitor as the datasheet says (here 1 µF ceramic each).
5. **Draw it** in KiCad (next chapter) and add a 100 nF decoupling capacitor near the load.

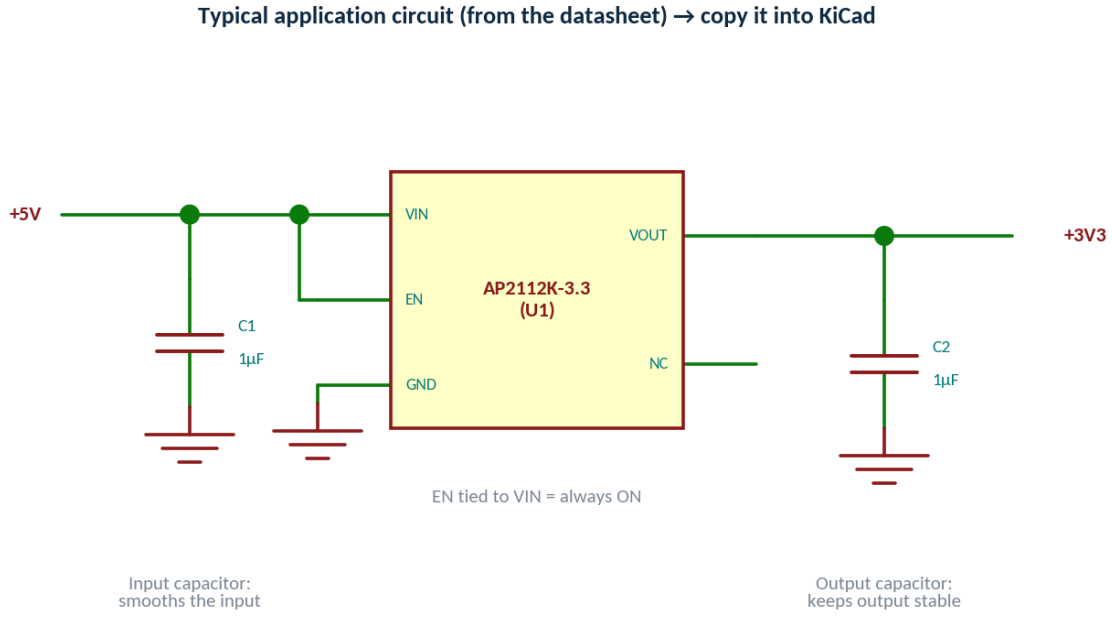


Figure 18. Typical application circuit of the example regulator. Values come from the datasheet.

Adjustable regulators: calculating the divider

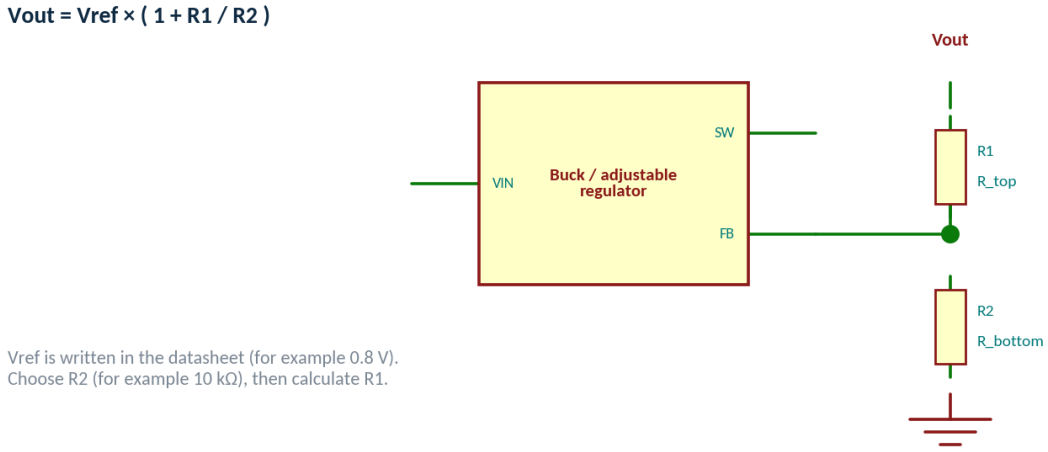


Figure 19. Feedback divider. The regulator keeps the FB pin at Vref, so the output is set by the two resistors.

Example

Vref = 0.8 V, we want Vout = 3.3 V, and choose R2 = 10 kΩ.

R1 = R2 × (Vout / Vref − 1) = 10 k × (3.3 / 0.8 − 1) = 10 k × 3.125 = **31.25 kΩ**.

Nearest 1 % value: **31.6 kΩ** → Vout = 0.8 × (1 + 31.6 / 10) = **3.33 V**. Good enough (+0.8 %).

4.9 Choosing parts wisely

- **Voltage derating:** use capacitors rated at least 2 × the working voltage. A 10 µF ceramic capacitor rated 6.3 V can lose more than half of its capacity when 5 V is applied (DC bias effect).
- **Current margin:** choose parts that can handle 1.5 – 2 × the current you need.
- **Availability:** check stock before you design. A great chip that nobody sells is useless.
- **Package:** as a beginner, choose packages you can solder (0603, SOIC, SOT-23). Choose QFN and BGA only when you must (Chapter 7).
- **Same-family parts:** try to use only a few resistor and capacitor values. It makes buying and assembly cheaper.

Datasheet checklist before you start the schematic

- Pin numbers and pin names understood?
- Typical application circuit found?
- Recommended input / output capacitors noted?
- Layout guideline page read?
- Package drawing and land pattern saved?

Check yourself

1. What is the difference between “absolute maximum” and “recommended operating” values?
2. Why is the “Typ” value not safe for design?
3. A 12 V to 5 V LDO supplies 200 mA. How much power turns into heat?
4. Calculate R1 for Vref = 0.6 V, Vout = 1.8 V, R2 = 10 kΩ.

05 New Project and Schematic Design in KiCad

Create a project, draw the circuit, name the nets, and run ERC

In this chapter you will learn

- Create a new project and understand the KiCad windows
- Page settings and the correct grid
- Place symbols, power symbols, wires and labels
- Hierarchical sheets, annotation, fields and the ERC check

5.1 Create a new project

1. Open KiCad. The **Project Manager** window appears.
2. Click **File › New Project...**.
3. Choose a folder, write the project name (for example *my_first_board*) and keep “Create a new folder for the project” checked.
4. KiCad creates three main files: **.kicad_pro** (project), **.kicad_sch** (schematic) and **.kicad_pcb** (PCB).
5. Double-click the **.kicad_sch** file or click the Schematic Editor button.

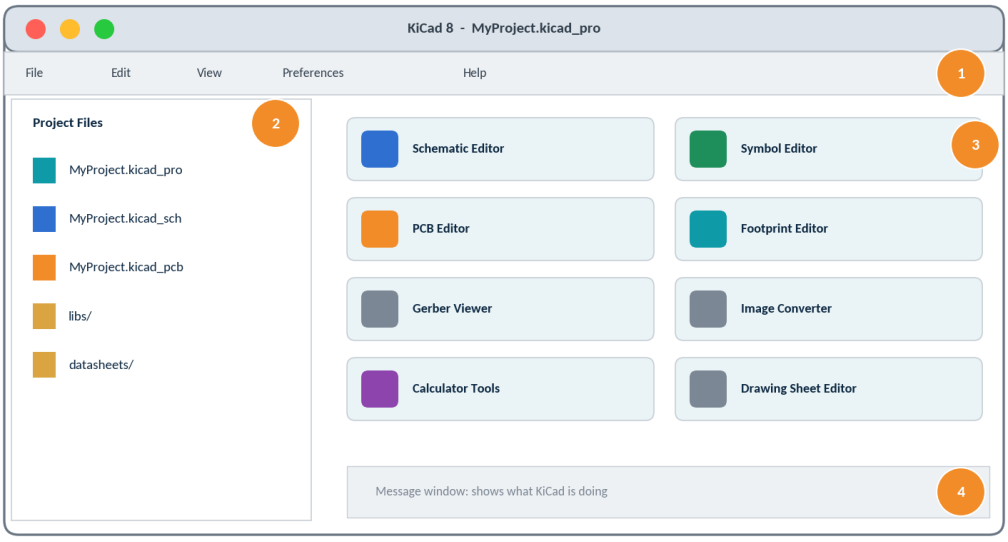


Figure 20. The KiCad Project Manager. (1) menu, (2) project files, (3) program buttons, (4) message window.

5.2 The Schematic Editor window

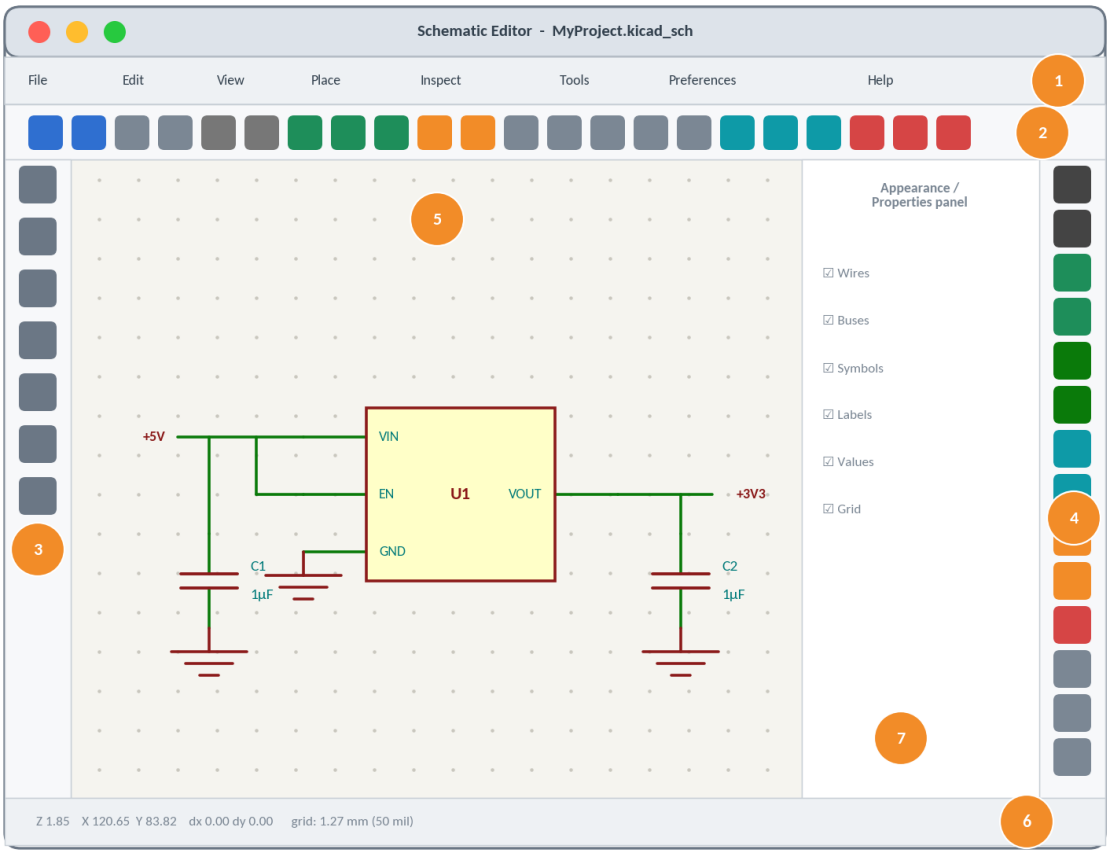


Figure 21. The Schematic Editor. Numbers are explained in the table below.

#	Area	What it does
1	Menu bar	File, Edit, View, Place, Inspect (ERC), Tools (annotate, fields, update PCB), Preferences
2	Top toolbar	Save, print, zoom, undo / redo, find, annotate, ERC, BOM, “Update PCB”, “Assign footprints”
3	Left toolbar	Display options: grid on/off, units (mm / mil), cursor style
4	Right toolbar	**Place tools:** symbol, power, wire, bus, label, global label, sheet, no-connect, text
5	Canvas	The white (cream) page where you draw
6	Status bar	Cursor position and current grid
7	Appearance / Properties panels	Choose which objects are visible, and edit properties of the selected item

5.3 Page settings

Open **File › Page Settings**. Choose the paper size (A4 or A3), write the **title**, **date**, **revision** and **company** (your name). Good title blocks make your documents look professional. For big circuits, use A3 or split the circuit into hierarchical sheets.

5.4 Use the 50 mil (1.27 mm) grid

> **Grid rule**
>
> Use the **50 mil (1.27 mm)** grid in the schematic editor. All KiCad symbol pins are on this grid. If you use a finer grid, wires may **look** connected but **not** be connected, and ERC shows "pin not connected".
>
> Set it with **Preferences › Preferences… › Schematic Editor › Grids** or the grid button in the top toolbar. Use a finer grid only for text or graphics, then return to 50 mil.

5.5 Placing symbols

1. Press **A** (or click the "Add a symbol" button on the right toolbar). The symbol chooser opens.
2. Type the name in the search box (for example "R", "C", "LED", "AP2112"). Double-click the symbol.
3. Click on the page to place it. You can place many copies. Press **Esc** when you finish.
4. Press **R** to rotate, **X** or **Y** to mirror, **M** to move, **G** to drag (wires stay attached), **Del** to delete.
5. Press **E** (Edit properties) on a symbol and write its **Value** (100nF, 10k, …). You can also press **V** to edit only the value.

The common symbols

Part	Search name	Notes
Resistor	R	Value 10k, 4.7k, 330 …
Ceramic capacitor	C	Non-polarised
Electrolytic capacitor	C_Polarized	Has + side
Ferrite bead	FerriteBead	
LED / Diode	LED, D, D_Schottky	
Connector	Conn_01x04 …	Pin headers, USB-C etc.
Ground and power	Press P	Power symbols: GND, +3V3, +5V

5.6 Power symbols

Press **P** to place a **power symbol** like GND, +5V and +3V3. All power symbols with the same name are connected together automatically, even on different pages. If a power rail comes from a connector or regulator pin, KiCad ERC may say *"Input Power pin not driven by any Output Power pins"*. Fix this by adding a **PWR_FLAG** symbol on that net. It tells KiCad: "this net really has a power source."

5.7 Wires, junctions, labels and no-connect flags

Tool	Key	Use
Wire	W	Click on a pin end, click again at the end. Wires connect only at their ends or at junctions.
Junction	J	A dot that joins crossing wires. KiCad adds it automatically at T-connections.
Label	L	Names a wire (net). Two wires with the same label are connected, even without a drawn wire.
Global label	Ctrl + L	Like a label, but works across all sheets.
Hierarchical label	H	Connects a sub-sheet to its parent sheet.
No-connect flag	Q	Put an X on pins you leave unused on purpose.
Text / note	T	Write notes (for example “USB pair: 90 Ω differential”).

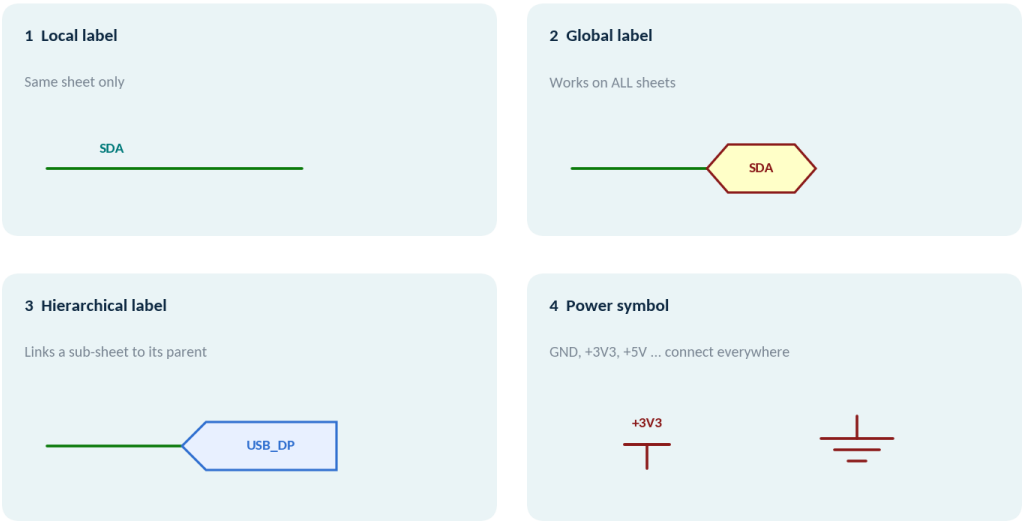


Figure 22. The four ways to connect nets without a wire.

> **Pro Tip: use net labels for readability**
>
> A schematic full of long crossing wires is hard to read. Draw a short wire from the pin and put a **label** (for example *USB_DP*, *I2C_SDA*). Give **meaningful names** to all important nets. In the PCB editor you will see these names and you can use them in net-class rules (Chapter 8).

5.8 Hierarchical sheets for bigger circuits

When the schematic is bigger than one page, divide it into **sub-sheets** by function: Power, MCU, USB, Ethernet, Sensors. Press S to draw a sheet box, give it a name and file name. Then press H inside the sheet to make hierarchical labels. The matching pins appear on the sheet box in the parent page and you connect them with wires.

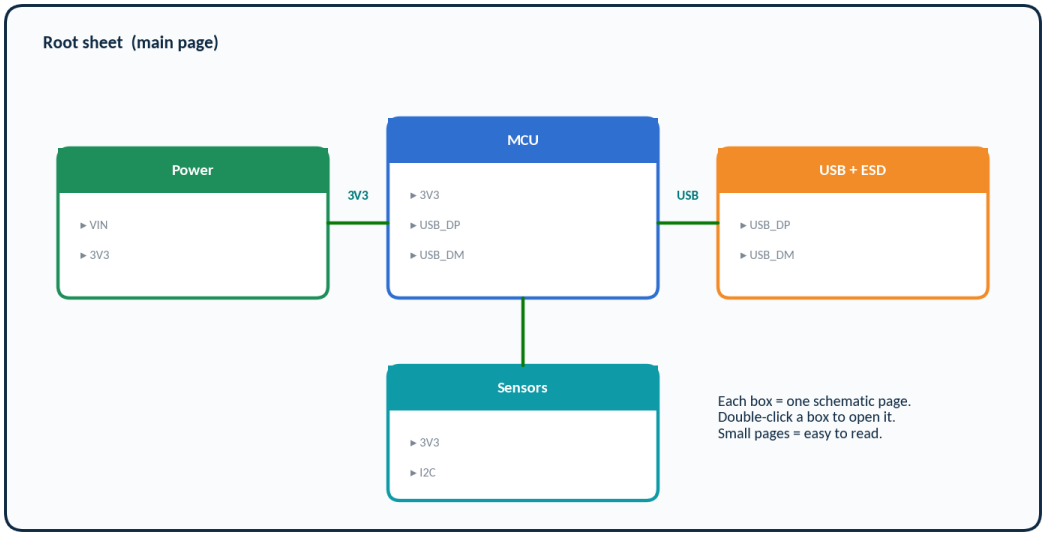


Figure 23. A root sheet with four sub-sheets. Each box is another schematic page.

5.9 Make a clean, readable schematic

- **Signals flow from left to right**: inputs on the left, outputs on the right.
- **Power at the top, ground at the bottom.**
- Draw decoupling capacitors **next to the IC pin** they belong to. When you later place them on the PCB, you will know which pin they serve.
- One function = one block. Put a text title above each block (for example “3.3 V regulator”).
- Write notes for special rules: “keep this trace short”, “90 Ω diff pair”.
- Fill the fields: **Value**, **Footprint**, **Datasheet**, and (for ordering) **MPN** and **LCSC part number**.

5.10 Annotation and fields

Annotation gives every part a unique name: R1, R2, C1, U1. Use **Tools › Annotate Schematic** (button in the top toolbar). Press “Annotate”. Annotate again after you add new parts. Duplicate names (two R1) cause serious errors.

Use **Tools › Edit Symbol Fields** to see and edit all parts in a **spreadsheet**. It is fast for filling values, footprints and part numbers.

5.11 ERC: Electrical Rules Check

ERC finds mistakes in the schematic **before** you make a PCB. Open **Inspect › Electrical Rules Checker** and click **Run ERC**. Fix all errors. Think carefully about warnings.

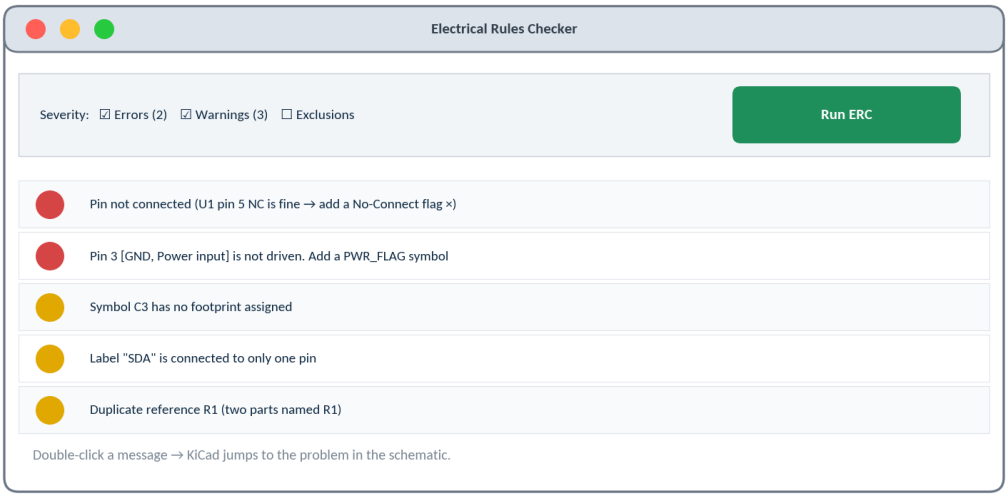


Figure 24. The ERC window. Double-click a line to jump to the problem.

Message	Meaning	How to fix
Pin not connected	A pin has no wire	Connect it, or place a No-Connect flag (Q) if it must stay open.
Input Power pin not driven	Power net has no output power pin	Add a PWR_FLAG symbol on the net.
Pins of type X and Y are connected	Two outputs are joined together	Check the circuit. Two outputs fighting is a real mistake.
Symbol has no footprint	Footprint field is empty	Assign a footprint (Chapter 7).
Duplicate reference	Two parts with the same name	Annotate again.
Label connected to only one pin	A typing mistake in a net name	Check the spelling (SDA vs SDa).

5.12 When a symbol does not exist

Sometimes the chip is not in KiCad libraries. First try to download its symbol (Chapter 6). If you cannot, open the **Symbol Editor**, choose **File › New Symbol**, draw a rectangle, add pins with `P`, give **pin name** and **pin number** exactly as in the datasheet, and save the symbol in your **project library**.

> **Pin numbers must be right**
>
> The pin number of the symbol must match the pad number of the footprint. A wrong number is the most expensive mistake: the board looks fine but does not work.

Check yourself

1. Why do we use a 50 mil grid in the schematic editor?
2. What is the difference between a local label and a global label?
3. What does PWR_FLAG do?
4. Why should you run ERC before making the PCB?

06 Footprints: Download, Import and Design

Where footprints come from and how to keep them in your own project library

In this chapter you will learn

- Symbol, footprint and 3D model: the three faces of a part
- The layers of a footprint
- Downloading footprints from SnapMagic and LCSC
- Adding them to a project-specific library and designing a footprint yourself

6.1 Symbol, footprint and 3D model

Each part has three parts in KiCad:

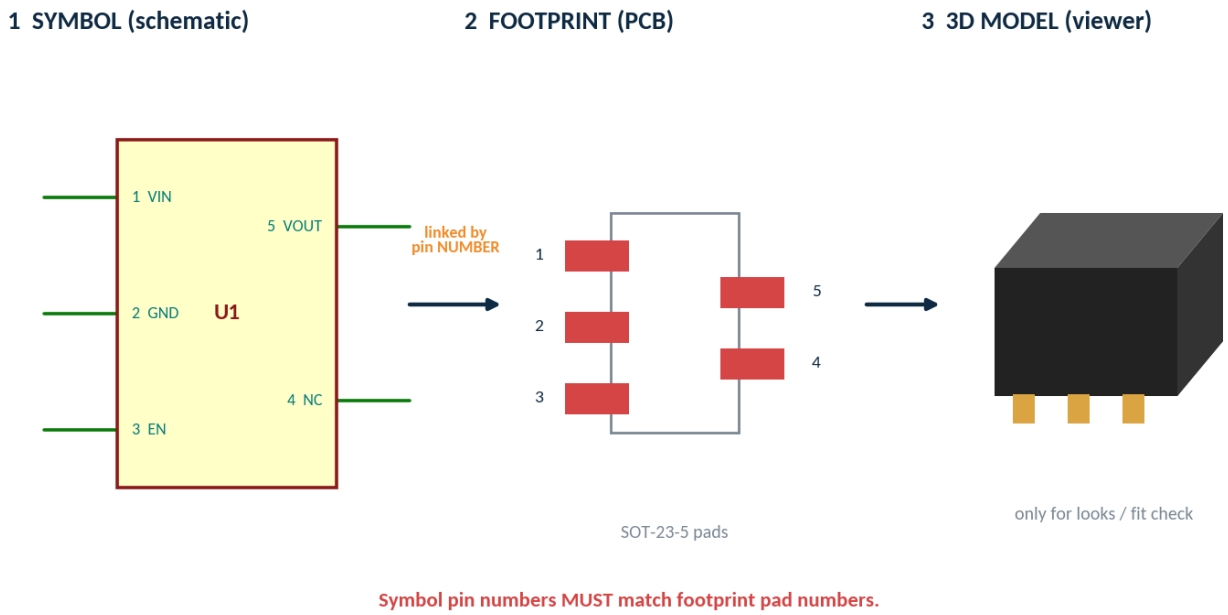


Figure 25. The symbol is for the schematic, the footprint is for the PCB, the 3D model is for the viewer.

- The **symbol** is a drawing for people. It shows pin names.
- The **footprint** is the exact copper shape on the PCB: pads in the right place and size. The factory uses it.
- The **3D model** is optional. It helps you check height and fit in a box.

Remember

A footprint is the most important file for assembly. If a pad is 0.2 mm in the wrong place, the chip may not be solderable. Always check the footprint against the datasheet drawing.

6.2 Anatomy of a footprint

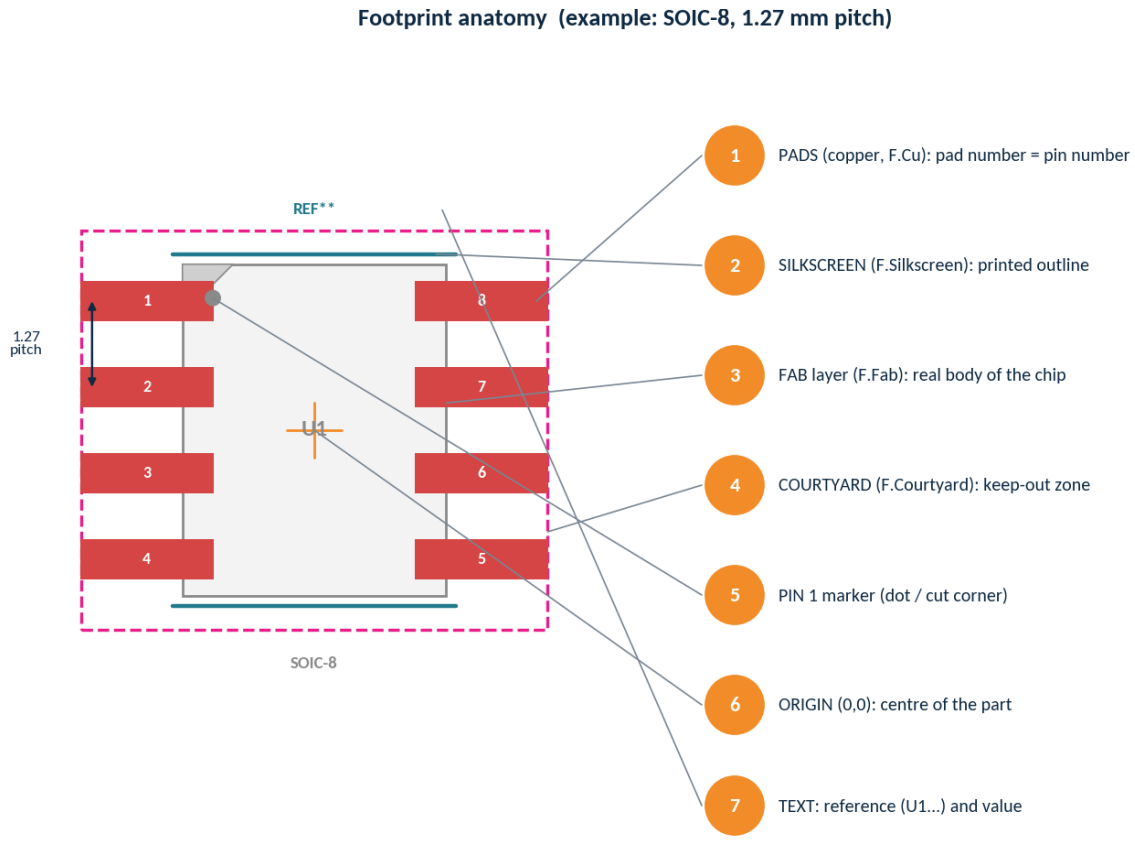


Figure 26. The layers of a footprint (example SOIC-8 with 1.27 mm pitch).

KiCad layer	What it holds	Rule
F.Cu	Copper pads	Size from the datasheet land pattern.
F.Paste	Solder paste openings (for stencil)	Made automatically from SMD pads.
F.Mask	Solder mask openings	Made automatically. Slightly bigger than pads.
F.Silkscreen	Outline and pin-1 mark, printed on the board	Never on top of pads.
F.Fab	Real body of the part	For your information and assembly drawings.
F.Courtyard	Keep-out line around the part	Other parts must not enter. Normal gap 0.25 mm around the part.

6.3 Where do I get footprints?

Source	Good for	Comment
KiCad standard library	Resistors, capacitors, LEDs, standard IC packages, connectors	**Always try this first.** It is checked by volunteers and very good.
SnapMagic (SnapEDA)	Almost any chip, connectors	Free account needed. Choose “KiCad” format. Always verify.
LCSC / EasyEDA	Any part sold by LCSC / JLCPCB	Very useful for assembly. Use a converter tool (see below).
Manufacturer website	Their own parts	Often the most correct, if KiCad files are offered.
Ultra Librarian	Many chips	Also gives KiCad files.
Draw it yourself	Rare parts	Use the datasheet land pattern (Section 6.6).

6.4 Download and import a footprint

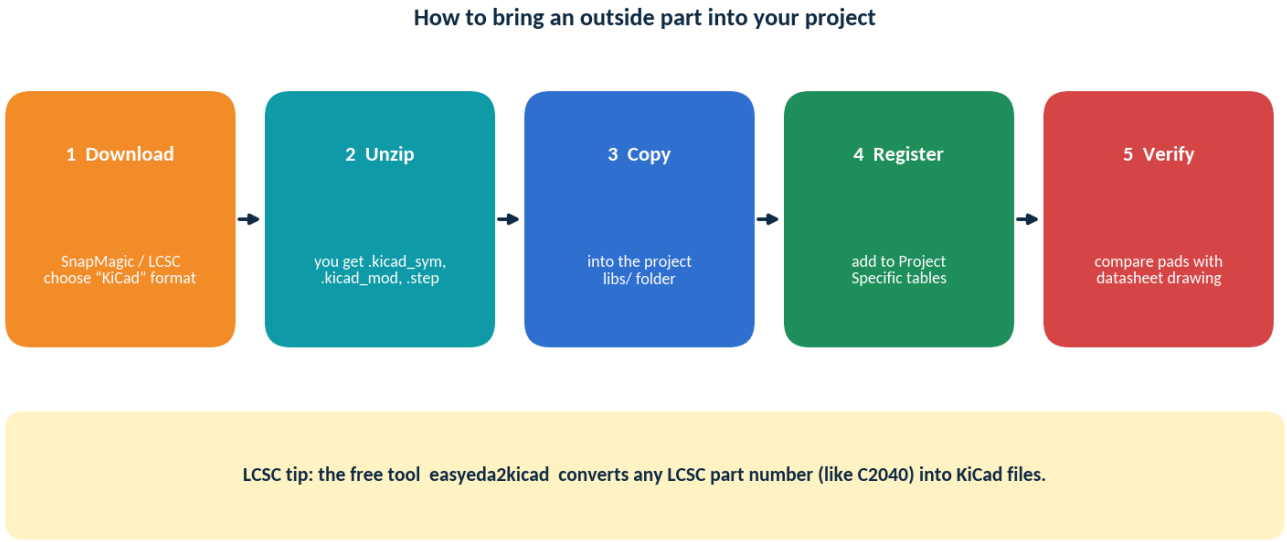


Figure 27. *The five steps to bring an outside part into your project.*

From SnapMagic (SnapEDA)

1. Search the exact part number on the SnapMagic website. Open the part page.
2. Click **Download Footprint / Symbol** and choose **KiCad** as the format. Download the zip file.
3. Unzip it. You usually get a **.kicad_sym** (symbol), a **.kicad_mod** (footprint) and a **.step** (3D model).

From LCSC (using easyeda2kicad)

Every LCSC part has a number like **C2040**. The free tool **easyeda2kicad** converts it. You need Python installed. Type in the command window:

```
pip install easyeda2kicad
easyeda2kicad --full --lcsc_id=C2040 --output libs/lcsc
```

It creates a symbol file, a footprint folder and a 3D model, all in the *libs* folder of your project. Replace C2040 with your part number.

> **Always verify downloaded footprints**
>
> Free footprints may have mistakes. Before you use one:
>
> - Compare the **pad size and pitch** with the land pattern drawing in the datasheet.
> - Check that **pad numbers** match the datasheet pin numbers.
> - Check the **pin 1** mark.
> - Open the 3D viewer: does the model sit on the pads?

6.5 The project-specific library

Keep the downloaded parts **inside your project**. Then your project works on any computer. If you only keep files in a global folder, a friend who opens your project will see missing footprints.

1. Inside your project folder, make a folder **libs**. Inside it make **myparts.pretty** (a footprint library is a folder that ends with *.pretty*) and **3d**.
2. Copy the **.kicad_mod** files into *myparts.pretty*. Copy the **.kicad_sym** file into *libs*. Copy .step files into *3d*.
3. In KiCad open **Preferences › Manage Footprint Libraries**. Click the tab **Project Specific Libraries**.
4. Click the **folder button** (Add existing library) and choose the *myparts.pretty* folder. Check that the path begins with **${KIPRJMOD}**.
5. Do the same for symbols: **Preferences › Manage Symbol Libraries** → Project Specific Libraries → add the *.kicad_sym* file.
6. Press OK. Now you can find your parts in the symbol chooser and the footprint list.

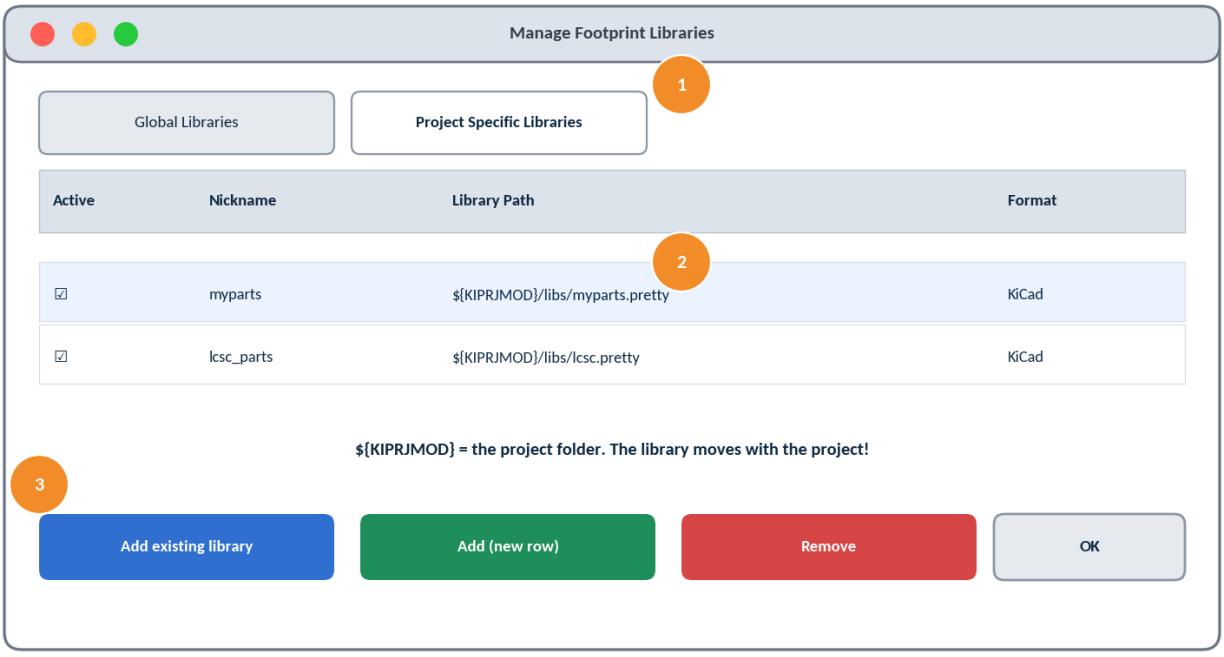


Figure 28. *Manage Footprint Libraries. (1) Project Specific tab, (2) the new library row, (3) button to add an existing library.*

Easy Words

- **${KIPRJMOD}** = "the folder of this project". KiCad replaces it with the real path.
- **Library table** = the list that tells KiCad which libraries to open.
- **Nickname** = the short name of a library (for example *myparts*).

6.6 Design your own footprint

When you cannot find a footprint, design it. You need the **package drawing** and **recommended land pattern** from the datasheet.

1. Open the **Footprint Editor** from the Project Manager. Choose **File › New Footprint**. Give a clear name, for example *SOT-23-5_1.6x2.9mm_P0.95mm*. Save it in your *myparts* library.
2. Set the grid to **0.05 mm** or **0.1 mm** (View › Grid).
3. Press **P** (Add pad). Click to place a pad. Press **E** to open **Pad Properties**: pad **number**, **type** (SMD or Through-hole), **shape** (rectangle, oval), **size**, and exact **position X / Y**.
4. Use **Edit › Create Array...** to make a row of pads with the correct pitch (for example 6 pads, pitch 1.27 mm).
5. Draw the **silkscreen outline** on F.Silkscreen. Keep it outside pads. Mark pin 1 with a dot or a line.
6. Draw the real **body outline** on F.Fab and the **courtyard** rectangle on F.Courtyard (0.25 mm around everything).
7. Add a 3D model in **Footprint Properties › 3D Models** (path **${KIPRJMOD}/libs/3d/part.step**), and set the **type** to SMD or Through-hole.
8. Run **Inspect › Footprint Checker** and save.

Simple rules for pad size when no land pattern is given

Part type	Pad rule
SMD lead (gull-wing)	Pad length = lead length + 0.3 – 0.5 mm outside + 0.1 mm inside. Pad width = lead width + 0.1 – 0.2 mm.
Chip resistor / capacitor	Pad extends 0.3 mm beyond the part's end metal on the outside, 0.2 mm on the inside.
THT pin	Hole diameter = pin diameter + 0.2 – 0.3 mm. Pad diameter ≥ hole + 0.6 mm (annular ring ≥ 0.3 mm).
Exposed pad (QFN)	Pad slightly smaller than the metal pad; add 4 – 9 small thermal vias (0.3 mm) to GND.

Pro Tip: test on paper

Print the footprint 1:1 on paper (File › Print in the Footprint Editor). Put the real chip on the paper. If the legs sit on the pads, the footprint is correct. This old trick has saved many boards.

6.7 Common footprint mistakes

- **Mirrored pins:** the datasheet shows a bottom view but you drew it from the top. Check “top view” or “bottom view” in the drawing title.
- **Wrong pitch or units:** 0.5 mm vs 0.05 inch. Double-check.
- **Pad numbers not matching** the symbol pin numbers.
- **Hole too small:** the pin does not fit. Add 0.2 – 0.3 mm to the pin size.
- **Pad too small or too big** → tombstoning (a small chip standing up) or solder bridges.

Check yourself

1. What are the three parts that belong to one component in KiCad?
2. Why do we copy downloaded files to a project-specific library?
3. What does ${KIPRJMOD} mean?
4. Which three checks must you do on every downloaded footprint?

07 Assigning Footprints and Knowing Packages

THT, SMD, resistor and capacitor sizes, SOT, SOP, QFN and DFN with pictures

In this chapter you will learn

- The difference between THT and SMD parts
- Chip sizes 0402, 0603, 0805, 1206 and more
- IC packages: SOT, SOP/SOIC, QFP, QFN, DFN, BGA
- How to assign footprints in KiCad

7.1 THT and SMD

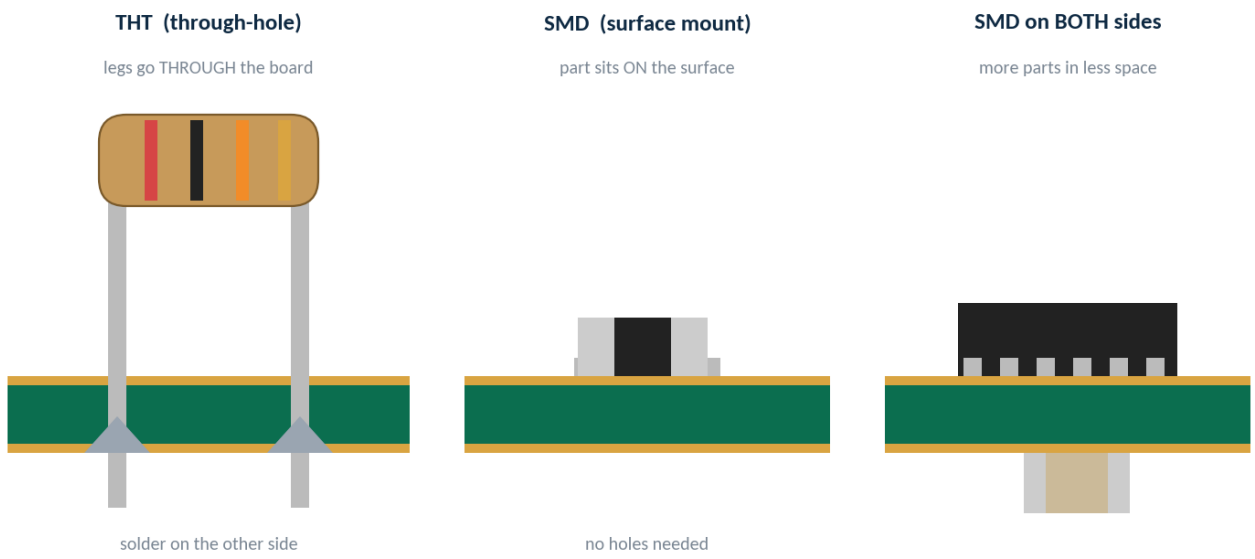


Figure 29. THT parts have legs through the board. SMD parts sit on the surface.

	THT (Through-Hole Technology)	SMD (Surface-Mount Device)
Mounting	Legs go through holes, soldered on the other side	Pads on the surface, no holes
Size	Bigger	Much smaller
Hand soldering	Very easy	Easy for 0603 or bigger; difficult for tiny parts

	THT (Through-Hole Technology)	SMD (Surface-Mount Device)
Strength	Very strong (good for connectors)	Good; use THT for parts that get pulled (USB, power jacks)
High frequency	Poor (long legs = inductance)	Very good
Cost in factory assembly	Higher	Lower (machines)
Use today	Connectors, big capacitors, transformers, learning	Almost everything else

7.2 Sizes of resistors, capacitors and inductors

Small rectangular SMD parts have a **size code** of four digits. The code is the length and width in **hundredths of an inch** (imperial). For example **0603** = 0.06 inch × 0.03 inch = 1.6 mm × 0.8 mm. There is also a **metric code**, which means millimetres in tenths: 0603 imperial = **1608 metric**.

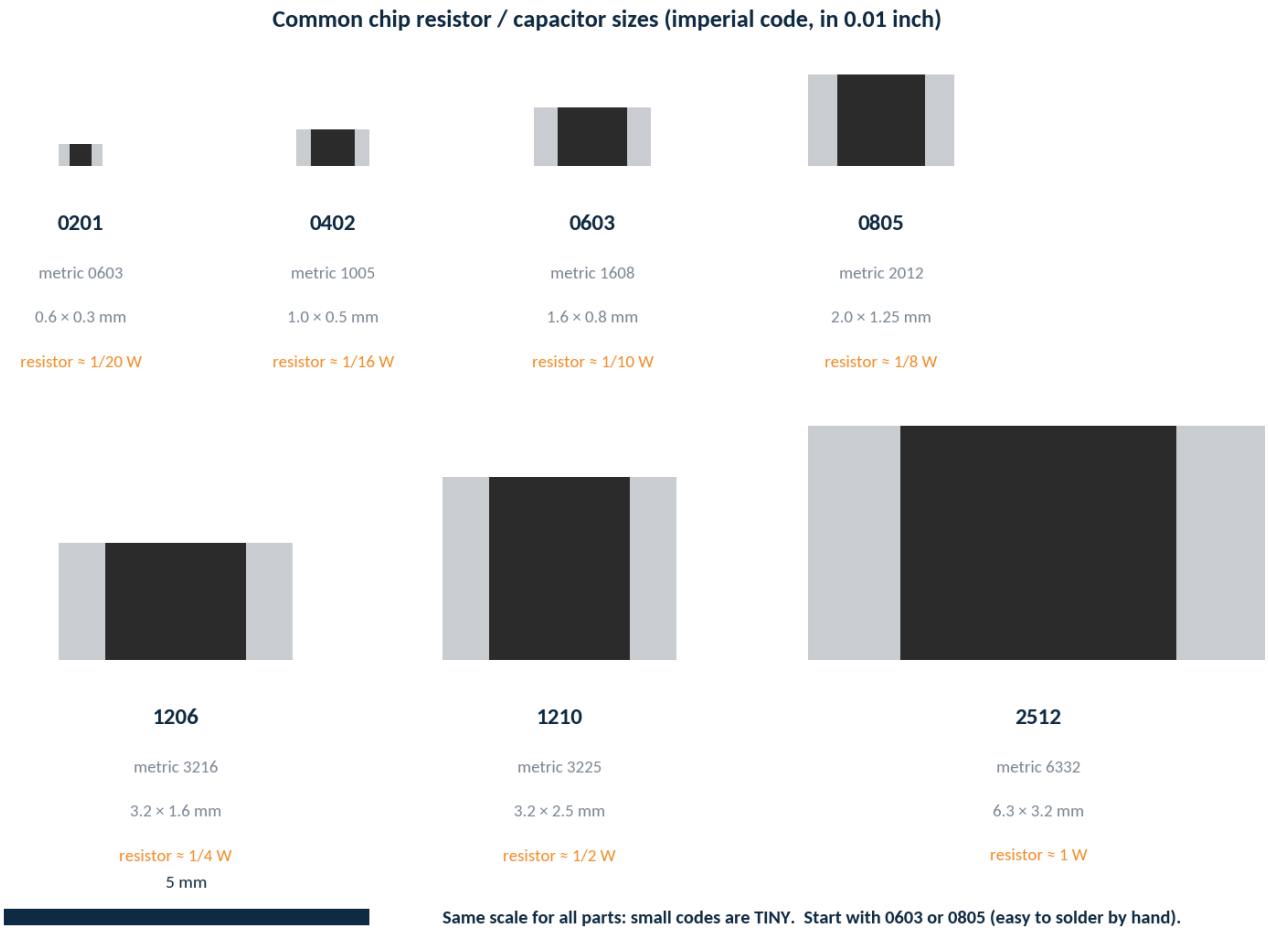


Figure 30. *All chip sizes are drawn on the same scale. The power numbers are for resistors.*

Imperial or metric?

Be careful: “0603” in imperial is 1.6 × 0.8 mm, but “0603” in metric is a tiny 0.6 × 0.3 mm part (called 0201 imperial). KiCad footprint names show both, for example **C_0603_1608Metric**: 0603 is imperial, 1608 is metric.

Imperial	Metric	Size (mm)	Hand soldering	Typical use
0201	0603	0.6 × 0.3	Very hard	Phones, wearables
0402	1005	1.0 × 0.5	Hard	Dense boards, decoupling
0603	1608	1.6 × 0.8	Easy	**Best for beginners**
0805	2012	2.0 × 1.25	Very easy	Beginners, power, higher voltage
1206	3216	3.2 × 1.6	Very easy	High voltage, power resistors, big capacitors
1210	3225	3.2 × 2.5	Very easy	10 – 22 µF ceramic capacitors
2512	6332	6.3 × 3.2	Easy	Current-sense and power resistors

Inductors and ferrite beads

- **Chip inductors and ferrite beads** use the same codes (0402, 0603, 0805). Check the **rated current**: small sizes carry only a few hundred mA.
- **Power inductors** are bigger blocks, named by body size, for example **4 × 4 mm**, **5 × 5 mm** or **6 × 6 mm**. Pick them by inductance and saturation current from the converter datasheet.

Capacitor packages

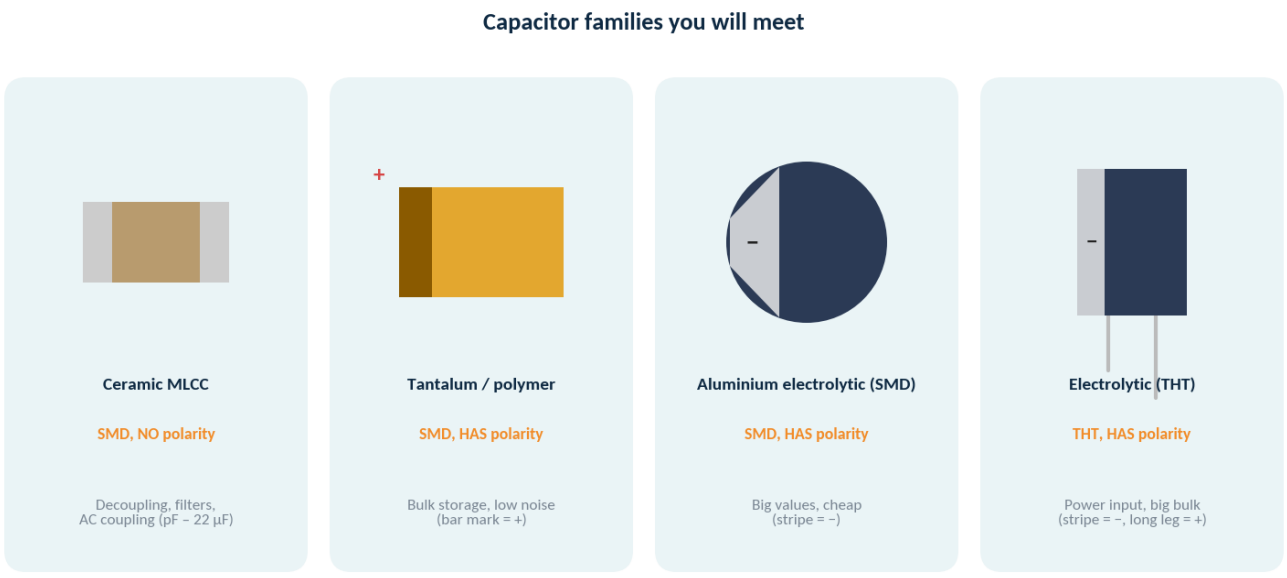


Figure 31. Types of capacitor packages. Polarised capacitors have a mark for + or −.

- **Ceramic:** use 0603 or 0805 (0402 for dense boards). No polarity.
- **Tantalum / polymer:** case sizes A (3216), B (3528), C (6032), D (7343). The stripe marks the **+** side.

- **Aluminium SMD can:** named by diameter × height, for example 6.3 × 5.4 mm. The stripe marks the − side.
- **THT electrolytic:** named by diameter and pin pitch, for example D5 mm, pitch 2.0 mm.

7.3 IC packages

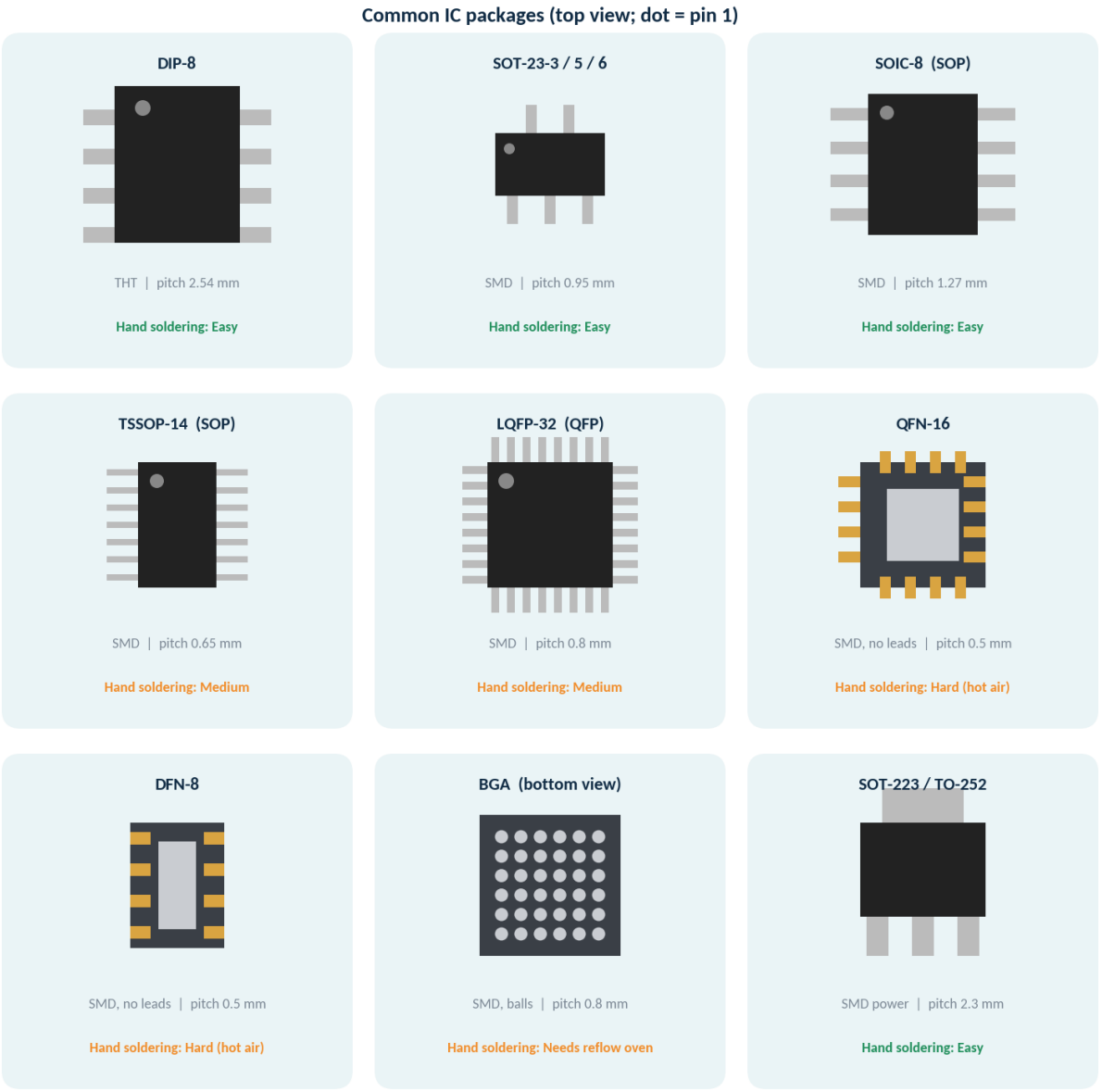


Figure 32. Common IC packages. Pitch is the distance between pin centres.

Package	Full name	Pitch	Notes
DIP	Dual In-line Package	2.54 mm	THT. Fits a breadboard. Old but good for learning.
SOT-23	Small Outline Transistor	0.95 mm	3 – 6 pins. Transistors, small regulators, op-amps. Easy to solder.
SOP / SOIC	Small Outline (Integrated Circuit)	1.27 mm	Pins on two sides. SOIC-8, SOIC-16. **Great for beginners.**

Package	Full name	Pitch	Notes
SSOP / TSSOP	Shrink / Thin Shrink Small Outline	0.65 mm	Same as SOP but thinner and with closer pins. Possible by hand with care.
QFP / LQFP	Quad Flat Package	0.5 – 0.8 mm	Pins on four sides, legs outside. Many microcontrollers. Can be hand soldered with a fine tip.
QFN	Quad Flat No-leads	0.4 – 0.65 mm	Pads under the chip, 4 sides. Has an **exposed pad** in the centre. Needs hot air or reflow.
DFN	Dual Flat No-leads	0.5 – 0.65 mm	Like QFN but pads on only **two sides**. Small, thin.
BGA	Ball Grid Array	0.4 – 1.0 mm	Balls under the chip. Needs an oven and often X-ray checking. Needs many layers.
SOT-223 / TO-252	Power packages	2.3 mm	Large tab for heat. Regulators, MOSFETs.

The exposed pad (thermal pad)

QFN, DFN and many SOIC power chips have a large metal pad on the bottom. It does two jobs: it carries **heat** to the PCB and it is often the **ground** connection. In the footprint, you must make a copper pad for it and connect it to the ground plane with several **thermal vias**. Follow the layout page in the datasheet.

7.4 Pin 1 and pin numbering

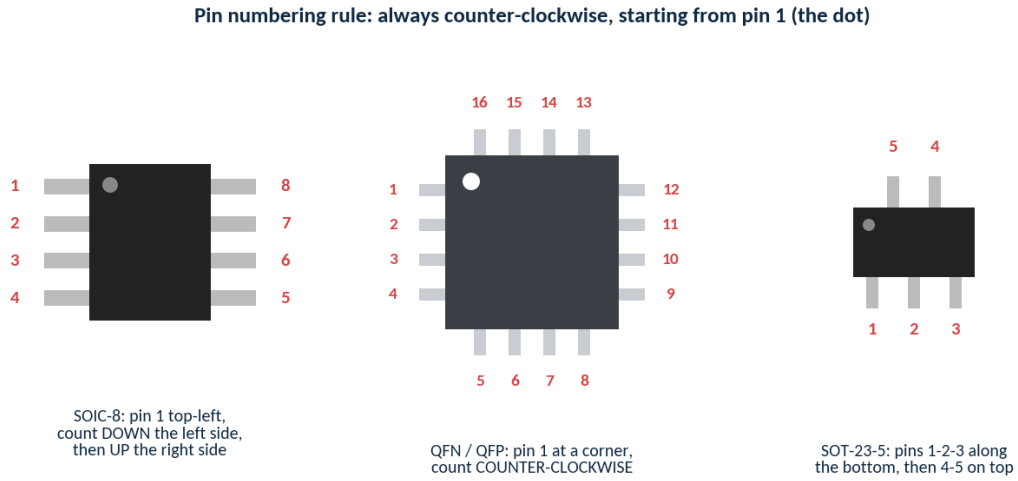


Figure 33. Pin numbers go counter-clockwise starting at pin 1.

Pin 1 rule

Look at the chip from the **top**. Find the dot, notch or cut corner. That is pin 1. Pins then go **counter-clockwise**. Always mark pin 1 on the silkscreen so that the assembler puts the chip in the right direction.

7.5 Assign footprints in KiCad

After the schematic is finished and ERC is clean, every symbol needs a footprint.

1. In the Schematic Editor click the **Assign Footprints** button (or **Tools › Assign Footprints**).
2. The window has three parts: **libraries** on the left, **your symbols** in the middle, and **footprints list** on the right with a preview.
3. Click a symbol (for example C1). Find the correct footprint (for example *Capacitor_SMD › C_0603_1608Metric*) and **double-click** it. The name appears in the middle column.
4. Use the **search box** and the filters to find footprints faster. To give the same footprint to many identical parts, select them all and assign once.
5. Check the **preview**. Click **Apply, Save Schematic & Continue**.
6. Your own parts are in your **project library** (for example *myparts*).

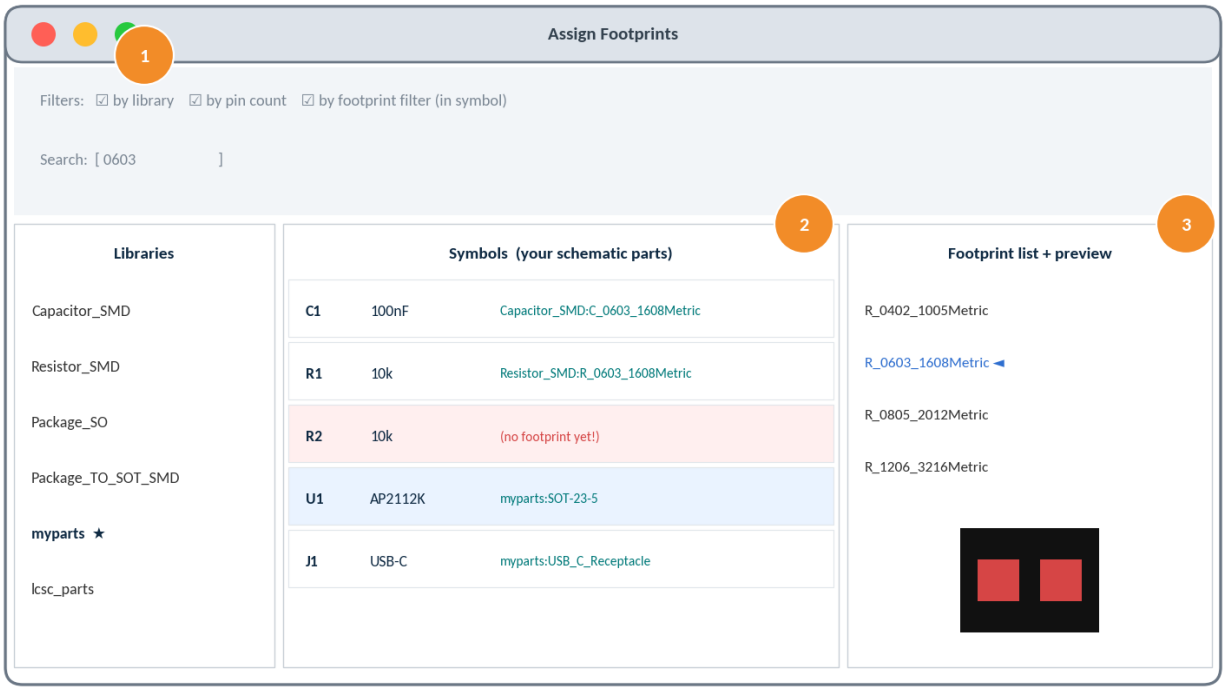


Figure 34. *The Assign Footprints window. (1) libraries, (2) symbols, (3) footprint list and preview.*

Other ways to assign

- Open the symbol properties **E** and write the footprint name in the **Footprint** field, for example *Resistor_SMD:R_0603_1608Metric*.
- Use **Tools › Edit Symbol Fields** and fill the Footprint column for many parts at once (copy and paste).

How to read a footprint name

Name	Meaning
C_0603_1608Metric	Capacitor, imperial size 0603, metric 1608
R_0805_2012Metric	Resistor, 0805
SOIC-8_3.9x4.9mm_P1.27mm	SOIC package, 8 pins, body 3.9 × 4.9 mm, pitch 1.27 mm
SOT-23-5	SOT-23 package with 5 pins
QFN-32-1EP_5x5mm_P0.5mm_EP3.1x3.1mm	QFN, 32 pins, 1 exposed pad, 5 × 5 mm body, pitch 0.5 mm, pad 3.1 × 3.1 mm
PinHeader_1x04_P2.54mm_Vertical	Header, 1 row, 4 pins, 2.54 mm pitch

7.6 Choosing footprints: a simple guide for beginners

Part	Footprint to start with
Resistor, capacitor (up to 1 µF), LED	0603 (or 0805 if you solder by hand)
Capacitor 10 µF / 22 µF	0805 or 1206 (check voltage rating)
Decoupling capacitor	0402 or 0603
Small regulator	SOT-23-5 or SOT-223
Op-amp, logic, EEPROM	SOIC-8 / TSSOP
Microcontroller	LQFP (easier) or QFN (smaller)
Connector that gets plugged in often	THT or SMD with strong mounting tabs
Crystal	3225 or 2520 (follow the chip guide)

Check before you update the PCB

- Every symbol has a footprint (no empty field).
- Polarised parts (diodes, tantalum, electrolytic, LEDs) have the right pin order. Check pin 1 / anode / cathode.
- Connectors have the right pitch and number of pins.
- You compared pad sizes with the datasheet for every new footprint.

Check yourself

1. What size in mm is a 0805 part?
2. Which package has pads on only two sides and no legs?
3. How do you find pin 1 on an SOIC chip?
4. What is the exposed pad and why do we add vias under it?
5. What does “SOIC-8_3.9x4.9mm_P1.27mm” tell you?

08 PCB Layout Setup: Page, Stackup, Constraints, Net Classes

Open the PCB editor and prepare the board before placing any part

In this chapter you will learn

- The PCB editor window and its buttons
- Board outline, grid and layers
- Stackup: which layer order is best
- Design constraints and net classes

8.1 Open the PCB editor and import the design

1. Open the **PCB Editor** from the Project Manager (double-click the .kicad_pcb file). The board is empty.
2. Press **F8** or click **Tools › Update PCB from Schematic**. A window appears. Click **Update PCB**.
3. All footprints appear in a pile, with thin lines (the **ratsnest**) showing which pins must be connected. Click on the canvas to drop them.
4. If you change the schematic later, run **Update PCB from Schematic** again. Your old placement is kept.

Easy Words

- **Ratsnest** = thin straight lines that show which pins are connected but not yet routed.
- **Footprint** = the pad pattern of a part (Chapter 6).
- **Net** = a group of pins that are connected.

8.2 The PCB Editor window

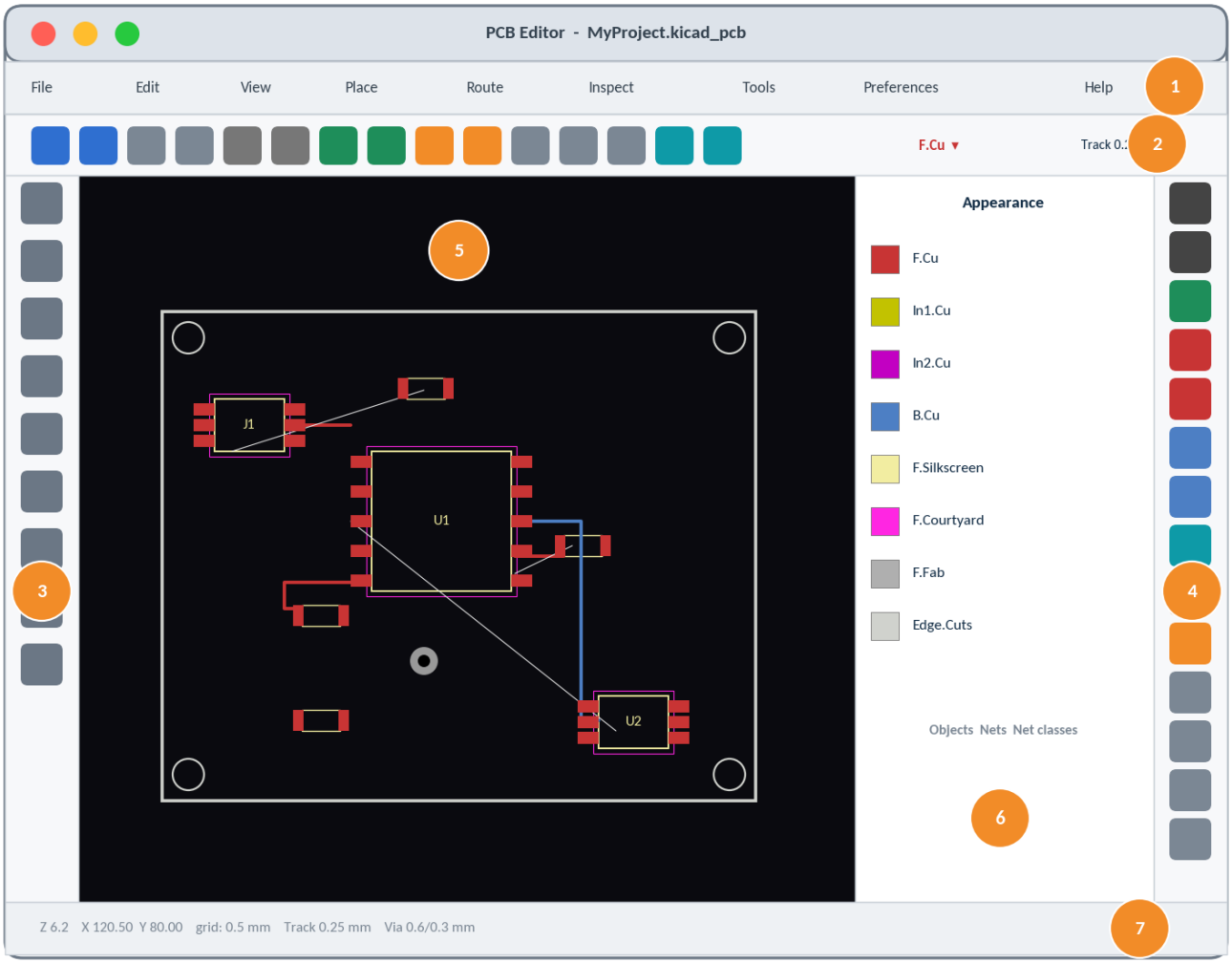


Figure 35. The PCB Editor. The black area is the canvas. Numbers are explained below.

#	Area	What it does
1	Menu bar	File (Board Setup, Plot), Edit, View (3D viewer), Place, Route, Inspect (DRC), Tools (Update PCB), Preferences
2	Top toolbar	Save, undo, zoom, Board Setup, Update PCB, Net Inspector, DRC, 3D viewer, active **layer selector**, track width and via size
3	Left toolbar	Display: grid, units, ratsnest, highlight, show/hide copper fill
4	Right toolbar	**Draw tools:** select, route track, route diff pair, add via, add zone, draw lines, add text, measure
5	Canvas	The board itself. The active layer is drawn on top
6	Appearance panel	Show or hide layers, objects and nets. Tab “Net classes”: colour nets. Very useful!
7	Status bar	Cursor position, grid, selected track width and via

8.3 The layers of a PCB in KiCad

Layer	Colour	Purpose
F.Cu	Red	Top copper
In1.Cu, In2.Cu …	Gold, purple	Inner copper layers (4-layer board: In1 = GND, In2 = POWER)
B.Cu	Blue	Bottom copper
F.Silkscreen / B.Silkscreen	Yellow / pink	Printed text and outlines
F.Mask / B.Mask	Purple	Openings in the solder mask (over pads)
F.Paste / B.Paste	Gray	Solder-paste stencil openings
F.Courtyard	Magenta	Keep-out line around parts
F.Fab	Gray	Real part outlines
Edge.Cuts	Yellow-gray	**The board outline**. Must be a closed shape

Select the active layer in the top toolbar, in the Appearance panel, or with `PgUp` / `PgDn`. When you route and press `V`, a via is placed and the active layer changes automatically.

8.4 Page settings, units and grid

- **File › Page Settings**: choose paper size (A4 / A3) and fill the title block, as in the schematic.
- Use **millimetres** (mm). Switch the unit with the button on the left toolbar or in **View › Units**.
- Choose the grid with **View › Grid**. Use **0.5 mm** or **1 mm** for placing big parts, **0.1 mm** or **0.05 mm** for routing and small parts.
- Right-click on the canvas to set **Grid Origin** if you want the board corner to be (0,0).

8.5 Draw the board outline

1. Select the layer **Edge.Cuts** in the layer selector.
2. Use **Place › Line** (or the rectangle tool) and draw a **closed shape**. Start with a rectangle, for example 60 × 40 mm. The end must touch the start exactly.
3. Round the corners with a small radius (2 – 3 mm). It looks nicer and avoids sharp corners that break.
4. Add **mounting holes**: place a footprint named *MountingHole_3.2mm_M3* in each corner, 4 mm from the edges.
5. Keep copper at least **0.3 mm** (better 0.5 mm) away from the board edge, and parts at least 1 mm.

> **Pro Tip**
>
> Decide the board size from the box (enclosure) or the connectors first. Put the connectors on the edges where users will plug cables. The outline should not change after you start routing.

8.6 Board Setup

Open **File › Board Setup...**. This window holds every important rule of your board. Do it **before** placing and routing.

A. Board Editor Layers

Here you choose the number of copper layers: 2, 4, 6 ... For a first fast board (USB, SPI, I2C) choose **4 layers** if the budget allows. For simple boards, 2 layers is fine.

B. Physical Stackup

The **stackup** is the order and thickness of copper and insulating layers. It decides the impedance of your traces, the crosstalk and the EMI. Always ask your PCB factory for **their stackup** (thickness of each prepreg and core) and enter it here. The numbers you enter are used for your own documents and when you ask the factory for impedance control.

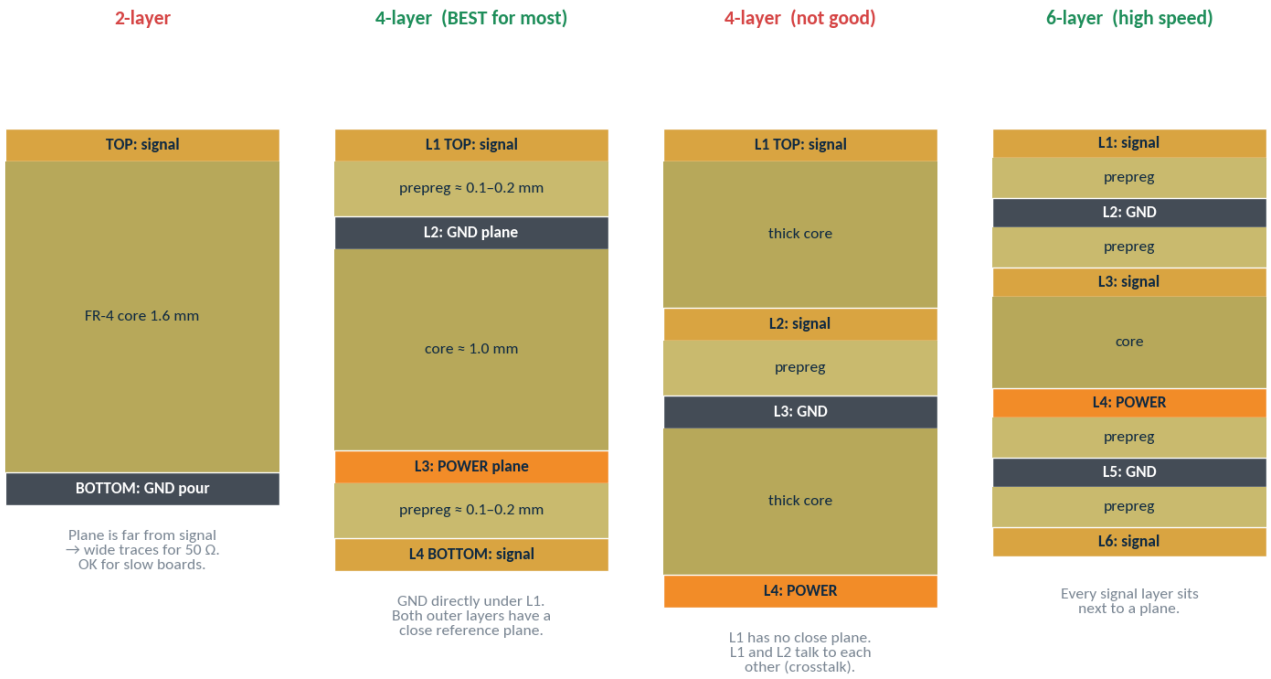


Figure 36. *Stackup examples. The best stackups put a ground plane right next to every signal layer.*

Which stackup is best? Ground plane above or below the signal?

The key idea is simple: **every signal layer should be next to a solid plane, with a thin insulating layer between them**. The plane is the return path (Chapter 2). The closer the plane, the smaller the loop, the lower the crosstalk and the narrower the trace for the same impedance.

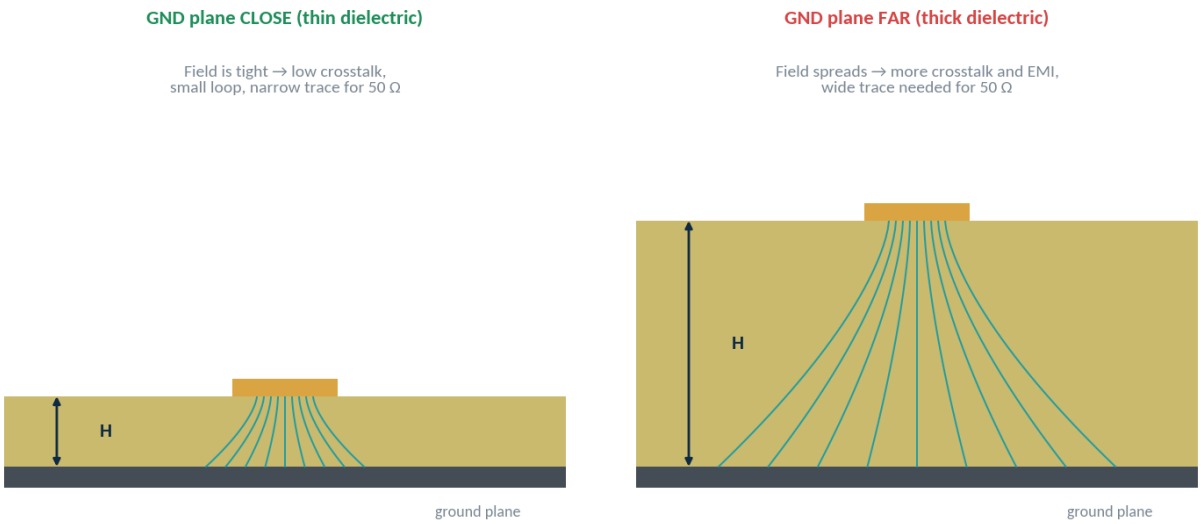


Figure 37. A thin insulation (small H) keeps the electric field tight between trace and plane.

Layers	Recommended order	Why
2 layers	Top: signals and short power. Bottom: **ground pour** (as much solid ground as possible)	Plane is far (1.6 mm). Use it for slow and medium speed boards.
4 layers	**L1 signal / L2 GND / L3 POWER / L4 signal**	Most popular. GND plane is directly **below** the top layer, so L1 has a close reference. L3 power plane is next to L4.
4 layers (alternative)	L1 signal / L2 GND / L3 GND / L4 signal	Power distributed with wide traces and zones. Excellent for EMI.
6 layers	L1 signal / L2 GND / L3 signal / L4 POWER / L5 GND / L6 signal	Every signal layer is next to a plane. Used for fast boards.

Answer in one line

Put the **ground plane immediately next to the signal layer** (directly under the top signal layer, and directly above the bottom signal layer). The distance between them (the prepreg, about 0.1 – 0.2 mm) should be **thin**. The thick core goes in the middle of the board.

Also: keep the stackup **symmetric** around the centre. A non-symmetric board can bend (warp) during soldering.

Avoid

- Two signal layers next to each other without a plane between them (they will have crosstalk with long parallel traces).
- A fast signal layer far from any plane (thick dielectric).
- Splitting the ground plane under fast traces.

C. Design Rules › Constraints

These are the **minimum sizes** that your PCB factory can make. If you break them, the factory cannot build your board (or charges more). The values below are safe for most cheap factories. **Always check the capability page of your own factory.**

Constraint	Safe beginner value	Typical factory minimum
Minimum clearance (copper to copper)	0.2 mm (8 mil)	0.1 – 0.15 mm
Minimum track width	0.2 mm	0.1 – 0.15 mm
Minimum via diameter (pad)	0.6 mm	0.45 – 0.5 mm
Minimum via hole (drill)	0.3 mm	0.2 – 0.3 mm
Minimum annular ring	0.15 mm	0.1 – 0.13 mm
Hole to hole clearance	0.5 mm	0.25 – 0.5 mm
Copper to board edge	0.5 mm	0.3 mm
Minimum through hole	0.3 mm	0.2 – 0.3 mm
Minimum silkscreen text	height 1.0 mm, width 0.15 mm	height 0.8 mm, width 0.15 mm

> **Why use safe values?**
>
> Factories can do smaller sizes, but each small size increases price, and mistakes become more likely. Safe values give you a reliable board on the first try.

D. Design Rules › Pre-defined Sizes

Here you list the track widths and via sizes you like to use, for example tracks: **0.2, 0.25, 0.4, 0.5, 1.0 mm**; vias: **0.6 / 0.3 mm** (pad / drill) and **0.8 / 0.4 mm**. Then you can pick them in the toolbar while routing.

E. Design Rules › Net Classes

A **net class** is a group of nets that share the same rules. Instead of setting the width of each trace by hand, you say: “all power nets use 0.5 mm; all USB nets use the 90 Ω differential width.” KiCad then uses those numbers automatically when you route.

Net class	Clearance	Track width	Via (pad / drill)	DP width / gap	Used for
Default	0.2 mm	0.2 mm	0.6 / 0.3	—	Normal signals (GPIO, SPI, I2C)
Power	0.2 mm	0.5 – 1.0 mm	0.8 / 0.4	—	+5V, +3V3, VBAT (width from current, Chapter 10)
HS_50	0.2 mm	for 50 Ω (e.g. 0.20 mm)	0.6 / 0.3	—	Clocks, RF, fast single lines
USB_DIFF	0.2 mm	0.16 mm	0.6 / 0.3	0.16 / 0.13 mm	USB 90 Ω pairs
PCIE_DIFF	0.2 mm	0.14 mm	0.6 / 0.3	0.14 / 0.10 mm	PCIe / HDMI / Ethernet pairs (values from your calculator)

In the **Netclass Assignments** table, tell KiCad which nets belong to which class. Use patterns with a star: for example pattern **/USB_D*** → class USB_DIFF; pattern **+5V**, **+3V3**, **VBUS** → class Power. Net names come from your schematic labels, so give good names there.

> **Where is the Net Classes page?**
>
> In most versions it is in Board Setup › Design Rules › Net Classes. In some newer versions it is moved to File › Schematic Setup › Project › Net Classes, because net classes belong to the whole project. The idea is the same.

F. Design Rules › Custom Rules (optional)

For advanced rules, such as a maximum skew inside a differential pair, you can write a small text rule in the **Custom Rules** page:

```
(version 1)
(rule "USB pair skew"
   (condition "A.inDiffPair('/USB_D')")
   (constraint skew (max 0.1mm)))
```

The rule means: for the pair named USB_D, the two traces must be within 0.1 mm of the same length. DRC will warn you if not. (Check the KiCad manual page “Custom Rules” for the exact syntax of your version.)

Check yourself

1. Which layer holds the board outline?
2. Draw the best 4-layer stackup. Where is the GND plane?
3. Why must the stackup be symmetric?
4. What is a net class and why do we use it?
5. Why should you use the factory’s capability numbers in Constraints?

09 Component Placement

Where to put each part. Good placement makes routing easy.

In this chapter you will learn

- Why placement is the most important step
- How to select a circuit in the schematic and move its parts together
- Placement rules for decoupling capacitors, regulators and connectors
- Useful tools: move, rotate, flip, align

Experienced designers say: **"Good placement is 70 % of a good PCB."** If the parts are in the right places, routing becomes easy and the board performs well. If placement is bad, no routing trick can save it.

9.1 Placement order

1. **Fixed parts first:** connectors, mounting holes, switches, LEDs, antennas. They are decided by the box and the user.
2. **The main IC** (microcontroller, FPGA, SoC) near the centre.
3. **Parts that must be close to the main IC:** crystal, decoupling capacitors, boot resistors.
4. **Power circuits** (regulators, inductors) near where power enters. Keep noisy switching parts away from sensitive parts.
5. **Everything else** (resistors, LEDs, small ICs) around, following the ratsnest lines.

9.2 Move a whole circuit together (step by step)

KiCad connects the schematic and the PCB. When you select parts in the schematic, the same parts are selected in the PCB. This is called **cross-probing**. Let us use a voltage regulator circuit as an example.

1. Open **both** the Schematic Editor and the PCB Editor.
2. In the **schematic**, draw a selection box around the regulator circuit: the regulator U1, its input capacitor, output capacitor and feedback resistors. (Or hold Shift and click each part.)
3. Look at the **PCB editor**. The same footprints are now **selected** (highlighted), even if they are scattered. If nothing happens, check that cross-probing is enabled in Preferences, and that the PCB editor is open.
4. In the PCB editor press `M` (Move) and click on an **empty area**. The **whole group** moves together. Click again to drop it.
5. Press **Esc** to clear the selection. Now move the group near the place where it belongs: for example near the power connector.

6. Zoom in. Place the parts **one by one**, as close as possible (Section 9.3). Use the ratsnest lines as guide.
7. Repeat for the next circuit: USB, crystal, LEDs.

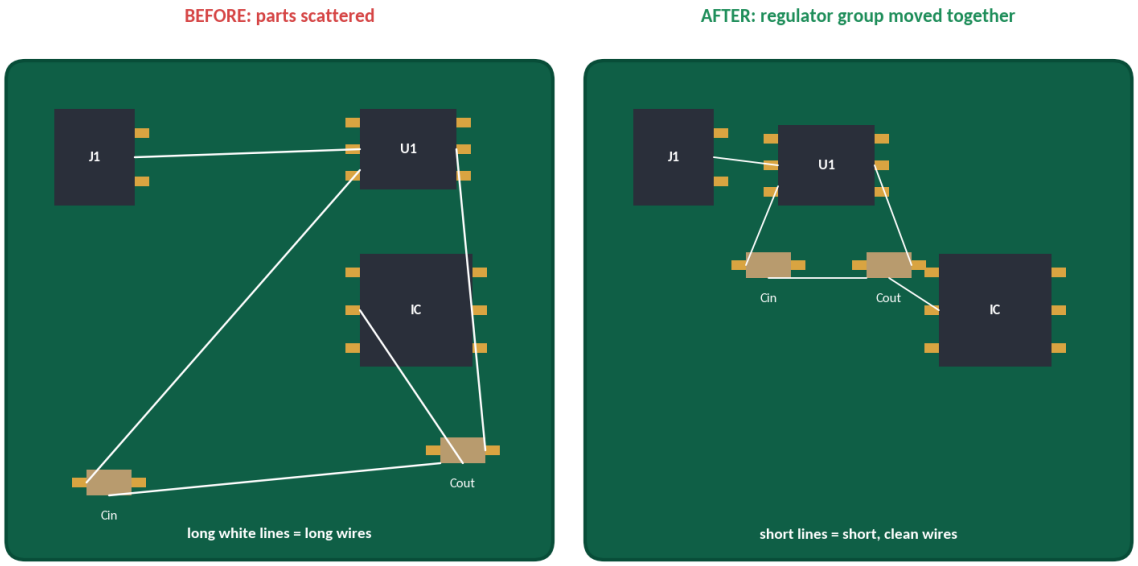


Figure 38. Before: scattered parts and long lines. After: the regulator group moved together, with short connections.

Pro Tips for selecting

- In the PCB editor, press **Ctrl** + **F** to find a part by its name (for example C5).
- Group a finished block with **Ctrl** + **G** so that it moves as one.
- Lock a finished part (right-click › Locked) so you do not move it by mistake.

9.3 Placement rules

Decoupling capacitors

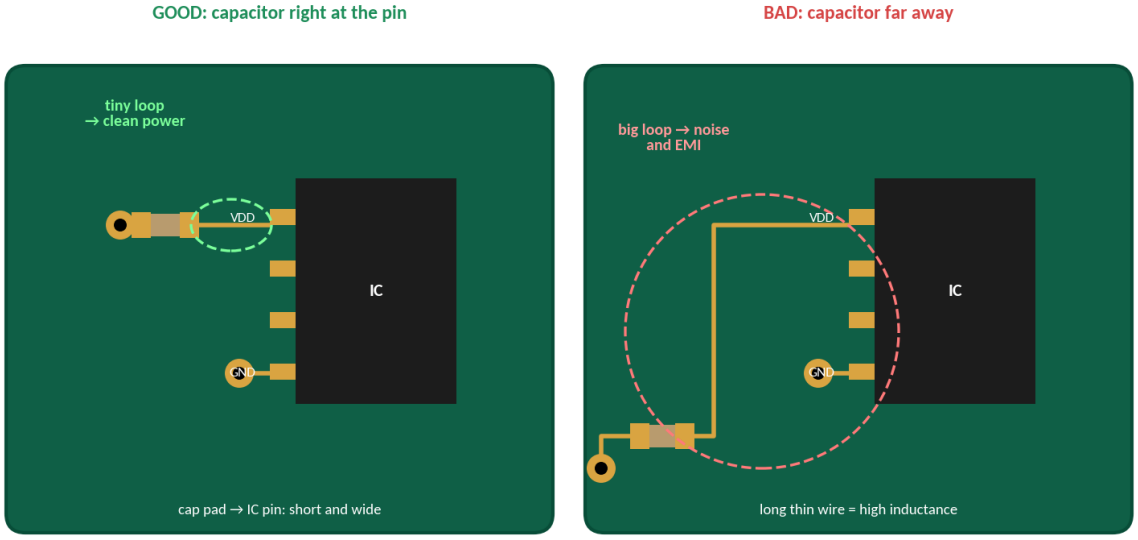


Figure 39. The capacitor must be right at the power pin, with a short path to the ground via.

- Place the 100 nF capacitor **as close as possible** to the IC power pin (touching distance, 1 – 2 mm).
- The power should go **through the capacitor pad** to the pin, not pass by it. The ground end of the capacitor gets its **own via** directly at the pad.
- Place the capacitors on the **same side** as the IC if possible. If the IC is a BGA or the top is crowded, place them on the back, directly under the IC.
- Order of size: the smallest capacitor closest to the pin, the bigger capacitor a little further.

Voltage regulators

- **Input capacitor** right at the input pin. **Output capacitor** right at the output pin. Both ground ends go to the same ground area with short paths.
- **Switching converters:** the loop "input capacitor → switch → inductor → output capacitor" carries high current. Make it **as small as you can**.
- **Feedback resistors** close to the FB pin, and the feedback trace away from the inductor and the switching node (noisy).
- Keep a **copper area** around the regulator (or the exposed pad) to get rid of heat.

Crystals and clocks

- Crystal as close as possible to the chip (less than 10 mm). Load capacitors next to the crystal. No other traces under the crystal. Surround it with a ground guard if possible.

Connectors and protection

- Connectors at the **edge** of the board. Make sure the cable and the plug can enter.
- **ESD / TVS diodes** and ferrite beads right next to the connector pin, **before** the signal goes into the board.
- AC coupling capacitors of a differential pair: side by side, same orientation, close to the connector or chip.

Analog, digital and heat

- Keep noisy parts (switching regulators, motor drivers, clocks) **away** from sensitive parts (ADC, sensors, radio, audio).
- Do not place hot parts (regulators, power transistors) next to heat-sensitive parts (crystals, sensors).
- Leave space for the tall parts and screws. Check the 3D view.

Assembly rules

- Keep the gap between part courtyards at least **0.3 – 0.5 mm**. For hand soldering use 1 mm.
- Rotate parts in steps of 90°. Put all pin-1 marks in the same direction when you can.
- Put all SMD parts on **one side** if possible. It is cheaper to assemble.
- Add **test points** for power rails, ground and important signals.
- Align resistors and capacitors in neat rows. It looks professional and saves routing space.

9.4 Useful placement tools

Tool	Key	What it does
Move	**M**	Move the selected parts. Click again to place.
Drag	**G**	Move a part and keep its tracks connected.
Rotate	**R**	Rotate 90°. (Edit › Rotate Counterclockwise)
Flip	**F**	Put the part on the other side of the board.
Edit properties	**E**	Exact X, Y position, orientation, lock, side.
Align / distribute	Right-click	Align selected parts by their edges or centres; distribute with equal spacing.
Measure	**Ctrl** + **Shift** + **M**	Measure distance between two points.
Find	**Ctrl** + **F**	Find a reference or net.

Placement checklist

- Every decoupling capacitor is touching its IC power pin.
- The ratsnest is short and does not cross a lot.
- Connectors are on the edges and accessible.
- Noisy and sensitive parts are separated.
- Space for traces and vias between parts (do not pack too tight).
- You can still read all part names (reference designators).

Check yourself

1. What is cross-probing? Describe the steps to move the regulator circuit together.
2. Where do you place a decoupling capacitor and why?
3. Which parts go on the board edge?
4. Why do we keep switching regulators away from sensitive analog parts?

10 Trace Width: Power, Signal and Impedance

How wide must a trace be? It depends on current for power and on impedance for fast signals.

In this chapter you will learn

- How to choose the width of a power trace from the current
- Single-ended impedance (50 Ω), microstrip, stripline, coplanar
- Differential pairs (90 Ω, 100 Ω): geometry, spacing, length
- How to find the width with calculators and use it in KiCad

There are two different reasons to choose a trace width. **Power traces** must be wide enough for the current. **Fast signal traces** must have the exact width that gives the right impedance.

10.1 Power traces: the width from current

Current heats a trace. If the trace is too thin it becomes hot, wastes voltage and can burn. The standard formula is **IPC-2221**:

IPC-2221 formula

$I = k \times \Delta T^{0.44} \times A^{0.725}$

I = current (A) · ΔT = temperature rise (°C) · A = cross-section area (mil²) = width × thickness · k = 0.048 (outer layer) or 0.024 (inner layer)

You do not need to calculate by hand. KiCad has a **Track Width** tool inside the **Calculator Tools**. This table was calculated with the formula for a temperature rise of 10 °C:

Current	Outer layer 1 oz (35 µm)	Outer layer 2 oz (70 µm)	Inner layer 1 oz
0.5 A	0.12 mm	0.06 mm	0.30 mm
1 A	0.30 mm	0.15 mm	0.78 mm
2 A	0.78 mm	0.39 mm	2.03 mm
3 A	1.37 mm	0.68 mm	3.56 mm
5 A	2.77 mm	1.38 mm	7.19 mm
10 A	7.19 mm	3.60 mm	18.71 mm

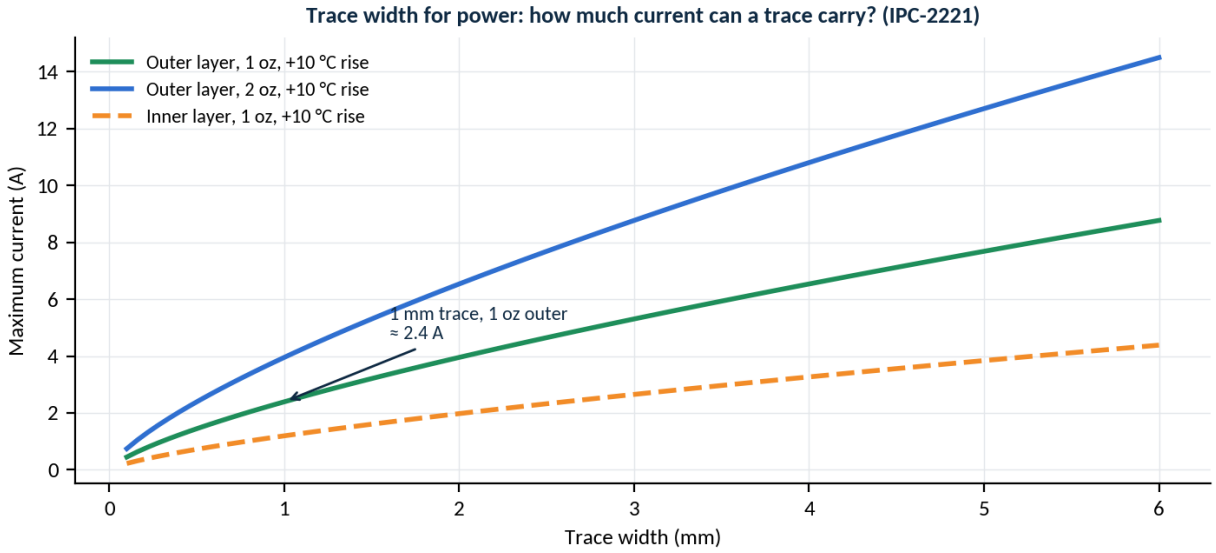


Figure 40. Maximum current vs. trace width (10 °C temperature rise).

Practical rules for power

- **Add a margin.** Design for 1.5 – 2 × the real current. Do not design at the limit.
- **Inner layers** carry less current than outer layers (they cannot cool as well).
- A **copper zone** (pour) or a **plane** is better than a trace. Use zones for 2 A and more.
- **Bottleneck rule:** the whole path is only as strong as its thinnest part. Check the pad, the via and the connector.
- **Vias:** one 0.3 mm via carries about 1 A. Use two or more vias in parallel for big currents.

Voltage drop

Copper has resistance. For 1 oz copper the resistance is about **0.5 mΩ per square** (one square = any length equal to the width). So R ≈ 0.5 mΩ × (length / width).

Example

A 3.3 V line, 100 mm long, 1 mm wide, 1 oz copper carries 2 A.

R = 0.5 mΩ × (100 / 1) = 50 mΩ. Voltage drop = 2 A × 0.05 Ω = **0.1 V** (3 % of 3.3 V).

If the line is only 0.3 mm wide: drop = 0.33 V, too much. Wider traces or a plane are needed.

10.2 Signal traces: width from impedance

For fast signals, the width is chosen to get the correct impedance Z0 (Chapter 2). The width, the thickness of the insulator under the trace (H) and the board material (εr) decide Z0. **You cannot choose the width freely.** If you choose a different width, the impedance changes.

Types of controlled-impedance traces

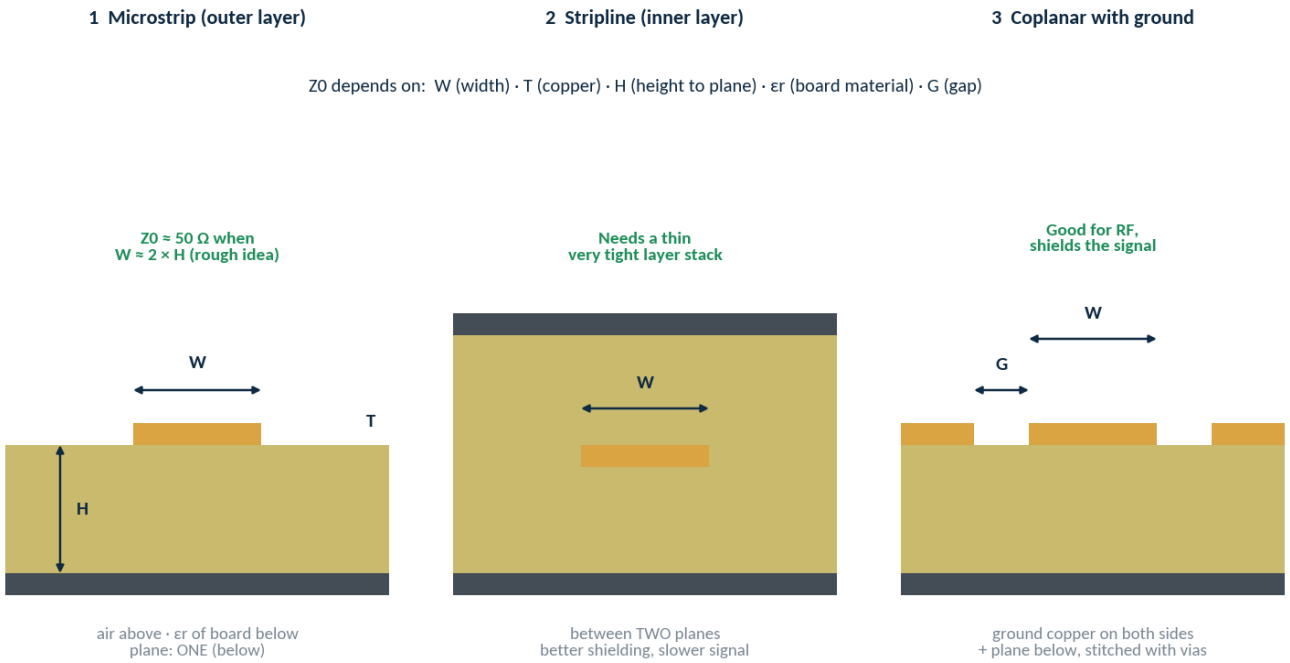


Figure 41. *Three common geometries. W = width, T = copper thickness, H = height to plane, G = gap to side ground.*

Type	Where	Description
Microstrip	Outer layer (top or bottom)	Trace on the surface, one reference plane below. Easy and common. Air above. Most USB, HDMI, PCIe traces use it.
Stripline	Inner layer	Trace between two planes. Better shielding, less EMI. Needs a special stackup. Signal is a bit slower.
Coplanar waveguide (with ground)	Outer layer	Ground copper on both sides of the trace plus a plane below. Good for RF and very clean signals.

What changes the impedance?

If you ...	Impedance becomes ...	Why
make the trace **wider** (W ↑)	lower	More capacitance to the plane
make the insulator **thicker** (H ↑)	higher	Plane is farther away, less capacitance
use a material with higher εr	lower	More capacitance
make the copper thicker (T ↑)	slightly lower	The trace is a bit bigger
bring a side ground closer (G ↓)	lower	More capacitance to the side ground

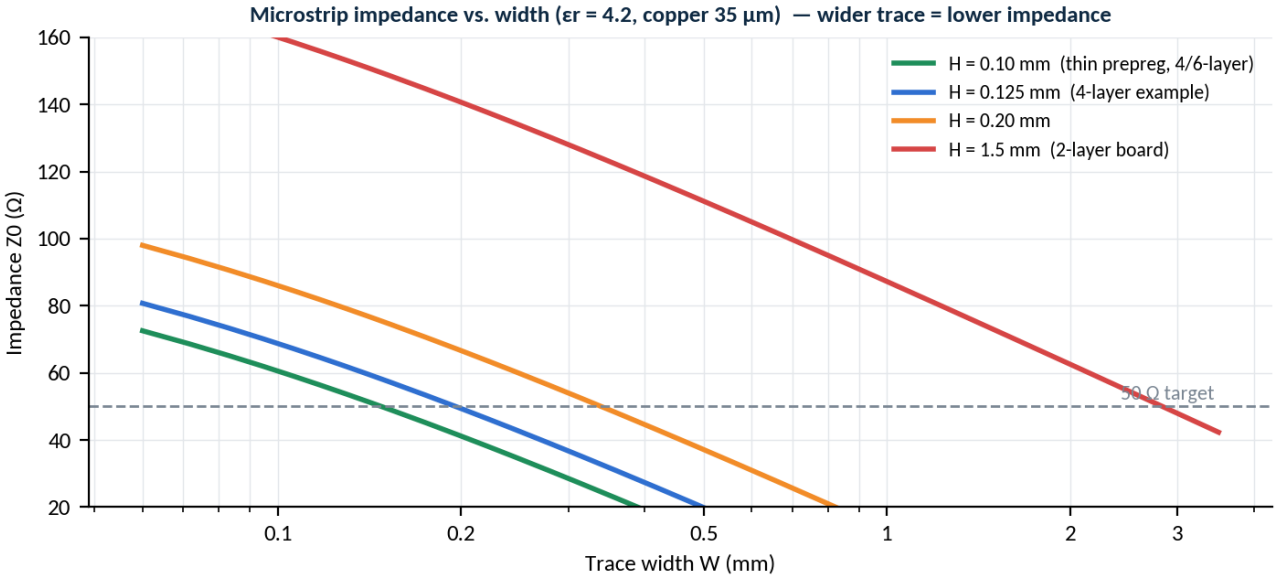


Figure 42. Microstrip impedance vs. width for different insulation heights H. Notice: the thin 4-layer insulation needs a narrow trace; the 2-layer board needs a very wide trace.

10.3 How to calculate the width

For learning, the following simple formula for microstrip gives a good estimate (valid for about 0.1 < W/H < 2):

Microstrip formula (estimate)

$$Z0 \approx 87 / \sqrt{(\varepsilon r + 1.41)} \times \ln(5.98 \times H / (0.8 \times W + T))$$

Example: 50 on a 4-layer board

H = 0.125 mm, εr = 4.2, T = 0.035 mm. Solve for W: W ≈ **0.196 mm** (about 0.2 mm or 7.9 mil).

The same 50 Ω on a **2-layer board** with H = 1.5 mm needs W ≈ **2.8 mm**! That is why fast signals are easier on a 4-layer board.

For real work, use a **calculator**:

- **KiCad PCB Calculator** (Project Manager › Calculator Tools › PCB Calculator › **TransLine**): choose Microstrip, Coplanar or Stripline, enter εr, H, T, and the target Z0, press “Synthesize”. It gives W.
- **Your factory’s calculator** (for example NextPCB or JLCPCB impedance calculators). It is better, because it uses the factory’s real stackup and material.

Example result from a factory calculator

A 4-layer 1.6 mm board (L1 signal / L2 GND / L3 power / L4 signal) with a 0.125 mm prepreg (type 2116) between L1 and L2 gave these values:

Target	Type	Trace width W	Gap S	Result
50 Ω	Single-ended, L1 over L2	7.89 mil (0.200 mm)	—	49.7 Ω
90 Ω	Differential, L1 over L2	6.45 mil (0.164 mm)	5.26 mil (0.134 mm)	90.2 Ω

Target	Type	Trace width W	Gap S	Result
90 Ω	Differential (tighter)	5.51 mil (0.140 mm)	4.06 mil (0.103 mm)	90.3 Ω

These numbers are only an example

Your factory may use a different prepreg, a different εr and a different copper thickness. Always calculate with **your factory's** stackup and write "impedance control: 50 Ω single, 90 Ω differential, layer L1" in your order.

10.4 Differential pairs

A **differential pair** is two traces that carry the **same signal with opposite signs**: one is P (+), the other N (−). The receiver looks only at the **difference** between them. Noise that hits both traces equally (common-mode noise) disappears in the subtraction. USB, HDMI, PCIe, Ethernet, SATA and LVDS all use differential pairs.

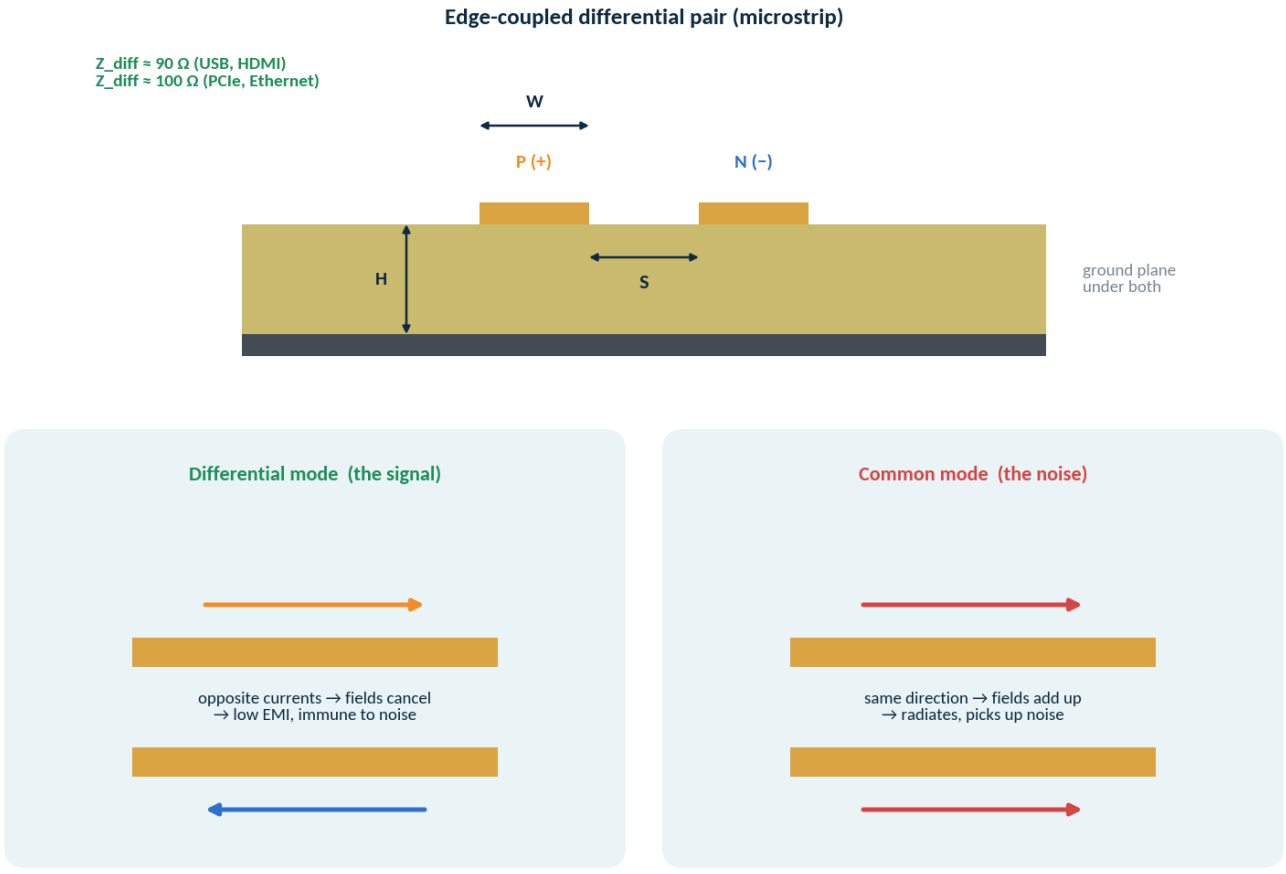


Figure 43. Cross-section of an edge-coupled pair and the two "modes". W = trace width, S = gap, H = height to plane.

Why are differential pairs good?

- **Noise immunity:** the same noise on both lines cancels at the receiver.
- **Low EMI:** the two currents are opposite. Their magnetic fields cancel.
- **Better for low voltage:** small voltage swings are possible, so very fast speeds.
- **Return path is the other trace** (partly) and the plane below. It is less sensitive to ground noise.

The geometry: W, S and H

The differential impedance **Zdiff** depends on W, H, εr and the gap **S**. When the traces are close, their fields overlap and Zdiff goes **down**. An estimate:

Zdiff estimate

$$Zdiff \approx 2 \times Z0 \times \left(1 - 0.48 \times e^{(-0.96 \times S / H)} \right)$$

Z0 is the single-ended impedance of one trace alone (from the microstrip formula).

Example with W = 0.164 mm, S = 0.134 mm, H = 0.125 mm, εr = 4.2: Z0 = 55 Ω → **Zdiff ≈ 91 Ω**. This agrees with the factory result above.

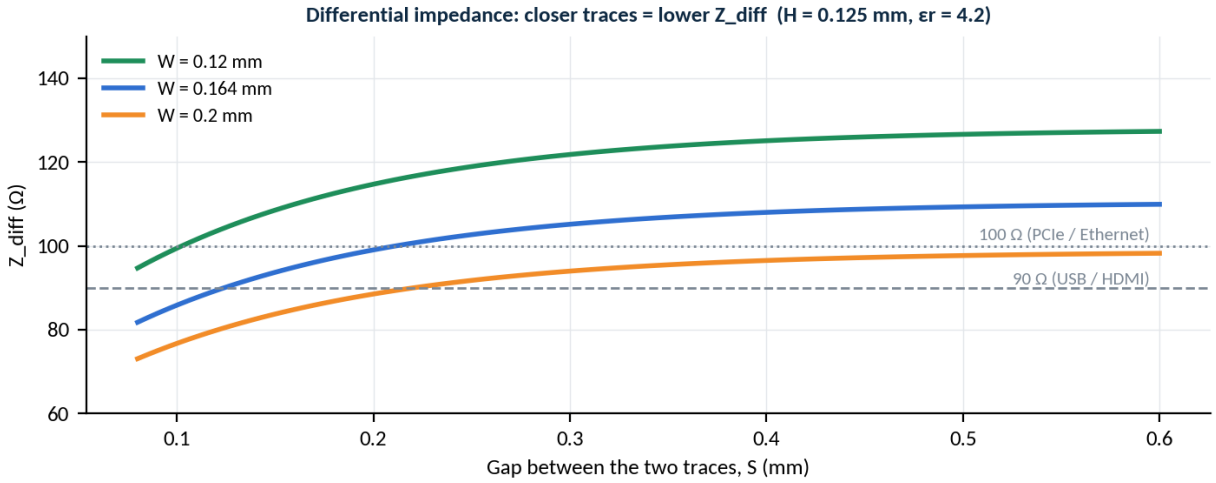


Figure 44. *Zdiff for three trace widths. Zdiff falls when the gap S gets smaller.*

Remember: two numbers decide a pair

W (width of each trace) and **S** (gap between them). Both come from the calculator for the target Zdiff (90 Ω or 100 Ω).

The gap S must stay **constant** along the whole pair. If S changes, Zdiff changes and a reflection happens.

Coupled or not?

Tightly coupled pairs (small S, about the trace width) are more immune to noise and are the normal choice for USB, HDMI and PCIe. **Loosely coupled** pairs (large S) are used where the pair must pass through a narrow area. In both cases, W and S must give the target Zdiff.

10.5 Typical impedances of common interfaces

Interface	Impedance	Type
USB 2.0 / 3.x	90 Ω	Differential
HDMI / DisplayPort	100 Ω	Differential
PCI Express	85 Ω or 100 Ω (see chip guide)	Differential
Ethernet (10/100/1000)	100 Ω	Differential
SATA, LVDS, MIPI	100 Ω	Differential
RF antenna lines, fast clocks	50 Ω	Single-ended
Video (analog)	75 Ω	Single-ended

10.6 Put the numbers in KiCad

1. Open **File › Board Setup › Net Classes**.
2. Add a class, for example **USB_DIFF**. Fill the **Track Width** (W) and the **DP Width** and **DP Gap** (differential pair width and gap) with the calculator results.
3. Add a pattern in the assignment table (for example **/USB_D***).
4. In the PCB editor press **6** (route differential pair). KiCad uses these numbers automatically.
5. For 50 Ω nets create a class with the calculated W and assign the nets.

Check yourself

1. What width do you need for a 2 A power trace on an outer layer, 1 oz copper (10 °C rise)?
2. Why is a wider trace not better for a 50 Ω signal?
3. What is the difference between microstrip and stripline?
4. Name two reasons why differential pairs are used.
5. What happens to Zdiff when two traces of a pair move closer together?

11 Routing: Traces, Vias, Power Zones and Planes

How to connect everything cleanly: short traces, 45° corners, matched pairs, good vias

In this chapter you will learn

- The best order for routing and the KiCad routing tools
- Short traces, trace capacitance and 45° corners
- Differential pairs, length matching and high-speed traces
- Vias, copper pours, power planes and ground stitching

11.1 Before you start routing

- Placement is finished and the ratsnest is short.
- Board Setup is done: stackup, constraints, net classes (Chapter 8).
- Ground and power planes are planned. On a 4-layer board, L2 will be GND and L3 power.
- Save a backup copy of the project.

11.2 Routing order

Route the **most difficult** signals first, while there is still free space. Easy signals can go around them.

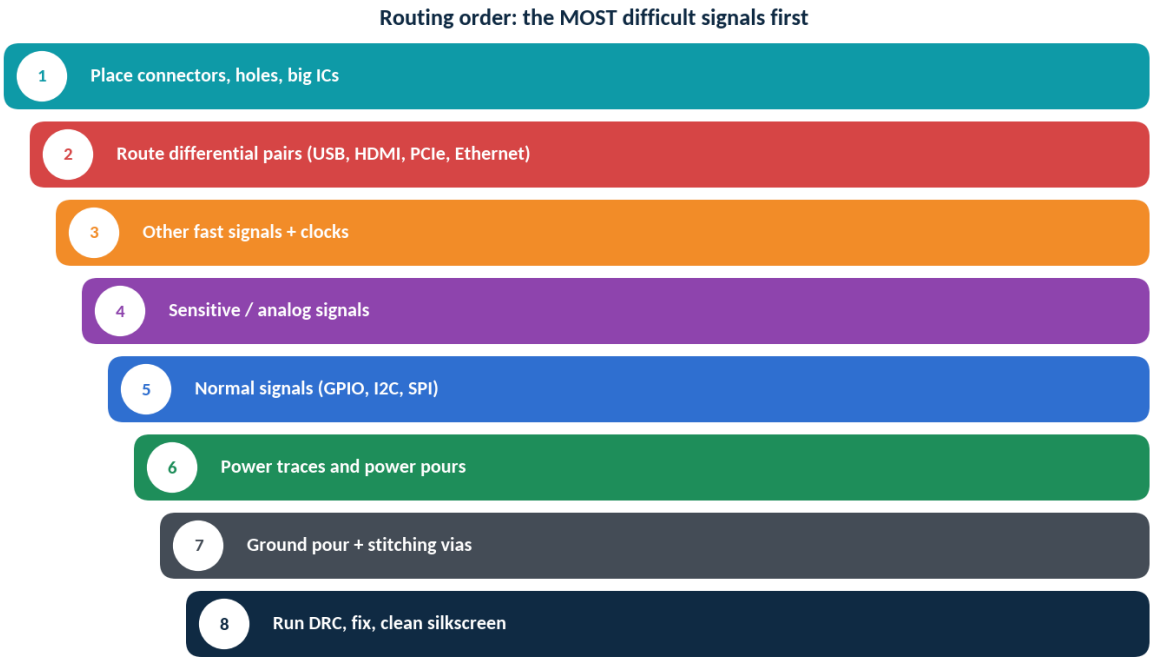


Figure 45. *A good routing order, from the hardest signals to the easiest.*

A common plan for a high-speed board is: (1) differential pairs such as USB, HDMI, PCIe and Ethernet; (2) other high-speed single lines and clocks; (3) sensitive analog lines; (4) normal signals; (5) power traces and pours; (6) ground pour and stitching vias. Route each group on the layer that has a good reference plane.

11.3 The routing tools of KiCad

Action	Key	Notes
Route single track	**X**	Click a pad, move, click to place corners. Click on the target pad to finish. Double-click ends the track.
Route differential pair	**6**	Click on one pad of the pair; KiCad routes both traces with the DP width and gap.
Place via while routing	**V**	Adds a via and switches to the other layer.
Switch layer	**PgUp** / **PgDn**	Change active layer (also places a via if routing).
Change posture (corner shape)	**/**	Changes which part of the corner is diagonal.
Routing mode	**Shift** + **R**	Cycle: Highlight collisions / Shove / Walk around. Beginners: Shove or Highlight.
Undo last segment	**Backspace**	Removes the last placed segment while routing.
Drag a track	**D**	Move a track and keep connections.
Tune single track length	**7**	Adds serpentine to make a trace longer.
Tune differential pair length	**8**	Adds serpentine to both traces of a pair.
Tune skew of a pair	**9**	Adds serpentine to the shorter trace of a pair.
Delete	**Del**	Delete the selected track.

Pro Tips

- Look at the status bar: the track width and via size change with the net class.
- If a track "jumps" to a strange place, use **Highlight collisions** mode and move slowly.
- Click on a pad with the mouse and press **X** to start routing exactly from it.
- Turn on **Net colours** in the Appearance panel to see important nets.

11.4 Keep traces short

Short traces are the first rule of good routing. Reasons:

- **Less capacitance.** A trace over a plane has about **1.2 pF per cm** (50 Ω line). A 10 cm trace has about 12 pF. It slows edges and loads the driver.
- **Less inductance.** About 3 nH per cm. A long trace resists sudden current changes.
- **Less EMI.** Short traces are poor antennas.
- **Less delay and skew** and less loss.
- **Less crosstalk.** There is less length to run alongside others.

Where do 1.2 pF/cm and 3 nH/cm come from?

A signal travels on FR-4 at about 5.9 ps per mm. For a 50 Ω line: $C = t_{pd} / Z0$ = 5.9 ps / 50 Ω = 0.118 pF per mm; $L = Z0 \times t_{pd}$ = 50 × 5.9 ps = 0.295 nH per mm.

These are approximate values, but good enough to feel how a trace "looks" electrically.

11.5 Route at 45°, not 90°

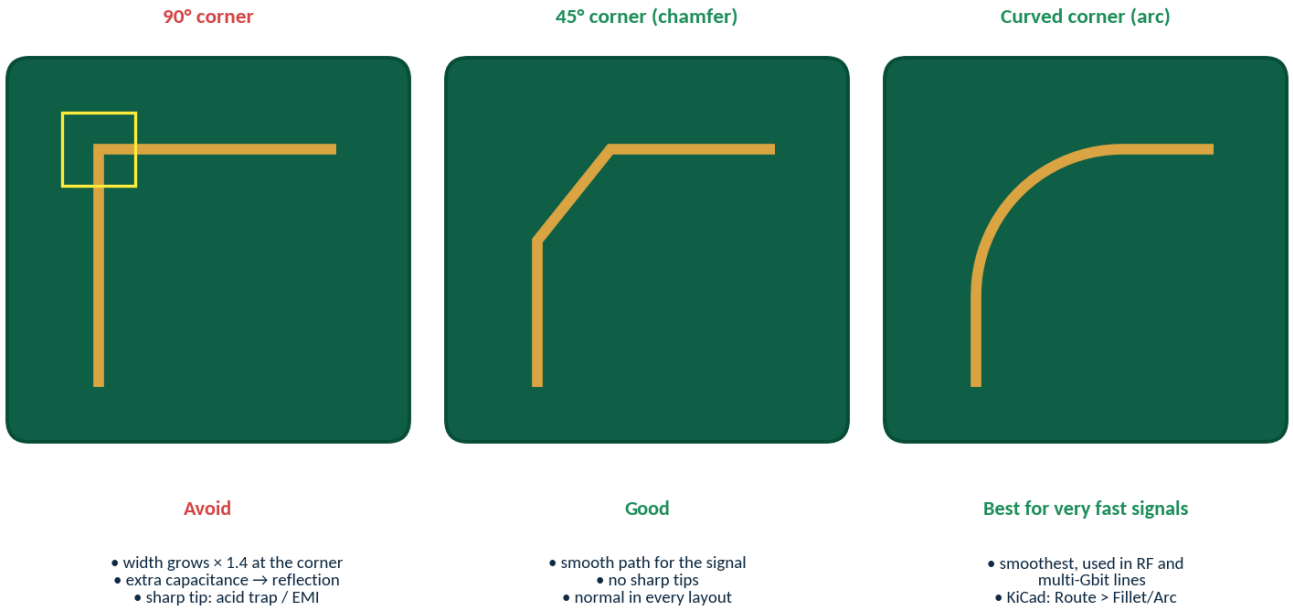


Figure 46. Compare the three corner styles.

Use **45° corners** or curves. Avoid 90° corners. The reasons:

- **Width and capacitance:** at a 90° corner the trace is wider by 1.4 times (the diagonal of a square). This is extra capacitance, so the impedance dips and a small reflection is produced. It matters most for fast signals.
- **Field concentration:** the electric field is strong at the outer tip of a sharp corner. It can radiate more.
- **Manufacturing:** a sharp inside angle can trap etching acid ("acid trap") and slightly change the width.
- **Looks and shorter path:** a 45° bend is shorter and cleaner.

> **Honest advice**
>
> For slow signals (LEDs, buttons) a 90° corner will work. But a good habit costs nothing. Use 45° always, so you do not forget when the signal is fast. For very fast or RF lines, use **curved (arc) corners**.

11.6 Trace widths while routing

- Use the width from the **net class**. Do not change it by hand for impedance-controlled nets.
- A short **neck-down** (narrower trace) to enter a fine-pitch pad is allowed. Keep it as short as possible, then return to the full width.
- Keep the width the same on the whole path. A sudden width change is an impedance step and gives a reflection.
- Add **teardrops** at pads and vias (Chapter 12). They make joints stronger.

11.7 Differential pairs: routing and length matching

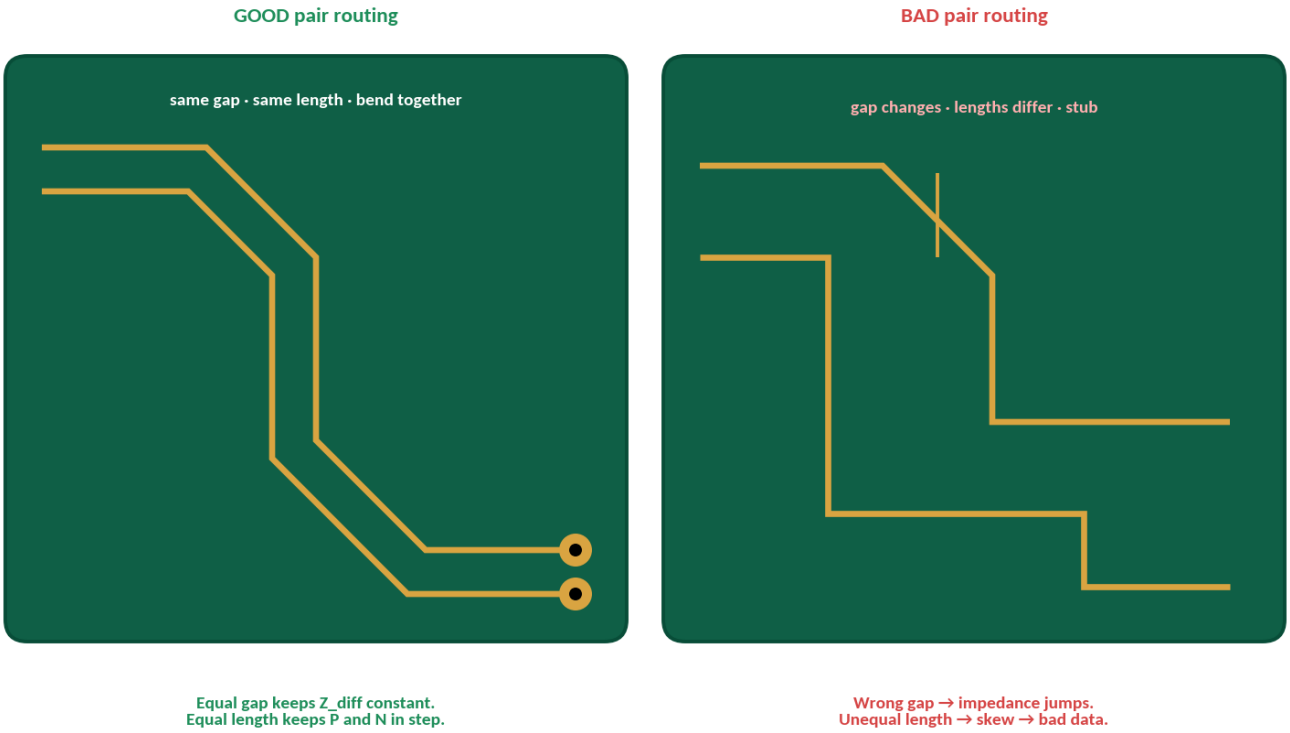


Figure 47. Good and bad pair routing.

Rules for a differential pair

1. **Set the net class** with correct W and gap (Chapter 10).
2. Press `6` and route from the connector or the chip. The pair follows your mouse. Keep the pair **together** for the **whole path**.
3. **Keep the gap S constant.** Spread the pair apart only where you must (at pads), and keep that part short.
4. **Keep equal length.** Bends make the outer trace longer. Use the same number of left and right bends so that lengths balance.
5. **Stay on one layer** and one reference plane. Avoid vias in fast pairs. If you must change layer, change **both traces together** with two vias side by side, and put **ground vias** next to them.
6. **No stubs** (small side branches) and no test points on the pair.
7. **Keep away** from other signals: at least 3 × W (more for clocks and switching noise). Keep away from the board edge and from holes.
8. **Do not cross a gap** in the reference plane. Do not route over a plane split.

Length matching (skew control)

When two traces of a pair have different lengths, the signals arrive at different times. This time difference is called **skew**. Skew turns the differential signal into common-mode noise and closes the “eye” of the data. A fast signal moves about 6 ps per mm, so 1 mm of extra length is 6 ps of skew.

What to match	Beginner target	Comment
Inside a fast pair (USB 3, PCIe, HDMI): P vs N	within **±0.127 mm (5 mil)**, tighter is better, for example 0.05 mm	Called **intra-pair skew**.
Slower pairs (USB 2.0, 100 Mb Ethernet)	within about 0.5 mm	More relaxed, but practise the habit.
Between pairs of one bus (lanes)	Follow the chip guide (often a few mm)	Called **inter-pair** skew.
Maximum pair length	Example from one reference design: USB 3 ≤ about 100 mm, PCIe ≤ about 75 mm	Limits depend on speed, material and connectors. Follow your chip vendor's guide.

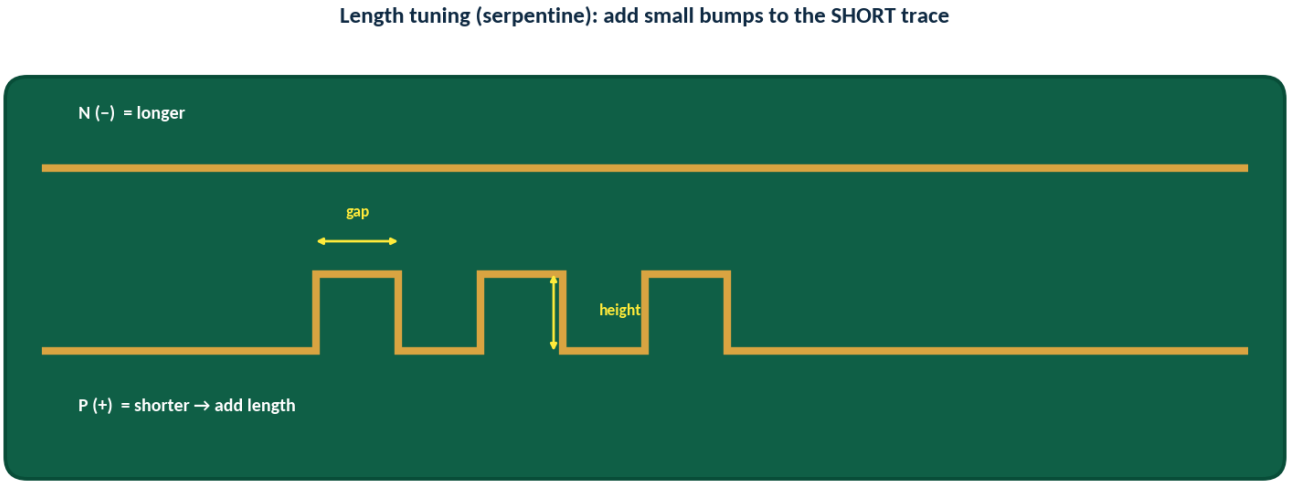


Figure 48. Add small bumps (serpentine) to the shorter trace, near where the mismatch happens.

Use **KiCad tuning tools**: press `8` to tune the length of the whole pair, or `9` to add length to the shorter trace of the pair to equalise it. Many designers find tool **9** works well. Click on the track, move the mouse to make the bumps, and finish. Check the result in **Inspect › Net Inspector**, which lists the length of every net.

Serpentine (meander) rules

- Keep the bumps small and tidy; do not make long, wide meanders.
- Spacing between meander legs ≥ 3 × W (to avoid crosstalk with itself).
- Add the extra length **near the place where the mismatch is made** (for example near a bend or a pad).
- Do not place meanders right at a via or a pad.

11.8 High-speed single-ended traces

- Give the trace a **solid reference plane**. Use the calculated width for 50 Ω.
- Keep it as short and direct as possible. No stubs, no T-branches (route "point-to-point" or "daisy chain").
- Add a **series resistor** (22 – 33 Ω) near the driver when the datasheet suggests it.
- Keep clocks away from other signals. Surround with ground if needed and add ground vias beside layer changes.

11.9 Vias: types, size and spacing

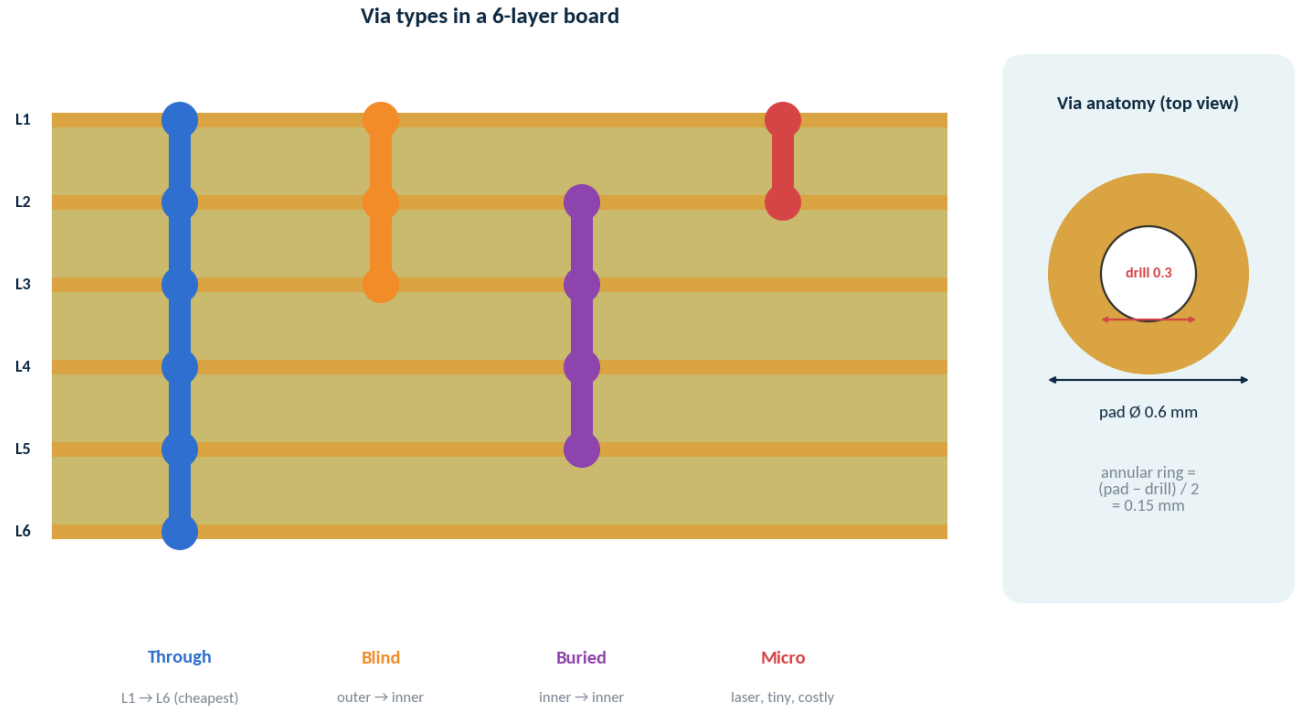


Figure 49. Via types and the parts of a via.

Via type	Connects	Cost	Use
Through	All layers	Cheapest	**Use this one.** 95 % of all boards.
Blind	Outer to inner layer	Higher	Dense boards (phones)
Buried	Inner to inner layer	High	Dense boards
Micro-via	Neighbour layers by laser	Highest	BGA with very fine pitch

Standard via: pad 0.6 mm, drill 0.3 mm. Use 0.8 / 0.4 mm for power. Each via adds a little inductance (about 1 nH) and a little capacitance, so **do not use more vias than needed on a fast signal**.

Distances: trace-to-via and via-to-via

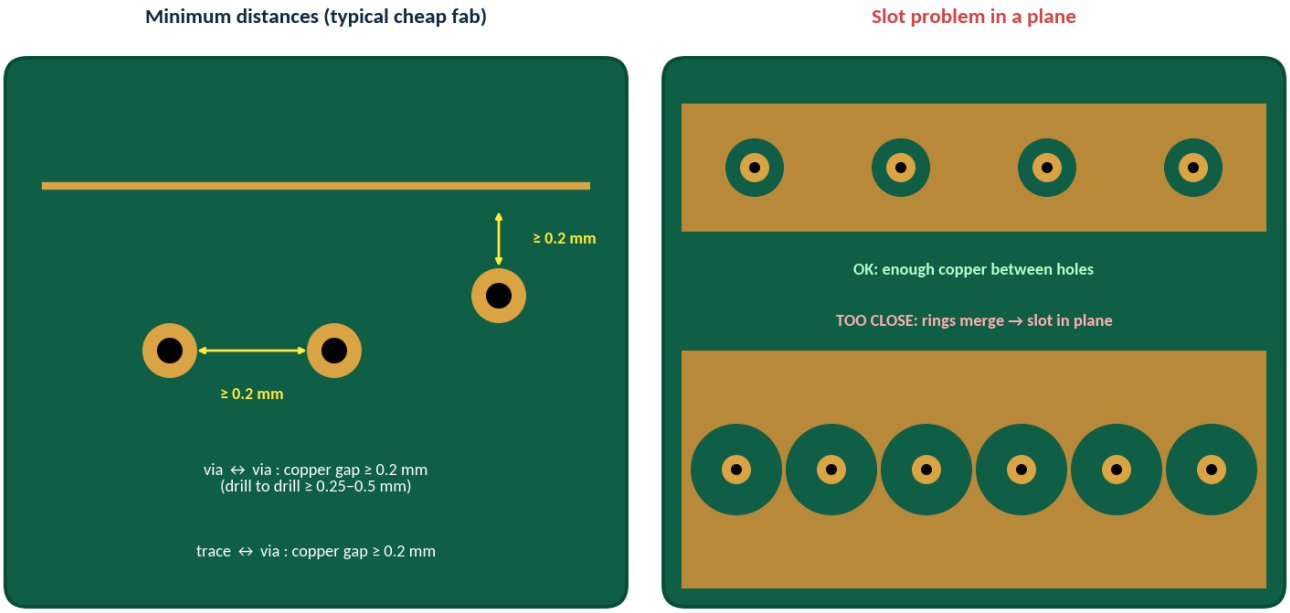


Figure 50. Keep the minimum clearances. Do not place vias in a tight row on a plane.

Distance	Value	Why
Trace to via (copper to copper)	≥ 0.2 mm (the board clearance)	Factory limit; avoids short circuits.
Via to via (copper to copper)	≥ 0.2 mm	Factory limit.
Drill hole to drill hole	≥ 0.25 – 0.5 mm	Holes too close break the board between them.
Via pitch in a row on a plane	≥ 2 × via pad (for 0.6 mm via: ≥ 1.2 – 1.5 mm)	Otherwise the clearance rings join and cut a **slot** in the plane.
Via near a pad	≥ 0.25 mm from the pad edge (avoid via-in-pad)	Solder can flow into the via and leave the joint empty.
Via to board edge	≥ 0.5 mm	Avoid broken copper at the edge.

Via in pad

Do not put a via inside an SMD pad unless the factory offers **filled and capped vias**. Normally, the solder runs down the hole and the part is not connected well. Put the via next to the pad and connect with a short trace.

11.10 Power routing: traces, zones and planes

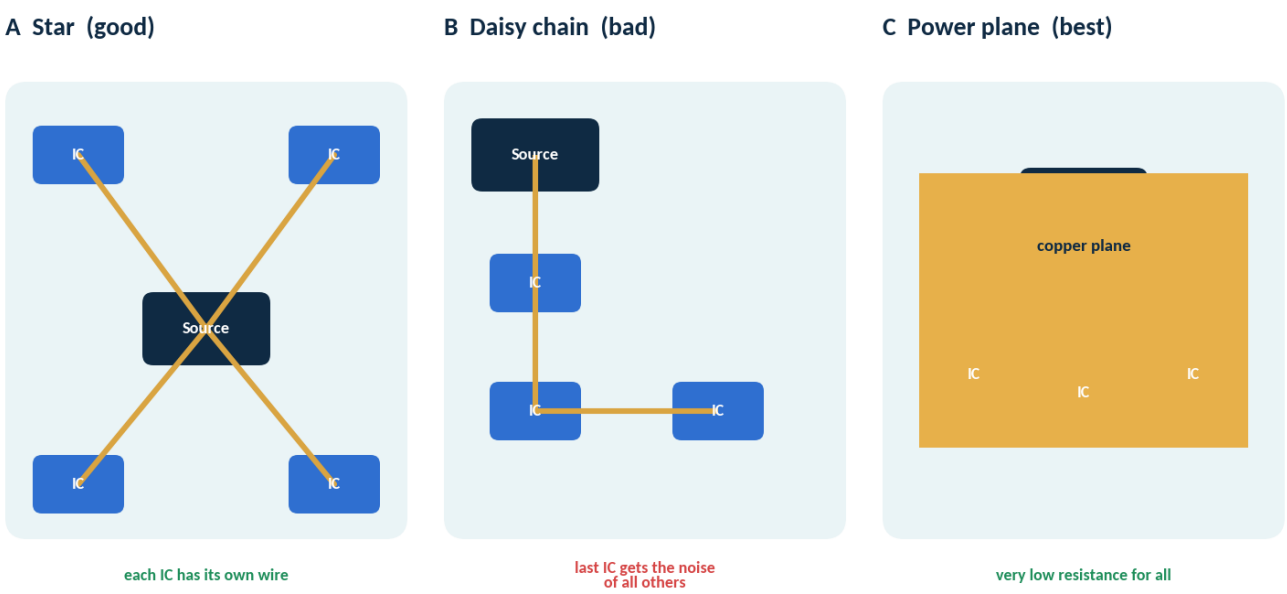


Figure 51. Power distribution styles. Star is better than a daisy chain. A plane is best.

- **Wide traces.** Use widths from Chapter 10. Do not use thin 0.2 mm lines for power.
- **Star distribution:** each IC gets its own branch from the source. This keeps noise of one chip away from the others.
- **Power zones (pours):** a copper area on a layer that carries a power rail. Use it for high-current rails or when one rail is needed in many places.
- **Power plane:** a whole inner layer (L3). On a 4-layer board it can be one rail (3.3 V) with **islands** of other rails (5 V, 1.8 V) in places where needed. Do not make island borders cross under fast traces.
- **Connect plane to pin with a short, fat via**, near the pin. Put the bypass capacitor on the way.
- **Ground:** use **many** ground vias. Every ground pin should have its own via to the plane right next to the pad.

11.11 Copper zones (pours)

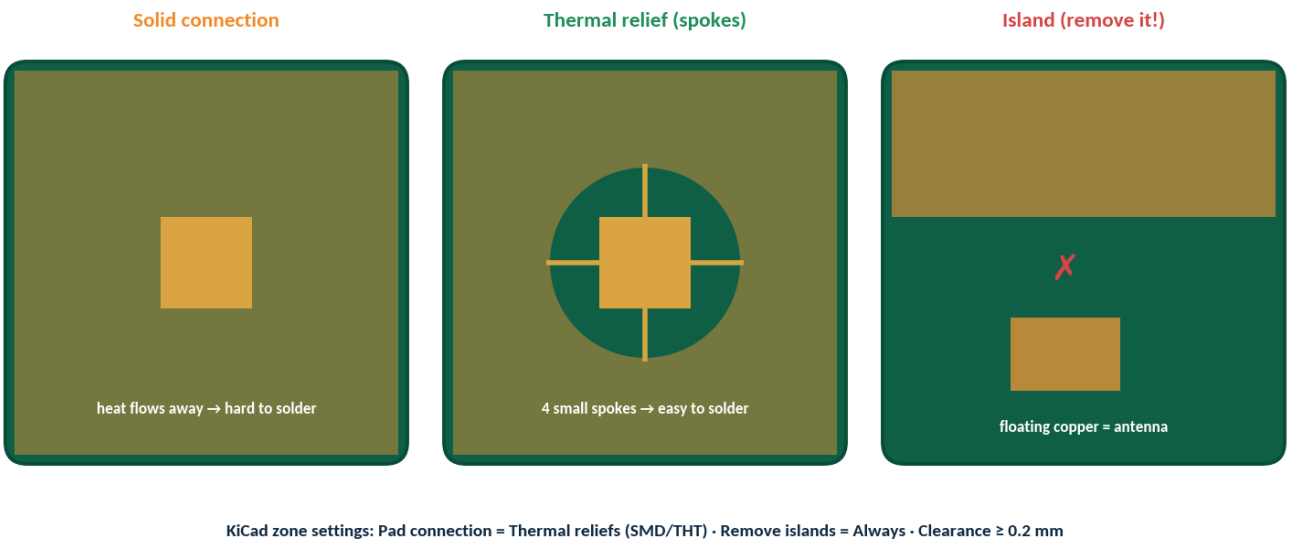


Figure 52. Zone connection types and the island problem.

1. Select the layer (for example F.Cu or In1.Cu). Press **Ctrl** + **Shift** + **Z** or click **Add Filled Zone**. Click inside the board and choose the net (for example GND).
2. In the dialog set: **Clearance 0.2 – 0.3 mm**, **Minimum width 0.2 mm**, **Pad connections: Thermal reliefs** (for easy soldering), **Remove islands: Always**.
3. Draw the zone outline around the board (stay inside Edge.Cuts) and finish with a double click.
4. Press **B** to **fill all zones**. (Press **Ctrl** + **B** to unfill.) Fill again after every routing change.
5. If two zones overlap, give the more important one a higher **priority**.

Pour rules

- On a 2-layer board, pour **ground on the bottom** and also on the top in free space, joined by many vias.
- **Remove islands** (floating copper). They work like small antennas and can collect noise.
- Use **thermal reliefs** on pads so that soldering is easy; use solid connections for power pads that carry high current.
- Avoid thin copper necks in a zone; they act as bottlenecks.
- Do not let a pour run under the crystal or antenna keep-out. Use a **Rule Area (keepout)** there.

11.12 Ground stitching vias

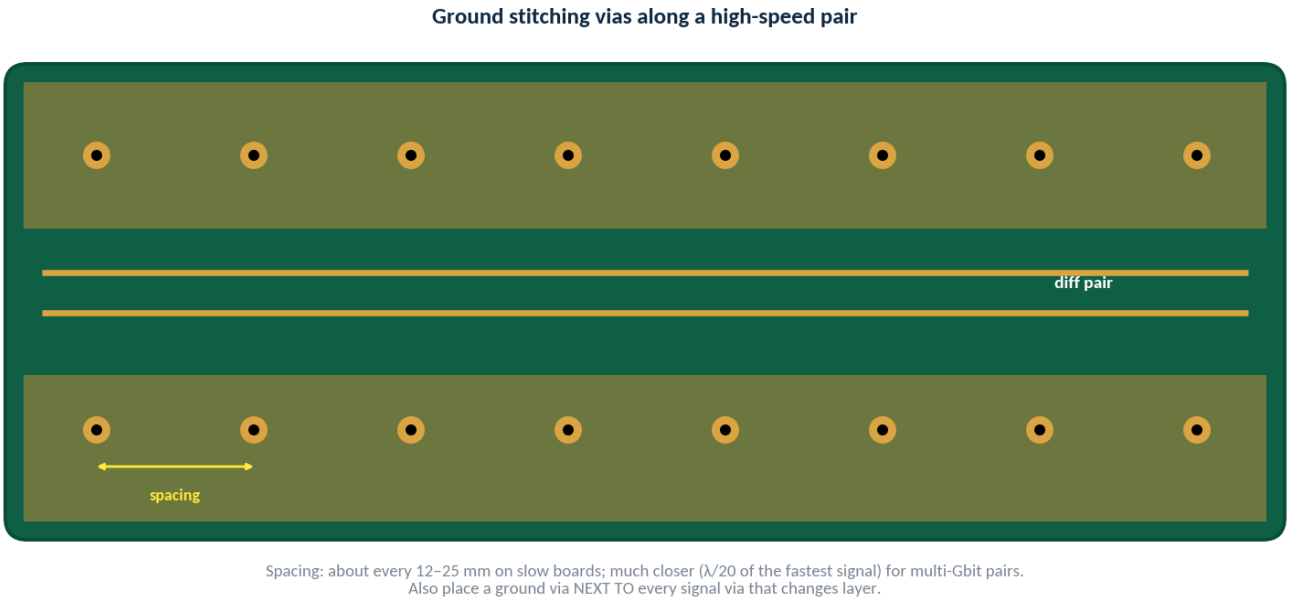


Figure 53. Stitching vias tie the ground areas together and make a good return path.

Ground stitching vias connect the ground copper of different layers. They give the return current a short, low-impedance path. Place them:

- **Along high-speed traces and differential pairs**, on both sides (a “via fence”).
- **Next to every signal via** that changes layers, so the return current can change layers too.
- **Around the board edge** and between separated ground areas.
- In the pour, spread vias **every 12.5 – 25 mm** (500 – 1000 mil) as a general rule. For multi-gigabit pairs, put them much closer: about 1/20 of the wavelength of the fastest signal (a few millimetres).

11.13 Other routing hints

- **Analog and digital:** route analog lines away from digital ones; do not let digital traces cross over analog ground areas.
- **Crystal:** keep traces as short as possible; no vias; no traces below it.
- **Decoupling capacitor traces:** as short and wide as you can; via at the capacitor pad.
- **Acute angles:** avoid very sharp angles between traces. They can trap acid.
- **Leave space** between pins: do not run a thin trace between two pads of a fine-pitch part unless needed.
- **Test points** on a stub: keep them short, never on a fast pair.

Routing checklist

- All nets are routed (ratsnest empty).
- Differential pairs: width and gap per class, matched lengths, no stubs.
- Fast traces have a solid plane below, no gaps crossed.
- Power traces are wide enough for the current.
- Corners are 45° or curved.
- Vias: standard size, spaced correctly, ground vias next to signal vias.
- Ground zones are filled, no islands.

Check yourself

1. Why do we avoid 90° corners on fast signals?
2. Explain intra-pair skew and how to fix it in KiCad.
3. How far apart should two vias be in a row on a plane? Why?
4. Why must you not place a via inside an SMD pad (usually)?
5. Where should ground stitching vias be placed?
6. In which order do you route different kinds of signals?

12 Finishing the Board and Sending to the Factory

Clean up, silkscreen, DRC, 3D check and Gerber files

In this chapter you will learn

- Clean up the layout, add teardrops and refill zones
- Silkscreen, fiducials, mounting holes and test points
- Design Rules Check (DRC) and the final inspection
- Generate Gerber, drill, BOM and position files; order the board

12.1 Clean up the layout

- **Edit › Cleanup Tracks & Vias**: removes duplicate tracks, dangling stubs and tiny pieces.
- **Edit › Edit Teardrops**: adds smooth "teardrop" shapes where traces meet pads and vias. They strengthen the joint and avoid broken traces.
- Press **B** to **refill all zones**. Always do this before DRC and before plotting.
- Check the corners of the board: are the outline and mounting holes correct?

12.2 Silkscreen

Silkscreen is the white text on the board. It helps you to build, test and repair the board.

- Text size: **height 1.0 mm, line width 0.15 mm** (or bigger). Smaller text is not readable.
- Place the **reference designators** (R1, C5, U2) so you can read them after soldering. Turn them in 90° steps and read in one or two directions only.
- Never put silkscreen on pads or over holes. KiCad can remove the part that touches pads when you plot.
- Mark **pin 1** of ICs and connectors, **polarity** of diodes and capacitors, and the names of connectors ("USB", "5V IN").
- Add the **board name, version and date**, and your name or logo. You can add text with the Text tool and a logo with the Image Converter.
- Hide the **values** (100nF, 10k) if there is no space. Keep references.

12.3 Fiducials, mounting holes and test points

- **Mounting holes:** M3 holes (3.2 mm) at the corners, with a ground ring or a keep-out for the screw head.

- **Fiducials:** small round copper marks (1 mm) used by the assembly machine to align. Add 2 or 3 on the board for assembled boards.
- **Test points:** for power rails and main signals. Label them in silkscreen.

12.4 Design Rules Check (DRC)

DRC checks your layout against the rules you set in Board Setup. Open **Inspect › Design Rules Checker**, tick **Refill zones before performing DRC**, and click **Run DRC**. Fix all errors. Think about each warning.

Message	Meaning	Fix
Clearance violation	Two copper objects are too close	Move one of them, or make the trace thinner for a short part.
Unconnected items	A net is not fully routed	Route the missing link, or check a missing zone fill.
Track width / via size too small	Smaller than the minimum	Use a bigger size.
Hole clearance / hole to hole	Two drills too close	Move the via or hole.
Copper edge clearance	Copper too close to board edge	Move away at least 0.3 – 0.5 mm.
Silkscreen overlap / over pad	Text on pads (warning)	Move the text.
Footprint does not match library	Footprint was edited	Update footprints from library, or accept.
Courtyards overlap	Parts are too close	Move parts apart.

> **Do not ignore errors**
>
> Never “exclude” or ignore an error you do not understand. A DRC error is the factory telling you: “this will not work.” Fix it.

12.5 Final visual check

- Open the **3D Viewer** (**View › 3D Viewer** or **Alt** + **3**). Rotate the board. Are the parts on the correct side? Is pin 1 right? Are the connectors at the edge? Does everything fit in the box?
- Print the layout **1:1 on paper** (File › Print). Place real parts on the paper to check footprints.
- Look at each layer one by one. Hide the others in the Appearance panel.
- Check one more time: **power and ground** nets for shorts using the Net Inspector. Highlight the GND net and look at its pieces.

12.6 Make the factory files

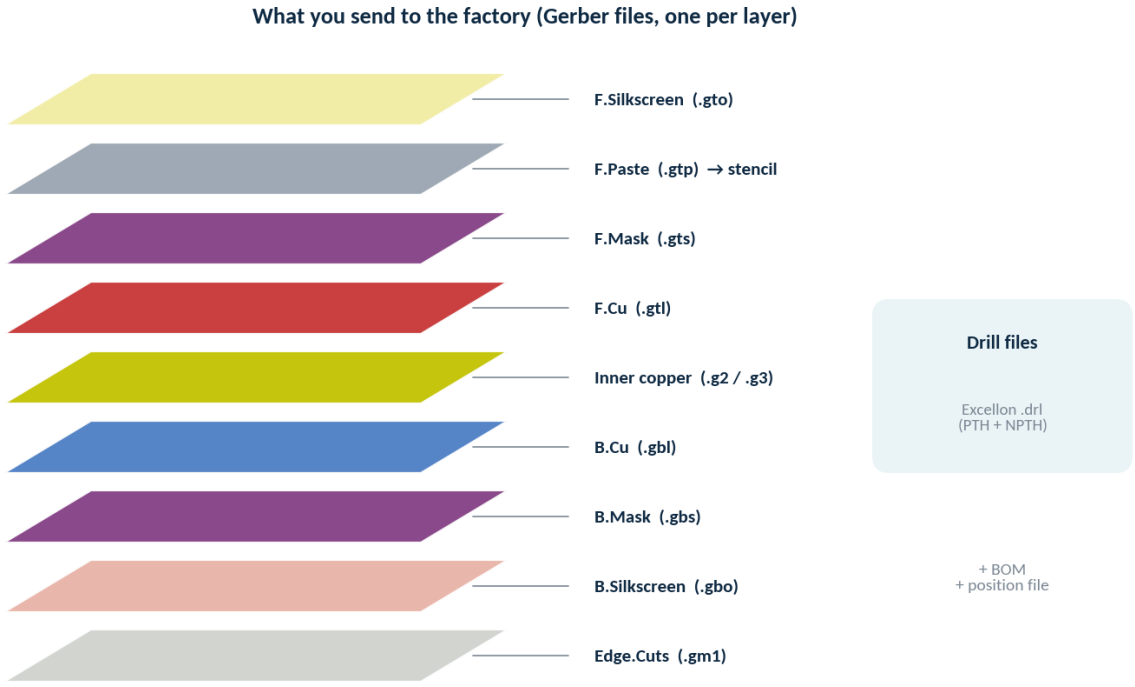


Figure 54. Each Gerber file is one layer. The factory builds the board from these files and the drill files.

Gerber files

1. Open **File › Plot** (in newer versions: **File › Fabrication Outputs › Gerbers**). Choose an output folder, for example *gerbers/*.
2. Choose the **layers**: F.Cu, all inner copper layers, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste (only if you order a stencil) and **Edge.Cuts**.
3. Keep the default options. Tick **Plot reference designators**. Do not plot values. Tick **Check zone fills before plotting**.
4. Click **Plot**.

Drill files

1. In the same window click **Generate Drill Files**.
2. Use **Excellon** format, unit **millimetres** and decimal format. Click **Generate Drill File**. (Follow the factory's instructions about "merge PTH and NPTH".)

For assembly (optional)

- **BOM** (Bill of Materials): **File › Fabrication Outputs › BOM** or the BOM tool in the schematic editor. It lists all parts with values, footprints and part numbers.
- **Position file** (component placement, .pos / CPL): **File › Fabrication Outputs › Component Placement**. It tells the machine where to put each part.

Collect the Gerber files and drill files into one **zip** file and send it to the factory.

Check the Gerber files before ordering

Open the Gerber files in the KiCad **Gerber Viewer** (Project Manager) or in the online viewer of the factory. Look at every layer. Check the outline, the drills and the silkscreen. This free step finds many mistakes.

12.7 Order the board

Choice	Typical beginner choice	Notes
Layers	2 or 4	Same as in your design.
Thickness	1.6 mm	Standard. If you use impedance control, tell the factory the thickness and stackup.
Copper	1 oz outer (35 µm)	2 oz for high current.
Surface finish	HASL (cheap) or ENIG (flat, for fine pitch)	Use **ENIG** for QFN, BGA and fine-pitch parts.
Solder mask colour	Any (green is cheapest)	Colour does not change electrical function.
Impedance control	Only if your design needs it	Write the target values and the layer in the order note.
Quantity	5 or 10	Prototypes are cheap.

Popular online PCB factories include JLCPCB, PCBWay and NextPCB. Check the price, the delivery time to Pakistan and the customs rules before you order.

12.8 After the boards arrive

1. **Look at the board:** check the outline, holes, silkscreen and finish.
2. **Solder in a good order:** small parts and ICs first, then bigger parts, connectors last.
3. **Before power:** measure with a multimeter (continuity mode) between each power rail and GND. There must be no short circuit.
4. **First power-up:** use a **current-limited lab supply** (for example 100 mA). Watch the current. If it is too high, switch off immediately.
5. **Measure rails:** 5 V, 3.3 V and others. Compare with the design.
6. **Bring-up:** program the microcontroller, test blocks one by one: LED, UART, I2C, USB…

12.9 Final checklist before sending to the factory

	Check
☐	ERC clean and schematic printed / saved as PDF
☐	All footprints checked against datasheets
☐	Board outline closed, size correct, mounting holes placed
☐	Stackup, impedance and net classes set
☐	Decoupling capacitors at every power pin
☐	Differential pairs matched and checked
☐	Zones refilled, no islands, stitching vias added
☐	DRC with 0 errors
☐	Silkscreen readable, pin 1 and polarity marked, version and date written
☐	3D view checked
☐	Gerber and drill files made and opened in a viewer
☐	Project backed up

Check yourself

1. Why do you refill all zones before DRC and plotting?
2. Which layers must be plotted for a 4-layer board?
3. What are teardrops and why do we add them?
4. List the steps for the first power-up of a new board.

Appendix: Quick Reference

Shortcuts, quick tables, common mistakes, glossary and practice projects

A. Schematic Editor shortcuts

Key	Action	Key	Action
A	Add symbol	W	Wire
P	Power symbol	L	Net label
Ctrl + L	Global label	H	Hierarchical label
S	Hierarchical sheet	Q	No-connect flag
T	Text	J	Junction
M	Move	G	Drag (keeps wires)
R	Rotate	X / Y	Mirror
E	Edit properties	V	Edit value
Del	Delete	Esc	Cancel tool
F8	Update PCB	Ctrl + Z	Undo

B. PCB Editor shortcuts

Key	Action	Key	Action
X	Route track	6	Route differential pair
V	Via / change layer	PgUp / PgDn	Switch layer
7	Tune single length	8	Tune pair length
9	Tune pair skew	D	Drag track
M	Move	R	Rotate
F	Flip to other side	E	Edit properties
B	Fill all zones	Ctrl + B	Unfill zones
Ctrl + Shift + Z	Add filled zone	Alt + 3	3D viewer
Ctrl + F	Find	F8	Update PCB from schematic

Shortcuts can differ

Keys may be different in your version or if you changed them. Look in Preferences › Hotkeys, or hover over a toolbar button, to see the key.

C. Quick trace-width card (1 oz copper, 10 °C rise)

Use	Width
Normal signal	0.2 – 0.25 mm
Power up to 0.5 A	0.3 mm
Power 1 A	0.4 – 0.5 mm
Power 2 A	1.0 mm (or a pour)
Power 3 A and more	Use a pour or plane
50 Ω single-ended (4-layer, H = 0.125 mm)	≈ 0.2 mm (check with your factory)
90 Ω differential (4-layer, H = 0.125 mm)	≈ 0.16 mm width / 0.13 mm gap (check)

D. Ten most common beginner mistakes

1. Wrong footprint: pads do not fit the real chip. **Always compare with the datasheet.**
2. Mirrored pins because of “top view” / “bottom view” confusion.
3. Forgetting decoupling capacitors, or placing them far from the pins.
4. Routing fast signals over a gap in the ground plane.
5. Using 90° corners and long stubs on fast lines.
6. Not checking the polarity of diodes, LEDs and electrolytic capacitors.
7. Using a grid that is too fine in the schematic, so that wires do not connect.
8. Ignoring DRC and ERC warnings.
9. Making the traces for power too thin.
10. Not backing up the project.

E. Glossary: easy words

Word	Simple meaning
AC coupling	Blocking the DC part of a signal with a capacitor
Annular ring	The copper ring around a drilled hole
BOM	Bill of Materials: the list of all parts
Courtyard	Keep-out area around a part
Crosstalk	Noise that jumps from one trace to the next
Decoupling	Placing small capacitors near chips to give them clean power
DRC	Design Rules Check: checking the PCB against the factory rules
EMI	Electromagnetic interference: unwanted radio noise
ERC	Electrical Rules Check of the schematic
Footprint	The copper pad pattern of a part on the PCB
Gerber	The standard file format for PCB layers
Impedance	How a trace “resists” fast signals (ohms)
Net / net class	A group of connected pins / a group of nets with the same rules
Plane	A whole layer of copper (GND or power)
Prepreg / core	Insulating layers inside a PCB
Ratsnest	Thin lines that show unrouted connections
Reflection	Signal bouncing back from a mismatch
Return path	The way the current comes back to its source
Skew	Time difference between two signals
Stackup	The order and thickness of PCB layers
Stitching via	A via that joins ground copper on different layers
Thermal via	A via that carries heat to the other side
Trace	A copper line on the PCB
Via	A plated hole joining layers

F. Useful resources

- KiCad documentation and forum: docs.kicad.org and forum.kicad.info
- KiCad downloads: kicad.org
- Symbols and footprints: SnapMagic (snapeda.com), LCSC (lcsc.com), Ultra Librarian
- easyeda2kicad (free converter): search “easyeda2kicad” on GitHub
- Impedance calculators: KiCad PCB Calculator, and the calculators of NextPCB / JLCPCB / PCBWay

- Standards (to read later): IPC-2221 (generic PCB design), IPC-2141 and IPC-2152 (impedance and current)

G. Practice projects: learn by doing

Level	Project	You will practise
1	LED blinker with 555 timer (2 layers)	Schematic, footprints, simple routing, Gerber
2	3.3 V regulator board with USB-C input	Datasheets, decoupling, power traces, ESD protection
3	Microcontroller board (STM32 / ESP32 / ATmega) with USB	Crystal, 90 Ω USB pair, ground plane, placement
4	4-layer board with USB 3 / Ethernet or HDMI	Stackup, controlled impedance, differential pairs, length matching

H. Learning from the course notes

This book also includes ideas from a high-speed carrier-board design course: use the 50 mil grid in the schematic; keep a 4-layer stackup (signal / ground / power / signal) with ground right below the top layer; use ground stitching vias every 12.5 – 25 mm and near high-speed traces and layer changes; tune the shorter trace of a pair with the skew tool; use a factory impedance calculator for the exact W and S; follow the reference design for AC coupling capacitors, ESD diodes and level shifters; and remember fiducials and mounting holes.

The best of people are those who bring the most benefit to others.

— A saying (hadith) of Prophet Muhammad, peace be upon him

No one is useless in this world who lightens the burden of it for anyone else.

— Charles Dickens (attributed)

Learn well. Then teach one friend.
Every board you help someone finish is a small light in the world.

DESIGNED BY

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